

RESPONSE TO REQUEST FOR PROPOSAL

Design, Development and Implementation

Of

Artificial Intelligence (AI) Based Detailed Project Report

Appraisal Platform

PARAKH

Technical Proposal Response

Tender No.: RITES/ITES/APR/2026

Submitted to:

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Date: 5th June 2026

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Executive Summary

GlobalLogic India Private Limited proposes a solution for PARAKH, a Structured DPR Appraisal and Knowledge Platform for RITES Ltd. PARAKH replaces the current manual, inconsistently-applied DPR review process with a structured, evidence-driven platform covering the full appraisal lifecycle across all infrastructure sectors and all 16 functional modules specified in Annexure 12 of the RFP document. The platform digitizes DPR authoring, automates completeness and technical validation, provides machine-generated advisory inputs to appraisers, and builds a searchable repository of historical DPRs and engineering benchmarks, all on infrastructure wholly controlled by RITES.

Three design constraints govern every architectural decision. Data sovereignty: all AI inference and RITES project data are confined to RITES-controlled infrastructure; no third-party cloud AI API processes RITES data at any stage. Zero vendor lock-in: the entire technology stack is open-source or open-standards, with all IP (source code, model weights, training datasets, ontology, and infrastructure-as-code scripts) vesting entirely with RITES on project completion. Deployment neutrality: the platform runs unchanged on cloud, NIC MeghRaj, or on-premises environments without architectural modification.

The contract runs 24 months: nine months of development across five milestones (M1 System Design at D+1, M2 Core Platform at D+3, M3 Sector specific modules at D+5, M4/Go-Live Fully Integrated and UAT approved platform at D+7, M5 Stabilized and Performance tuned platform with refined AI models at D+9), followed by 15 months of Operations and Maintenance. The Project Manager is permanently onsite at RITES Corporate Office, Gurugram, from Day 1. VAPT is conducted at pre-UAT and pre-Go-Live stages; costs are included in GlobalLogic's quoted bid price. All licenses and subscriptions cover a minimum of three years from Go-Live.

1. Understanding of the Assignment

GlobalLogic's reading of the assignment, the problem PARAKH solves, and the scope boundaries are set out here. Section 1.7 (Scope Interpretation) and Section 1.8 (Key Assumptions) define what is in scope and what dependencies GlobalLogic expects RITES to fulfil.

1.1 Background and Context - RITES Ltd. and the DPR Ecosystem

RITES Ltd. is one of India's premier multi-disciplinary infrastructure consultancy organizations, operating across 13 sectors including railways, roads and highways, bridges and tunnels, urban transport, power, water supply, airports, ports, inland waterways, telecommunications, buildings, environment, and industrial infrastructure. With over five decades of advisory and project management experience, RITES appraises Detailed Project Reports (DPRs) across public and private sector clients, playing a critical gatekeeping role in determining project viability before capital commitment. The quality, consistency, and speed of DPR appraisal directly influences project outcomes, public investment efficiency, and national infrastructure delivery timelines.

A DPR is the foundational document for any infrastructure project. It encapsulates the technical design, financial projections, environmental impact assessments, geotechnical surveys, traffic or demand studies, and socio-economic justifications that collectively determine project feasibility. Given the multi-sector scope and the complexity of individual DPRs, which can span hundreds of pages and involve inputs from engineers, economists, environmentalists, and GIS specialists, appraising them

manually is inherently resource-intensive, inconsistently applied, and difficult to scale. PARAKH directly addresses this by digitizing, structuring, and automating the appraisal lifecycle.

1.2 Problem Statement - Gaps in Current DPR Appraisal Practices

The current DPR appraisal process at RITES, and across the broader infrastructure sector, suffers from well-documented structural gaps. First, DPRs arrive in unstructured or semi-structured formats, making systematic review labor-intensive and dependent on individual expert availability. Second, validation is largely experience-driven, leading to inconsistency in quality thresholds applied across sectors, geographies, and appraisal teams. Third, there is no accumulated knowledge base: learnings from past DPRs, benchmark data, and historical project outcomes are not captured in a structured, queryable form, preventing evidence-based appraisal. Fourth, cross-section consistency checks (e.g., alignment between financial estimates and technical specifications) are manual and error-prone. Fifth, the absence of anomaly detection capabilities means that outlier values in cost estimates, load assumptions, or demand projections may pass undetected, increasing project risk.

The net effect of these gaps is that appraisal cycles are long, quality varies across teams, and the expertise RITES' domain experts have built over decades stays locked in individuals rather than being captured where others can draw on it. PARAKH is designed to resolve each of these gaps through a structured, rule-driven, and machine-assisted platform.

1.3 Objectives of the PARAKH Platform

GlobalLogic's interpretation of the PARAKH platform objectives, drawn from the RFP and its annexures, is as follows:

- **Structured Authoring:** Enable structured, template-driven preparation of DPRs across all 13 sectors, ensuring completeness and standardization at the point of submission.
- **Automated Validation:** Automate completeness validation, technical parameter validation, cross-section consistency checks, and anomaly detection against benchmarks and engineering standards.
- **Institutional Knowledge:** Establish a Data Lake of historical DPRs, benchmark datasets, engineering standards (RDSO, IRC, BIS, NBC, AAI, ISO, ADB), and geospatial data to support evidence-based appraisal.
- **Appraisal Decision Support:** Provide machine-generated advisory inputs during appraisal review, where all recommendations require explicit reviewer action before influencing the appraisal record, and final appraisal authority rests with RITES domain experts.
- **Interoperability:** Expose a governed, API-driven integration layer compatible with GIS platforms, PM GatiShakti, BIM systems, and other national digital infrastructure.
- **Transparency and Governance:** Maintain full auditability, traceability, and explainability of every validation outcome and AI recommendation.

1.4 User Personas

The platform supports a broad range of users across the DPR lifecycle. Three primary personas are defined in detail, with all remaining roles consolidated under configurable system profiles. Access across the platform is governed by a Role-Based Access Control (RBAC) system, providing fine-

grained control over module access, chapter visibility, sector permissions, and appraisal actions, so each user interacts only with the areas and functions relevant to their role.

Three key stakeholders. Three distinct needs. One platform to align them.

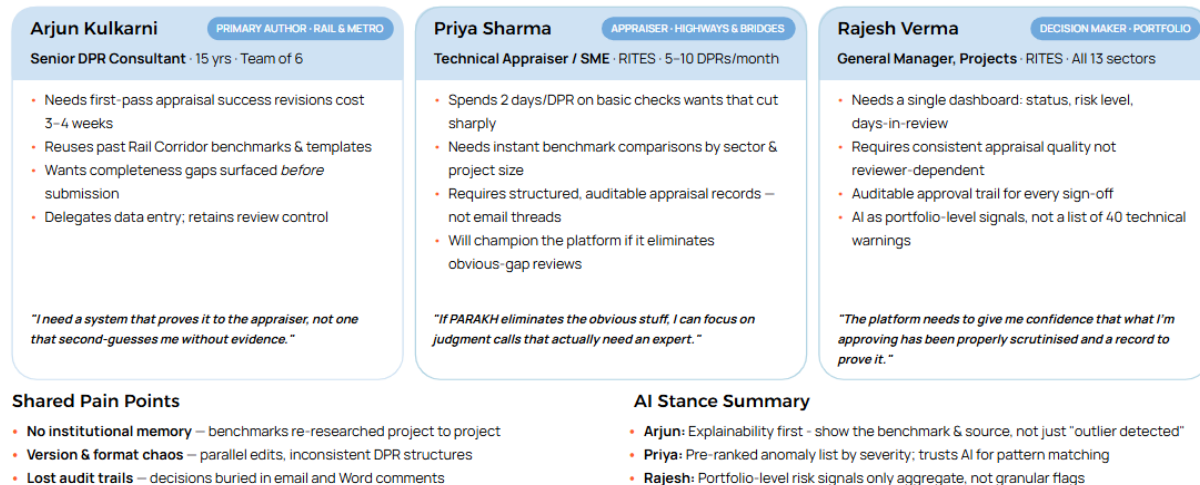


Figure 1.4 : User Personas

Other Roles:

- **Administrator** - Oversees platform configuration and governance. Manages users, roles, templates, workflows, and system settings.
- **Leadership / Decision Maker** - Provides strategic oversight and final approvals. Reviews DPR summaries, risks, and recommendations via dashboards.
- **DPR Author** - Prepares and submits DPRs. Uses templates, inputs project data, and leverages AI assistance for drafting.
- **Technical Expert (Analyst/Engineer)** - Provides domain-specific inputs (engineering, GIS, BIM, etc.). Supports DPR preparation with technical validation and analysis.
- **Appraiser / Evaluator** - Evaluates DPRs for quality, feasibility, and compliance. Uses AI-supported appraisal and provides recommendations.
- **Reviewer / Approver (Intermediate)** - Performs structured reviews before final decision-making. Ensures completeness and coordinates feedback loops.
- **Financial Analyst** - Focuses on cost estimation, financial viability, and funding structures. Validates financial assumptions within DPRs.
- **Compliance / Auditor** - Ensures adherence to policies, standards, and regulatory frameworks. Reviews DPRs against guidelines from govt. bodies.
- **Data & Insights Analyst** - Analyzes platform data and generates reports. Supports dashboards, trends, and decision-making insights.
- **Viewer / Stakeholder** - Has read-only access to DPRs and dashboards. Includes external stakeholders and monitoring agencies.

1.5 Workflow - User Journey

The user journey maps each persona interactions across the PARAKH platform, tracing every touchpoint from initial DPR creation through to final appraisal decision and tracking. The journey is structured across five stages, each representing a distinct phase of interaction with the platform.

Stage 1: DPR Creation The DPR Author initiates the process by selecting a sector-specific template and filling chapter fields. Chapters are assigned to relevant roles, with each assignee completing their designated sections. An internal review loop ensures chapters are approved or returned for revision before the DPR progresses, culminating in a saved draft ready for submission.

Stage 2: Submission Once the DPR reaches 100% completion, it is locked in the builder and formally submitted for appraisal, preventing further edits and ensuring submission integrity.

Stage 3: DPR Appraisal The appraisal stage offers two paths based on the review requirement. The AI Path runs an automated appraisal across six scoring dimensions, generating per-chapter suggestions for appraiser review. The Manual Path enables chapter-by-chapter review with structured comments and targeted flagging. Both paths converge at a single review stage, where AI-generated flags and human observations are considered together before a decision is reached.

Stage 4: Decision Leadership reviews the consolidated appraisal findings and issues one of three outcomes: Approved, Rejected with reason, or Returned with comments. Each decision triggers an immediate status update, with returned or rejected DPRs unlocked in the builder for the author to revise and resubmit.

Stage 5: Tracking The DPR Registry provides all stakeholders with real-time visibility into the status timeline, a full change history with actor attribution, and downloadable PDF reports, ensuring the entire lifecycle is transparent, traceable, and auditable.

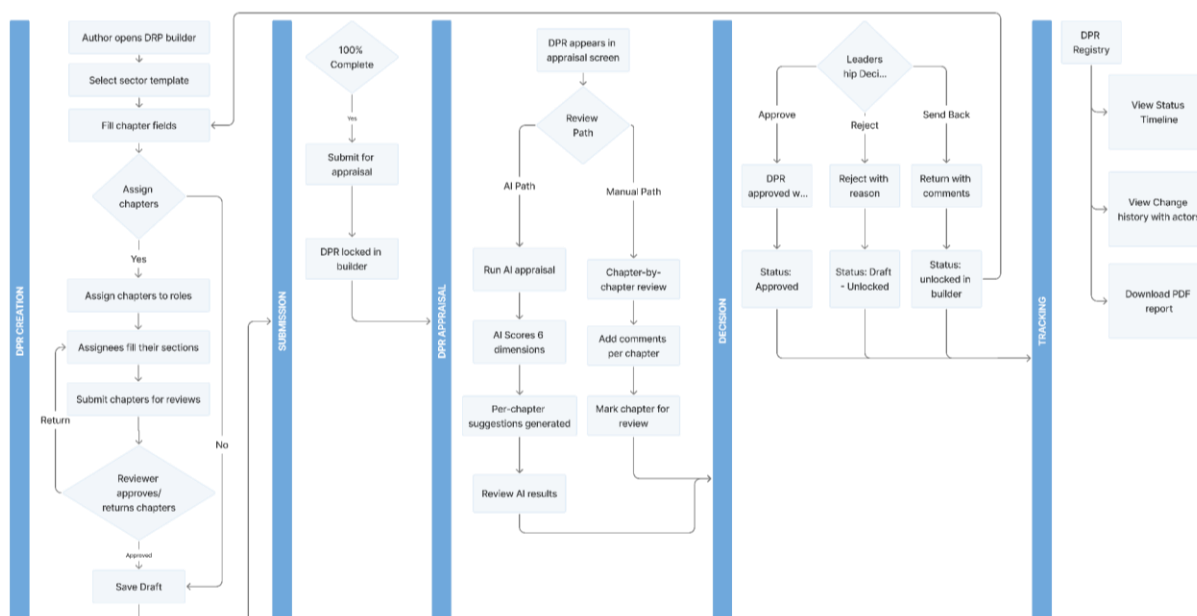


Figure 1.5 : User journey flow diagram

1.6 Feature Map

Figure 1.6 - PARAKH Platform Feature Map. The diagram below presents the high-level feature map of the PARAKH platform, organised across six functional modules operating within a Workflow Management layer and supported by a centralised Data and Knowledge Layer.

The platform covers the full DPR lifecycle: from structured template creation and guided authoring, through automated and manual appraisal, to governance-controlled decision outputs including appraisal reports, risk scores, and benchmark insights. External integration with BIM tools and PM GatiShakti (GIS) spans the authoring and appraisal layers, bringing spatial and engineering data into both DPR preparation and validation.

All modules operate on top of a shared Data & Knowledge Layer that ingests, governs, and serves historical DPR data, sector benchmarks, and reference datasets, forming the intelligence foundation of the platform.

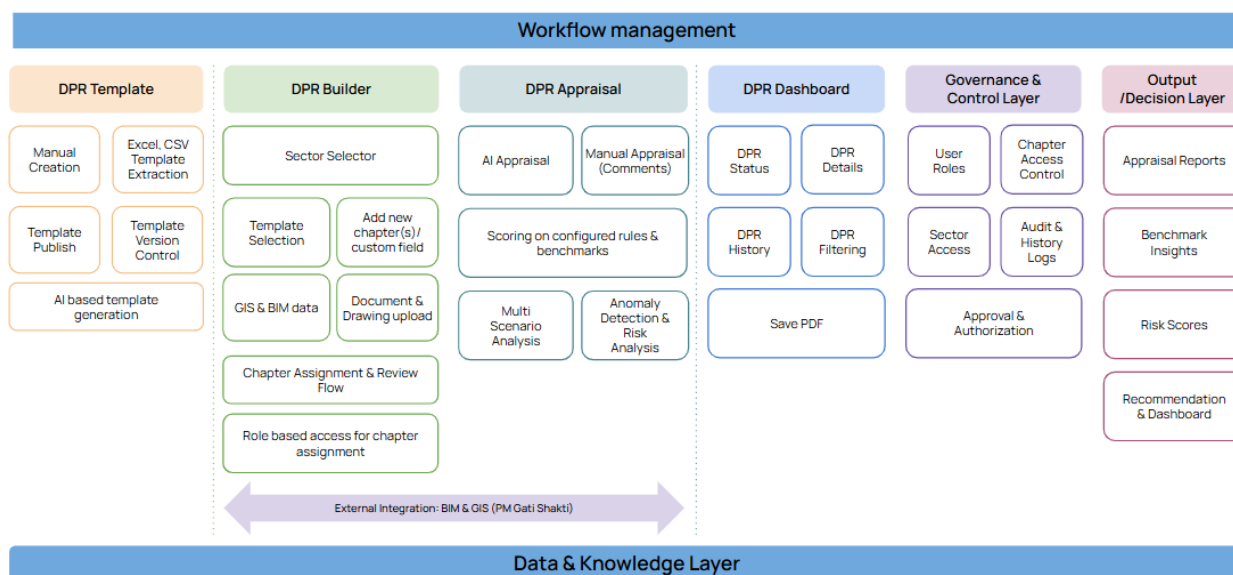


Figure 1.6 : Feature Map

1.7 Scope Interpretation and Coverage

GlobalLogic interprets the scope of work as covering full platform lifecycle responsibility: requirements finalization, system design and architecture, development, data lake implementation, integration, testing, deployment, training, and 15 months of operations and maintenance. Annexure 12 of the RFP is treated as the authoritative technical and functional baseline for design, build, and acceptance. All 16 functional modules described in Annexure 12 are in scope, spanning governance and RBAC, master data management, DPR template engine, authoring interface, validation engines (completeness, technical, consistency, anomaly), workflow management, reporting, simulation, and the historical benchmarking repository. The platform shall be designed from Day 1 to support all 13 infrastructure sectors, with extensibility to additional sectors.

GlobalLogic will also undertake bulk ingestion and extracting the data from the historical DPR data (including OCR-based processing of legacy PDF/scanned documents) for appraisal, engineering parameter library construction, and benchmark dataset assembly.

1.8 Key Assumptions and Clarifications

1.8.1 Scope & Requirements Assumptions

- Annexure 12 of the RFP is the authoritative functional and technical specification where any clause in the main RFP body references 'Annexure X', GlobalLogic treats this as referring to Annexure 12.
- Any requirements discovered post-contract that are not explicitly or implicitly covered in the RFP will be treated as change requests.
- RITES will nominate Subject Matter Experts (SMEs) across civil, structural, transport, finance, environment, and other relevant domains for knowledge transfer, template validation, and UAT to GlobalLogic domain SMEs, with reasonable turnaround commitments to avoid blocking engineering milestones.
- The DPR repository mentioned in the RFP (~1,000+ historical DPRs in PDF, Word, and scanned formats) will be made available in digital form to the delivery team within the first 30 days of contract commencement (D+30) for Data Lake ingestion and AI model training.
- RITES concurrence will be required for any risk acceptance on medium/low vulnerabilities that cannot be resolved before Go-Live.
- The key decisions of the architecture are to be discussed with the RITES team before the implementation starts.
- The decision of which features to be enabled on the mobile devices (in the PWA app) can be discussed with the RITES product team and finalized during the discovery phase.
- The proposed architecture may change in future after discussion with RITES architecture team during discovery phase

1.8.2 Milestone Assumptions

- RITES will review and provide written feedback on deliverables within 5 business days of their submission. Delays in feedback beyond this period, attributable to RITES, will not count against the delivery timeline.
- User Acceptance Testing (UAT) at Milestone 4 (D+7 Months) is assumed to have a duration of 4 weeks. RITES will nominate UAT participants within 2 weeks of notification, and UAT will be conducted in a controlled environment provisioned by the delivery team.

1.8.3 Infrastructure & Hosting Assumptions

- Post-UAT, RITES will provide and bear the cost of hosting the production infrastructure (cloud, private cloud, or on-premise). GlobalLogic will conduct all development, testing, and UAT on GlobalLogic-provisioned environments. Infrastructure sizing recommendations and IaC scripts will be provided to RITES if required.

1.8.4 Integrations Assumptions

- Access to national platforms (PM GatiShakti APIs, GIS data endpoints, RDSO/IRC/BIS standard repositories) will be facilitated by RITES for integration engineering.
- Integration with BIM tools (Autodesk, Bentley, Revit etc.) and engineering systems is assumed to be based on standard file format exchange (IFC, GeoJSON) and API-level connectivity.



Deep integration requiring proprietary SDK licensing will be flagged during Milestone 1 and may require a change request if not originally scoped.

- The delivery team will provide a sandbox/developer environment to facilitate third-party integration testing and onboarding of external agencies nominated by RITES.

1.8.5 Data & AI/ML Assumptions

- Historical DPR data provided by RITES for Data Lake ingestion may be incomplete, inconsistently formatted, or in varying quality (digital, scanned, partially hand-written). The delivery team will make best-effort attempts at extraction using OCR and NLP techniques, but data quality limitations may affect the accuracy of AI models, especially in early milestones.
- Historical DPR data containing GIS and BIM information is assumed to be available in PDF format only. Source documents are assumed to be legible and machine-readable degraded or low-resolution scanned drawings will be flagged during processing. GIS coordinates and spatial references are assumed to appear as text annotations within PDF documents, no coordinate transformation, projection conversion, or external GIS source integration is in scope. Engineering drawings are assumed to follow standard title block conventions. BIM information is assumed to be 2D drawing representations only, 3D BIM model processing is not in scope. All content is assumed to be in English. Total document volume is currently unknown, infrastructure cost and extraction effort will be confirmed during the discovery phase..
- Achievement of final-version accuracy (e.g., ≥95% text extraction, 0% hallucination for validated reference library) is contingent on the quantity, quality, and diversity of historical DPR data made available.
- All AI/ML models for the PARAKH platform will be hosted as customized LLMs on approved cloud infrastructure. Production inference will run on cloud-hosted fine-tuned models with no dependency on third-party managed inference APIs for core DPR appraisal workloads. RITES DPR data will only be processed within the agreed cloud environment under RITES-approved data governance controls. During GlobalLogic's internal Development and QA phases, which operate on GlobalLogic-provisioned infrastructure and process only synthetic or mock data, managed AI inference services may be used to accelerate development velocity; these environments contain zero real RITES data. All AI/ML model fine-tuning uses open-source tooling (GaLore for DAPT and QLoRA via Unsloth/PEFT for IFT) on spot GPU instances within GlobalLogic's controlled cloud infrastructure. The customized fine-tuned model will be deployed and served on cloud infrastructure approved by RITES.
- The proposal bid accounts for LLM inference token costs for Dev, QA & UAT environments based on rates prevailing at the time of submission. Given that AI token pricing is dynamic and subject to change in either direction, any significant deviation from the estimated costs will be brought to RITES's attention for separate discussion and agreement

1.8.6 SLA and Penalty Assumptions

- SLA obligations (uptime, latency, AI accuracy, incident response) as defined in Clause 6.3 of the RFP become effective after the stabilization phase (end of Milestone 5 / D+9). SLAs are not applicable during development, UAT, or stabilization phases
- Liquidated damages / penalties under Clause 6.4 will apply only to delays and failures attributable to the delivery team. Delays caused by RITES (e.g., delayed SME availability, delayed approvals, delayed infrastructure provisioning) will be formally documented via Delay Notices and excluded from penalty calculations.



- The project team includes a full-time onsite Project Manager at RITES Corporate Office, Gurugram. Other members including domain experts, architects will be onsite when required.

2. Proposed Solution Overview

GlobalLogic proposes PARAKH as a structured appraisal intelligence platform that replaces RITES's current manual, judgement-dependent DPR review process with a systematic, evidence-driven one. The architecture is built on three non-negotiable constraints: data sovereignty, with all AI inference and project data confined to RITES-controlled infrastructure; zero vendor lock-in, achieved through open standards and open-source components with all IP and model artefacts vesting entirely with RITES; and deployment neutrality, with an architecture that runs unchanged on cloud, NIC MeghRaj etc. The platform covers the full DPR appraisal lifecycle: structured authoring with sector-specific templates across all infrastructure domains, automated completeness and technical validation against applicable engineering standards and codal references, machine-generated advisory inputs that augment rather than replace appraiser judgement, benchmarking and comparative analytics against a curated historical DPR repository, and workflow management covering submission, review, appraisal, and final decision.

All AI-generated outputs carry explainability artefacts and are subject to mandatory human review before any appraisal decision is recorded ensuring accountability remains with RITES appraisers at all times. The platform is built on a modern, open-source technology stack selected for government-grade security, long-term maintainability, and interoperability with India's national digital infrastructure. The complete technology stack, component rationale, and licensing details are provided in Annexure A; the unpriced infrastructure Bill of Materials is in Annexure B; and the detailed architecture, deployment, and AI design are covered in Sections 3 through 8.

2.1 Enterprise Architecture Vision

GlobalLogic's architecture for PARAKH is a distributed, open-standards platform. Each domain concern, authoring, validation, analytics, integration, and AI inference, runs as a distinct, independently deployable service with well-defined interfaces rather than sharing a monolith. Each service is deployable, scalable, and maintainable in isolation, so RITES or a nominated agency can enhance, scale, or replace any component without dependency on a single vendor or GlobalLogic's ongoing involvement.

The enterprise architecture is organized in three horizontal planes:

- **Presentation Plane:** device-agnostic frontend applications and developer/integration interfaces
- **Application Plane:** microservices implementing business logic, validation engines, AI/ML inference, and workflow orchestration
- **Data Plane:** data Lake, transactional databases, geospatial stores, vector indexes, and caching layers

These planes are decoupled through a managed API Gateway and an event streaming backbone, which supports asynchronous processing, real-time data exchange, and resilient operation.

2.2 Alignment with Digital Public Infrastructure (DPI) Principles

PARAKH is positioned as a component of India's Digital Public Infrastructure for infrastructure governance, consistent with the India Stack approach. The platform meets DPI requirements on four dimensions: openness through OpenAPI-compliant interfaces; interoperability through standards-based data exchange using OGC, IFC, and canonical data models; scalability through horizontal scaling via container orchestration platform; and no proprietary lock-in through open-standards-only component selection. All APIs will be published as machine-readable OpenAPI 3.0 specifications, making the platform accessible to any authorized national platform or agency system. The data architecture supports future federation with the PM GatiShakti National Master Plan portal and other Digital India initiatives.

2.3 Architectural Philosophy

- **Modularity:** Each functional module (validation engine, authoring interface, analytics, etc.) is a discrete microservice with its own data store where appropriate, deployable and scalable independently.
- **Data Centricity:** The Data Lake is the single source of truth for all DPR data, benchmarks, standards, and AI model training data. All modules read from and write to governed data contracts.
- **Openness:** No proprietary protocols, data formats, or frameworks are used where open alternatives of equivalent capability exist. Event streaming platform, geospatial tile and feature server, identity and access management platform, relational database, object storage layer, and search and analytics engine are examples of this commitment.
- **Auditability by Design:** All validations, scoring outputs, and AI recommendations carry a full audit trail traceable to the rule, standard, or model version that generated them.

2.4 Human-in-the-Loop Design Philosophy

A foundational design constraint for PARAKH is that the platform must not replace expert judgment. Every AI-generated recommendation whether a validation flag, anomaly alert, benchmark comparison, or completeness warning is presented as advisory input to RITES domain experts. The appraisal authority, scoring decisions, and final sign-off remain with RITES-designated reviewers. The platform enforces this through workflow controls: no AI-generated output transitions a DPR to the next appraisal stage without an explicit human approval action. This design prevents automation bias, maintains institutional accountability, and satisfies audit requirements.

2.5 Vendor Neutrality and Avoidance of Lock-In

GlobalLogic commits to a zero-lock-in architecture. All data will be stored in open, portable formats (Parquet/Iceberg for the Data Lake; relational database-compatible schemas for transactional data; GeoJSON/Shapefile for geospatial data; IFC for BIM). All business logic will be documented and transferable. No component will require GlobalLogic's ongoing involvement for operation, maintenance, or enhancement. RITES will receive complete source code, IaC scripts, container images, and runbooks at each milestone and on project completion.

2.6 Compliance with Indian and Global Standards

- OWASP Top 10 (web application), OWASP API Security Top 10, OWASP ASVS Level 2

- ISO 9001:2015 (Quality Management), ISO 27001:2022 (Information Security), ISO 19650 (BIM Information Management)
- OGC standards: WMS 1.3, WFS 2.0, WMTS, WCS (geospatial services)
- IFC 4.x (BIM data exchange)
- OpenAPI 3.0 (API documentation and conformance)
- CERT-In empanelled VAPT standards; SSDLC practices
- Digital Personal Data Protection (DPDP) Act 2023; applicable GoI data protection guidelines and the IT Act, 2000

Make in India Policy (MII): GlobalLogic India Private Limited is an Indian-registered company and confirms compliance with applicable Government of India Make in India guidelines for procurement of goods and services per Correction Slip No. 3 to RITES Standard REOI Document. Platform components are selected in conformance with the MII preference policy where applicable; indigenously developed capabilities (IndicBERT fine-tuning, Indian engineering standards integration) and open-source components with active Indian community contributions are prioritised throughout the solution.

MeitY/NIC Guidelines: The platform will be developed and deployed in compliance with applicable Ministry of Electronics and Information Technology (MeitY) and National Informatics Centre (NIC) guidelines for Government of India IT systems, including security, hosting, and data handling requirements. Standardisation Testing and Quality Certification (STQC): The PARAKH web application and APIs will undergo STQC testing prior to Go-Live in accordance with MeitY/STQC requirements. Guidelines for Indian Government Websites (GIGW): The PARAKH web portal will comply with GIGW 2.0 guidelines covering accessibility (WCAG 2.1 AA), multilingual support, browser compatibility, and usability standards mandated for GoI portals.

3. Proposed Platform Architecture - PARAKH

This section covers the technical architecture: the system layers, microservices structure, event-driven integration backbone, security design, and scalability approach. The rationale for each architecture decision and the specific open-source components chosen are included for RITES technical reviewers.

3.1 High-Level System Architecture Overview

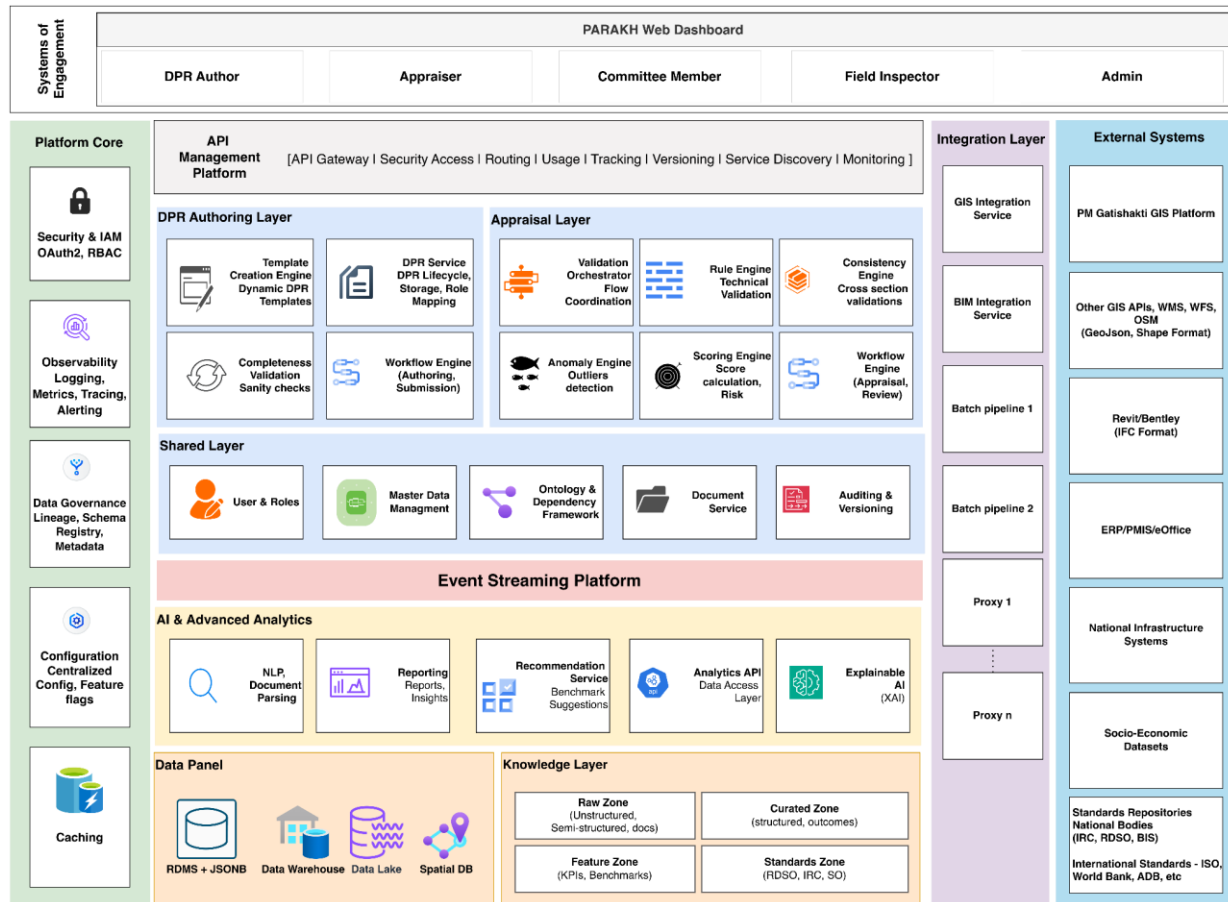


Figure: 3.1 - High Level System Architecture

The PARAKH platform architecture consists of various layers with different cross-cutting bands. The API and Access Layer is the single, secured entry point for all traffic human users, web/mobile apps authorised agency integrations, and batch connectors. The API Gateway handles routing, throttling, and TLS termination. The Auth Service issues and validates JWT tokens, enforces multi-factor authentication where required, and federates against RITES LDAP/Active Directory. The User and Roles component resolves the caller's identity to a platform RBAC role and sector scope, which flows downstream as injected request headers. Nothing enters the platform without passing through this layer (refer to Figure 3.1).

The **DPR Authoring Layer** hosts all capabilities that support a DPR Author from template selection to submission. The Template Creation Engine generates dynamic, sector-aware DPR forms from configurable chapter/section/field definitions covering all infrastructure sectors as configuration profiles, not separate codebases. The DPR Service manages DPR lifecycle state, storage, and role-to-section mapping. The Document Service handles file upload (IFC, PDF, DWG), virus scanning, and metadata tagging. The Workflow Engine coordinates authoring-stage transitions and SLA timers. The Ontology and Dependency Framework within this layer activates conditional fields and validation triggers based on site conditions (terrain, seismic zone, hydrology). Auditing and Versioning records every field edit with actor, timestamp, and value creating a tamper-evident history from the first keystroke.

The **Appraisal Layer** contains the platform's automated quality assurance pipeline, orchestrated by the Validation Orchestrator that sequences engines and aggregates their findings into a consolidated advisory report. The Completeness Validation engine checks mandatory field coverage, conditional field resolution, and attachment completeness. The Rule Engine applies DSL technical validation rules sourced from RDSO, IRC, BIS, MoRTH, and NBC standards (after discussing with RITES domain SME's) emitting clause-level rule traces for every finding. The Consistency Engine reconciles cross-section data (financial vs. technical vs. schedule). The Anomaly Engine applies Isolation Forest ML models and statistical IQR/z-score methods against the benchmark corpus to surface statistical outliers. The Appraisal Workflow Engine governs the downstream state machine

(Submitted → Under Appraisal → Pending Committee → Approved / Returned)

The Scoring Engine computes the structured appraisal score and risk profile that forms the basis of the committee's recommendation.

The **AI and Advanced Analytics Layer** provides intelligence services that inform but never replace the human appraiser. The NLP and Document Parsing service extracts entities, cost figures, and standard references from uploaded free-text documents using NLP processing library, Transformer models. The Recommendation Service compares submitted DPR parameters against historical benchmarks and suggests corrections. The Analytics API exposes the data access layer for dashboards and external reporting consumers. The Reporting service generates appraisal summary reports and KPI views for RITES management. Explainable AI (XAI) produces feature attribution-based feature importance charts and natural-language explanations for every ML-generated flag.

The **Knowledge Layer** is organised into four zones that form PARAKH's reference corpus. The Raw Zone stores ingested content in its original form: historical DPR PDFs, IFC models, GeoJSON files, and socio-economic datasets. The Curated Zone holds the canonicalized, structured representations of all DPR entities after transformation; this is where the Canonical DPR Data Model resides. The Feature Zone aggregates KPIs, sector-wise benchmark distributions (P10/P50/P90), and ML training feature vectors. The Standards Zone holds the structured parameter libraries extracted from national standards bodies (RDSO, IRC, BIS, NBC, AAI) and international references (ISO, ADB), the authoritative source for all rule-engine and benchmarking lookups. The standards will be validated with RITES domain SME's team during the discovery phase.

The Data Panel beneath the Knowledge Layer provides the polyglot persistence tier: a relational store (RDBMS with JSONB for flexible schema extensions) for transactional data and structured DPR content; a Spatial Database (geospatial database extension) for geospatial queries and GIS constraint analysis; a Data Lake (open table format on object storage layer) for the four Knowledge Layer zones; and a Data Warehouse for pre-aggregated analytics served to dashboards and management reports. The Shared Layer provides common technical services consumed by all functional layers above it. Master Data Management (MDM) holds versioned sector unit rates, lookup masters, and organisational hierarchies. The Caching layer (distributed cache) accelerates repeated lookups including GIS polygon queries, rate-schedule reads, and auth token introspection. The Ontology and Dependency Framework (also surfaced in the Authoring Layer) is maintained here as a shared service so that both authoring-time field activation and appraisal-time rule evaluation draw from the same OWL-based entity graph. Auditing and Versioning provides the platform-wide immutable event log.

The **Integration Layer** manages all connectivity to external systems whether they are open source or proprietary. Outbound API connectors reach PM GatiShakti (WMS/WFS for GIS constraint layers),



other GIS APIs (NRSC, Sol, State GIS in GeoJSON/Shapefile formats), BIM authoring tools, enterprise systems (ERP/PMIS/eOffice), National Infrastructure Systems, and Standards Repositories (IRC, RDSO, BIS, ISO, ADB). Batch and streaming connectors ingest Socio-Economic Datasets and historical DPR archives. All external calls are mediated through the Integration Layer with circuit-breaker, retry, and graceful-degradation patterns.

Three cross-cutting bands span all layers vertically. Observability provides centralised logging, metrics, tracing, and alerting across every service. Security and IAM (OAuth2/RBAC enforced at every layer boundary, secrets management system for secrets, mTLS between services via service mesh) maintains the platform's zero-trust posture. Data Governance ensures that data flowing through every layer is traceable, quality-checked, and auditable. A Configuration band (centralised config service, feature flags) allows operational behaviour to be changed without redeployment. The thirteen infrastructure sectors railways, roads, bridges, urban transport, power, water, airports, ports, inland waterways, telecom, buildings, environment, and industrial are configuration contexts within this architecture, not separate layers or services.

3.2 Three-Layer Logical Architecture

3.2.1 Layer 1: DPR Preparation Layer

This layer encompasses all capabilities that enable project proponents to prepare, structure, and submit DPRs. It includes the Template Engine (an engine provides configurable chapter/section/field structure per sector), the Authoring Interface (structured data entry, document upload, form validation), the Dictionary and Standards Repository (controlled vocabulary, units, codes), the Ontology and Dependency Framework (context-aware rules for conditional fields), and Version Control and Submission Workflow management. The frontend is a Progressive Web App (PWA) built in frontend framework with TypeScript, supporting desktops, tablets (including iPad), and mobile devices too as per recommendations from RITES. Offline-capable read access is supported for field use in low-bandwidth conditions.

3.2.2 Layer 2: Appraisal and Evaluation Layer

This layer implements the core appraisal intelligence of PARAKH. The Completeness Validation Engine checks every mandatory field and section against the approved template. The Technical Validation Engine applies rule-based and dependency-driven checks across all sectors, validating engineering parameters against RITES-approved standards and benchmarks. The Consistency Engine performs cross-section reconciliation, verifying that financial cost estimates align with technical quantities, that scheduling assumptions are consistent with resource projections, and that environmental impact assessments correspond to project footprint. The Anomaly Detection Engine uses statistical and ML-based methods to flag outliers in cost ratios, traffic volumes, soil parameters, and other DPR inputs against historical benchmarks. All engine outputs feed into a configurable Review Workflow that manages observations, revision cycles, and appraisal scoring.

3.2.3 Layer 3: Knowledge and Data Layer (Data Lake)

The Data Lake is the platform's reference corpus. It ingests and stores historical DPRs (structured and unstructured), engineering parameter libraries, Indian standards and engineering norms, international standards and reference frameworks, geospatial and GIS datasets, socio-economic and demographic data, and environmental datasets. This layer underpins benchmarking, ML model training, and



standards-driven validation. The storage tier uses S3-compatible object storage for raw and processed data, open table format (an open table format with built-in schema evolution) for analytical workloads, distributed data processing framework for batch and streaming data processing, and a relational database with geospatial database extension for transactional and geospatial data. workflow orchestration engine orchestrates all ingestion, transformation, and enrichment pipelines.

3.3 Microservices and API-First Architecture

PARAKH decomposes its backend platform with REST API framework for high-throughput REST and gRPC endpoints; asynchronous task framework for async and scheduled background jobs) and enterprise application framework for the stateful DPR Lifecycle and Workflow services into the following initially-identified bounded-context microservices (additional services will be defined during detailed design), each owning its own data store and exposing a versioned API contract: Identity & Auth, DPR Lifecycle, Validation & QC, AI Advisory, GIS & Spatial, Document & BIM, Workflow & Notification, and MDM & Knowledge. Each service is packaged as an OCI-compliant container image, deployed on a container orchestration platform, and managed through a GitOps pipeline. No service accesses another service's database directly; all cross-service communication is mediated either through the API Gateway (synchronous) or event streaming platform (asynchronous).

API Gateway serves as the single external ingress. It validates every inbound JWT against the IAM platform's endpoint, enforces rate limits per consumer tier, applies CORS policies, and injects downstream headers derived from token claims. Services read these headers directly for RBAC enforcement, removing the need for repeated token introspection inside the service layer. The API surface is protocol-differentiated by workload type. REST/JSON is used for all CRUD operations exposed to browser and mobile clients. gRPC with Protocol Buffers is used for high-throughput, latency-sensitive calls between the Validation Engine and the AI Advisory Service, where schema rigidity and binary efficiency matter. GraphQL is exposed for dashboard and reporting queries where clients need to compose cross-entity views without multiple round trips.

3.4 Event-Driven Architecture (EDA) and Streaming

The event streaming platform forms the event streaming backbone of PARAKH. All platform services that need to react to state changes consume from event streaming platform topics rather than polling databases or calling upstream services synchronously. This decoupling means that computationally expensive operations i.e. IFC file parsing, AI advisory generation, and GIS spatial analysis run at their own pace without blocking the user-facing request thread.

The platform defines a governed event taxonomy. Core topics include: dpr.submitted, dpr.state.changed, ifc.uploaded, ifc.parsed, qc.advisory.generated, dpr.approved, dpr.returned, gis.constraint.checked, field.sync.requested, and field.sync.conflict. Each topic name encodes the owning domain and the event name; consumer group naming follows a <service>.<topic> convention to prevent accidental cross-consumer interference. Producers must register a schema version before publishing; consumers deserialise against the registered schema and reject malformed events at the boundary. Schema evolution follows backward-compatible rules new optional fields are permitted, field removal requires a new major schema version with an overlap migration period.

All events published to the event streaming platform are governed by a schema registry). Event schemas are defined in Apache Avro format and registered in the registry before any producer is permitted to publish a topic. Consumers reject events that fail schema validation at deserialization



time, preventing malformed payloads from propagating downstream. Schema evolution follows a BACKWARD_TRANSITIVE compatibility policy every new schema version must remain backward-compatible with all prior versions on the same topic, guaranteeing that consumers on an older schema version continue to function correctly during rolling deployments and staged rollouts.

DPR state transitions use transactional event delivery (exactly-once semantics) to guarantee that state updates in the database and the corresponding streaming event are committed atomically. For advisory generation and notification delivery, at-least-once semantics with consumer-side idempotency keys are sufficient and offer higher throughput. Failed consumer events are routed to per-topic dead-letter queues (*.dlq); the Operations Dashboard surfaces DLQ depth as a first-class SLA metric, and an automated reprocessing workflow retries DLQ messages after root-cause remediation.

3.5 Integration Architecture

The API Gateway is the single entry point for all external API consumers: frontend applications, RITES enterprise systems, third-party integrators, and the developer sandbox. It provides OAuth2/OIDC authentication, rate limiting, request logging, API versioning, and developer portal capabilities. All API interactions are logged to the central audit system with request metadata, user identity, and response codes. The API Gateway enforces OWASP API Security Top 10 controls including authentication on all endpoints, input schema validation, rate throttling per consumer, and TLS 1.3 for all traffic.

3.6 Security Architecture

Identity and access management is anchored on identity and access management platform, configured as an OIDC provider with a SAML 2.0 bridge for any RITES enterprise SSO integration. Identity and access management platform federates against RITES's LDAP/Active Directory to validate credentials and resolve group membership, AD groups are mapped to RBAC roles at login time. JWT access tokens are short-lived (15 minutes) with claims sub, roles, sectors, and org_unit. Refresh tokens are issued as HttpOnly cookies to prevent JavaScript access. Multi-factor authentication (TOTP) is enforced by policy for all Admin and Committee roles.

The secrets management system manages all secrets and credentials, relational database credentials are dynamic, each service instance requests a short-lived database username and password (TTL: 1 hour) from Vault's database secrets engine; credentials are automatically revoked on lease expiry and rotated without service restarts. Transport security follows a layered model. TLS 1.3 is enforced on all endpoints exposed beyond the cluster boundary.

3.6.1 Authentication and Authorization

Identity and access management platform serves as the central Identity Provider (IdP). This enables RITES to integrate its existing directory services (Active Directory/LDAP) as an upstream identity source while identity and access management platform manages token issuance and session governance. RBAC is enforced at the API Gateway level and within each microservice, ensuring that users can only access resources and perform actions permitted by their assigned roles. Role definitions are configurable by RITES administrators without requiring code changes. Service-to-service authentication uses short-lived JWT tokens with private key signing.

3.6.2 OWASP API Security Compliance

The platform is designed to comply with OWASP Top 10, OWASP API Security Top 10, and OWASP ASVS Level 2. Static code analysis, dependency vulnerability scanning, and dynamic application security testing are integrated into the CI/CD pipeline. VAPT is conducted at pre-UAT and pre-Go-Live stages, covering application, API, data, infrastructure, and AI/ML layers.

3.6.3 Encryption in Transit and at Rest

All data in transit is encrypted using TLS 1.3. Data at rest is encrypted using AES-256 for database tablespaces, object storage buckets, and backup archives. Encryption keys are managed through a secret management system, with key rotation policies aligned to RITES security requirements. Sensitive fields in the database (PII, authentication credentials) are encrypted at the application layer in addition to storage-level encryption.

3.7 Scalability and High Availability Design

All stateless PARAKH services, the Validation Engine, AI Advisory Service, GIS Service, Document Service, and Workflow Engine are deployed with horizontal pod autoscalers (HPA). Scale-out is triggered on CPU utilisation exceeding 70% or on custom event streaming platform consumer-lag metrics published to metrics collection system; scale-in is gated by a five-minute stabilisation window to prevent thrash during transient spikes. Pod Disruption Budgets (PDB) guarantee a minimum of two replicas remain available during rolling updates and node drain operations. The primary data store is a relational database, deployed in a primary-standby topology with synchronous streaming replication. The standby in the same availability zone receives every WAL record before the primary acknowledges a write, guaranteeing zero data loss on primary failure. Separate read replicas (asynchronous) serve all reporting, dashboard, and analytics queries isolating the OLAP workload from the transactional primary.

The distributed cache provides the caching layer. The GIS Service caches PM GatiShakti API responses keyed on the SHA-256 hash of the project boundary polygon with a TTL of one hour - eliminating repeated external API calls for the same project area. The API gateway uses distributed cache for rate-limit counters and for identity and access management platform token introspection caching, reducing identity and access management platform load on high-concurrency pages. Session state for the frontend framework 18 SPA is kept server-side in distributed cache with short TTL and automatic expiry.

The object storage layer (S3-compatible) holds all large binary assets, IFC models, PDF reports, DWG drawings, and field-visit photographs. Object storage layer is configured with erasure coding across multiple nodes, providing resilience against single-node loss. For large file uploads (IFC up to 2 GB), the Document Service issues presigned URLs that allow the client to upload directly to object storage layer without routing the byte stream through the application tier, eliminating an unnecessary network hop and reducing application-pod memory pressure during concurrent uploads. service mesh circuit breakers are configured with a 5-second open window: if an upstream service returns 5xx errors above a threshold, the circuit opens and callers receive an immediate failure (with a meaningful error message) rather than waiting for a timeout. Event streaming platform consumer-lag alerting (metrics collection system + alerting system) fires a warning when any topic's consumer lag exceeds the SLA threshold, so operations staff can scale the relevant consumer deployment before user-visible impact occurs. All pods expose HTTP liveness and readiness probes; container orchestration platforms will



not route traffic to a pod until its readiness probe succeeds, and will restart any pod whose liveness probe fails.

Database high availability is achieved through database streaming replication with automatic failover. The event streaming platform is deployed as a multi-broker cluster with configurable replication factors. The data lake's object storage tier (object storage layer) is deployed in distributed erasure-coding mode for fault tolerance. All services are designed for at least N+1 redundancy. Load balancing uses NGINX Ingress Controller with session affinity where required. Target availability and recovery objectives (SLA, RTO, RPO) will be defined and agreed with RITES during project initiation.

3.8 Device-Agnostic and Responsive UI Architecture

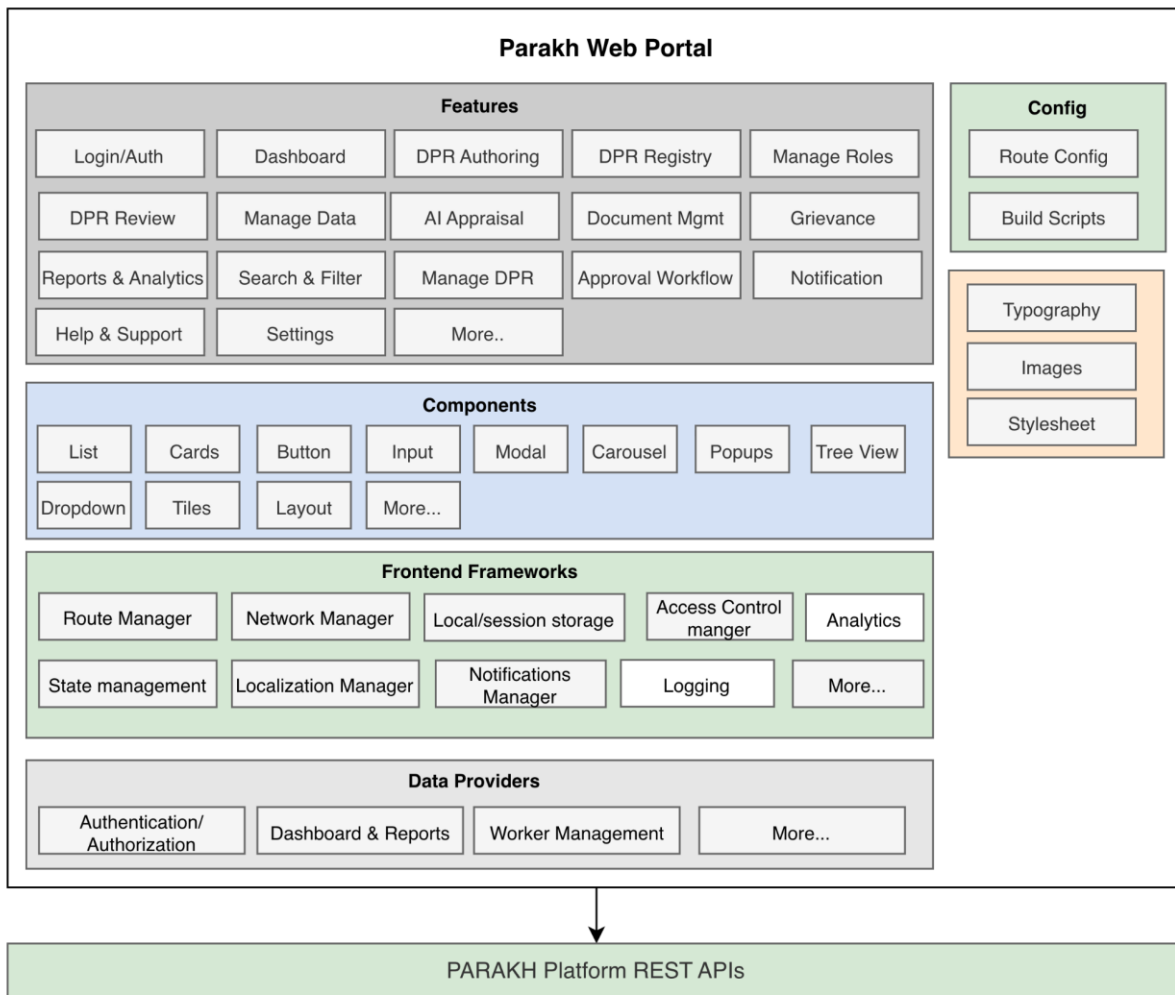


Figure 3.8. Frontend UI Architecture

The frontend is a single-page application (SPA) built with frontend framework and TypeScript, using a component library designed for responsive, accessible use across desktop, tablet (including iPad), and mobile form factors. Maps and GIS visualization are powered by open-source mapping library JS, an open-source mapping library supporting OGC tile services. PWA capabilities (service worker, manifest) enable offline read access and background synchronization for field use in low-bandwidth

environments. Accessibility compliance targets WCAG 2.1 Level AA. English support is built into the i18n framework from the outset.

Mobile Distribution and PWA Deployment Strategy

The PARAKH user interface is developed as a Progressive Web App (PWA) using React.js 18, ensuring a single codebase delivers a native-like experience across all devices. This strategy maximizes deployment velocity and feature parity while addressing the requirement for Android and iOS store distribution.

- **PWA Core Capabilities:** The PWA architecture leverages Service Workers for offline caching of essential resources (templates, lookup tables, and the UI shell), enabling read-only access and limited data capture in low-connectivity or offline environments (e.g., field surveys). The web manifest allows users to "install" the app directly from the browser onto their device home screen, bypassing traditional app stores for general enterprise use.
- **Deployment to Android Store (Google Play):** For deployment to the Google Play Store, the PWA will be wrapped using a **Trusted Web Activity (TWA)**. TWA allows the verified PWA URL to run full-screen within a native Android wrapper, ensuring the app is discoverable on the store while minimizing maintenance overhead.
- **Deployment to iOS App Store (Apple App Store):** For deployment to the Apple App Store, the PWA will be encapsulated using a native wrapper that embeds the web application within a **WKWebView**. This approach provides the native iOS shell required for distribution through the App Store and ensures the PWA meets Apple's store submission guidelines.

4. Data Lake and Data Management Architecture

This section describes the Data Lake, PARAKH's reference corpus of historical DPRs, engineering standards, geospatial data, and benchmark datasets. Evaluators assessing the AI/ML foundation, standards ingestion (RDSO, IRC, BIS, NBC, AAI), and long-term knowledge accumulation should read this alongside Section 6.

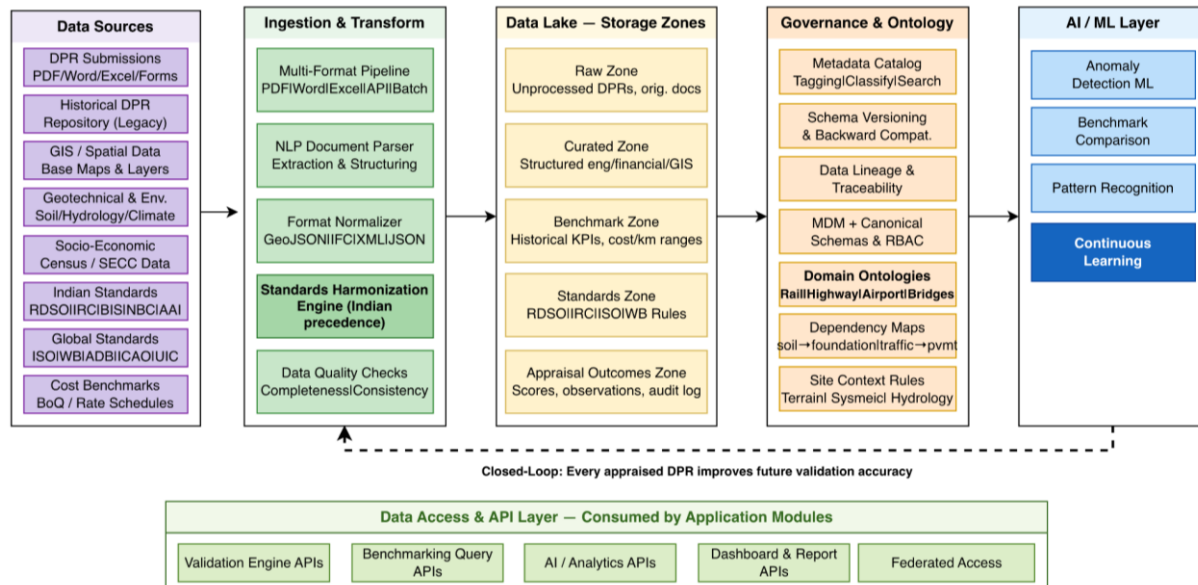


Figure 4. Data Architecture

4.1 Data Lake Design Principles and Objectives

The PARAKH Data Lake is a governed, structured repository built on distributed data processing framework (batch and micro-batch transformation), workflow orchestration engine (DAG orchestration), event streaming platform (event-driven ingestion), and open-source BI platform (open-source self-service BI and analytics dashboards for RITES management). It converts historical DPR data, engineering standards, geospatial datasets, and socio-economic information into queryable, benchmark-ready datasets. The design follows a medallion architecture: Bronze layer (raw ingested data, immutable), Silver layer (cleaned, validated, and standardized data), and Gold layer (curated analytical datasets, benchmark tables, and ML feature stores). Each layer is independently evolvable, and every analytical output traces back to its source records through the lineage metadata attached at ingestion.

4.2 Data Sources and Ingestion Framework

4.2.1 Structured Data Sources (DPRs, Engineering, Financial)

Structured DPR data is ingested through the PARAKH Authoring Interface via structured forms, validated API payloads, and Excel/CSV batch uploads. Financial models, cost estimates, and BOQ data are ingested from standardized templates. Engineering parameter values (spans, loads, gradients, material quantities) are captured in a canonical schema aligned with the sector-specific parameter libraries.

4.2.2 Geospatial and GIS Data Sources

Geospatial data is ingested from OGC-compliant WMS/WFS services (PM GatiShakti, NRSC, Survey of India, state GIS portals), uploaded GeoJSON/Shapefile/KML files, and geospatial database extension-stored spatial layers. The geospatial tile and feature server serves as the internal OGC service layer, exposing WMS, WFS, WMTS, and WCS endpoints. Spatial data is stored in geospatial

database extension with appropriate SRID configurations and indexed for fast spatial query performance.

4.2.3 Socio-Economic, Environmental and Geotechnical Data

Census data (Population Census, SECC), demographic datasets, environmental baseline data (land use, forest cover, protected areas from FSI/MoEFCC), and geotechnical reference datasets (soil classification zones, seismic zones per BIS, flood plains) are ingested as reference datasets during Phase 3. These datasets enrich DPR context validation.

4.2.4 Legacy and Historical DPR Ingestion

Historical DPRs in PDF format, including scanned and non-searchable variants, are processed through an Intelligent Document Processing (IDP) pipeline. This pipeline applies a layout-aware document intelligence engine for text and structure extraction, with an OCR fallback for degraded scans; spatial NLP for named entity recognition and section identification; and multimodal transformer models fine-tuned on infrastructure domain vocabulary for parameter extraction and table parsing. Extracted data is mapped to canonical schemas, validated through automated quality gates with manual review for low-confidence records, and loaded into the Silver and Gold layers. There can be complexities regarding the extraction of GIS & BIM data from the historical DPRs. This can be later looked at once the historical DPRs are received and then a detailed strategy can be discussed with the RITES team for this to convert to a structured format (if required).

4.3 Data Structuring and Transformation Framework

Distributed data processing framework handles all batch and micro-batch transformation workloads. Spark jobs are orchestrated by workflow orchestration engine, with DAGs versioned in source control alongside application code. Transformation logic includes schema normalization, unit conversion (applying the Standards Dictionary rules), deduplication, outlier flagging for manual review, and benchmark computation. Schema evolution is managed through open table format's built-in schema versioning, maintaining backward compatibility as new sectors and parameters are added.

4.4 Canonical Data Models and Ontology Framework

A Canonical DPR Data Model is the normalised, sector-agnostic schema that governs how all DPR content is represented, stored, validated, and analysed within PARAKH. Its purpose is to ensure that every DPR regardless of sector, project type, or authoring tool is persisted in a consistent, machine-readable form that all platform engines (validation, anomaly detection, benchmarking, reporting) can process without sector-specific branching logic. The model sits in the Curated Zone of the Knowledge Layer and is the single source of truth to which raw submissions are transformed during ingestion.

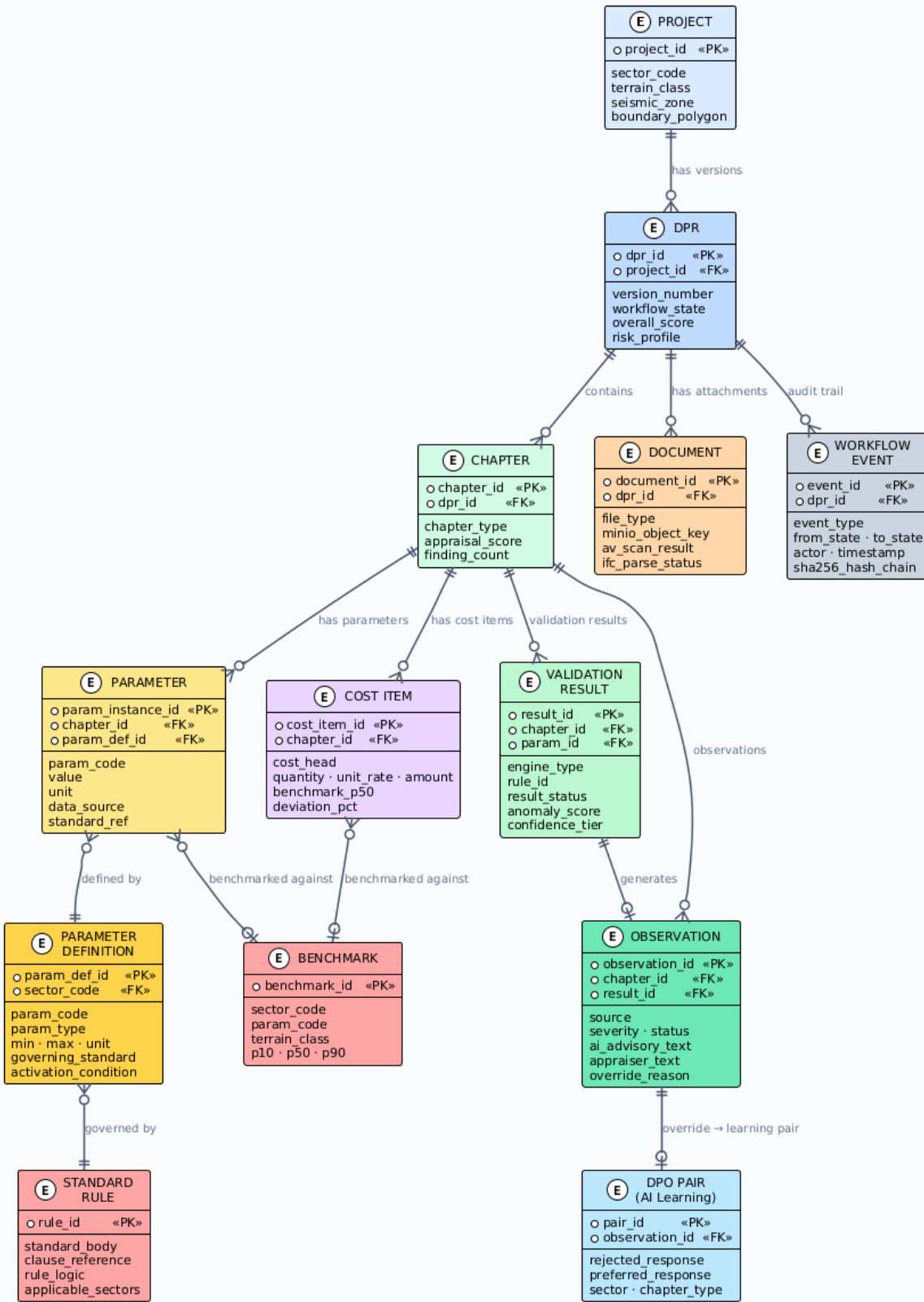


Figure 4.4 - Canonical Data Model

The model is organised around eight core entity types. The project is the root entity, carrying the project identifier, name, sector code (one of the 13 RITES sectors), project type, geographic context (district, state, terrain class, seismic zone, hydrological region), proponent organisation, and geospatial bounding polygon. DPR is the versioned document entity linked to a Project, holding the template version used, the current workflow state (Draft, Submitted, Under Appraisal, Pending Committee, Approved, or Returned), submission timestamp, and the SHA-256 hash chain anchor for audit integrity. Section and Chapter entities represent the hierarchical DPR structure, each carrying a completeness score and a list of child parameters. Parameter is the atomic unit of structured DPR content: a typed (numeric, text, boolean, date, geometry) value identified by a param_code drawn from the Standards Zone, accompanied by unit, data source (manual entry, BIM-derived, or GIS-derived), confidence score, and the governing IS standard reference clause.

The **Ontology and Dependency Framework** extends the canonical model with semantic relationships. Modelled in OWL (Web Ontology Language), it defines entity relationships (a Bridge DPR has-a Foundation type that requires-a Geotechnical Investigation parameter), conditional activation rules (if terrain_class=Hilly then enable Snow Load Parameter), and cross-entity validation constraints (the sum of all CostItem amounts must reconcile with the DPR's stated Total Project Cost within a configurable tolerance). These ontological rules are evaluated at authoring time (to guide the author) and again at appraisal time (to verify consistency), using the same rule definitions, ensuring no gap between what the author was asked to fill and what the appraiser checks. The ontology is stored in the Shared Layer and versioned alongside the parameter catalogue so that historical DPRs are always validated against the rule version in force at their submission date not the current version.

Interoperability follows directly from the canonical model design. External consumers authorised agency APIs, PM GatiShakti, or future national DPI platforms interact with PARAKH through the Data Access Layer (Section 4.10), which serialises canonical model entities to JSON-LD (linked data format preserving semantic context), OpenAPI-compliant REST payloads, or GeoJSON for spatial entities. The OpenAPI specification published on the Developer Portal is generated directly from the canonical schema, so the API contract and the internal data model stay in sync. PARAKH's Data Lake is therefore consumable by RITES's own analytics infrastructure or any future national data integration initiative without requiring bespoke export transformations.

4.5 Metadata, Data Catalog and Lineage Management

The open-source data catalog provides asset discovery, metadata management, data lineage visualization, and quality dashboards. Every dataset, table, and API endpoint in PARAKH is registered in the catalog with owner, classification, schema, and lineage information. Data lineage is tracked from raw ingestion through transformations to analytical outputs and AI model inputs, giving auditors a complete trace of how any benchmark value or validation rule was derived.

4.6 Data Quality, Governance and Stewardship

Data quality is enforced at every stage of the ingestion pipeline through a data validation framework (open-source data validation framework). Quality dimensions include completeness, accuracy (within valid ranges per sector standards), consistency (cross-field rules), and timeliness (freshness of reference datasets). Quality failures trigger automated alerts and quarantine of affected records, preventing contaminated data from reaching the Gold layer. RITES retains ownership of all data

governance decisions: quality thresholds, benchmarking logic, and interpretation frameworks are defined and approved by RITES.

4.7 Master Data Management (MDM)

A centralized MDM store manages the reference entities that anchor the entire platform: Organizations and departments; project types and sector classifications; geographic hierarchies (district, state, zone, corridor); material and equipment catalogs; cost index and inflation reference tables; and approval authority hierarchies. All master data is versioned, change-controlled, and accessible via a read-only API to all microservices, ensuring consistent reference data across the platform.

4.8 Global and National Knowledge Layer

4.8.1 Ingestion of Indian Standards (RDSO, IRC, BIS, NBC, AAI)

Indian engineering standards are ingested as structured parameter libraries: RDSO manuals and specifications for railway infrastructure; IRC codes (IRC:6, IRC:21, IRC:78, etc.) for road and bridge design; BIS standards (IS codes) for structural, electrical, and mechanical parameters; NBC 2016 for building design; AAI circulars and ICAO standards (Annex 14, Annex 9) for airport infrastructure geometry, clearances, and pavement parameters; MoPSW (Ministry of Ports, Shipping and Waterways) guidelines and IS codes for port, jetty, and waterway design parameters; NDMA (National Disaster Management Authority) guidelines together with BIS IS 1893 (seismic), IS 875 (wind loads), and SP 13 (flood management) for multi-hazard design parameters; and MoEFCC environmental assessment parameters (forest diversion, coastal regulation zone conditions, biodiversity impact). All standards are parsed, structured, and linked to the validation rule engine so that parameter checks reference specific standard clauses, and Indian standards take precedence over global standards where clauses conflict (Standards Harmonization Engine).

4.8.2 Ingestion of Global Standards

Global benchmarks for project appraisal frameworks, ADB project preparation guidelines, ISO standards relevant to infrastructure, and IEC electrical standards are ingested and mapped to the corresponding Indian standards. The Standards Harmonization Engine resolves conflicts and precedence between Indian and global standards, with RITES' domain authority as the final arbiter. This needs to confirm scope, ingestion sources, and precedence rules with RITES before finalising.

4.8.3 Standards Harmonization Engine

The Standards Harmonization Engine maintains a cross-reference matrix between Indian and global standards for each sector and parameter type. When a validation rule references a parameter (e.g., minimum bridge clearance, maximum gradient, minimum pavement design life), the engine resolves the applicable standard version, jurisdiction, and value range, applying Indian standards as primary with global benchmarks as secondary reference.

4.8.4 Continuous Knowledge Update Framework

A scheduled pipeline monitors authoritative sources (RDSO website, IRC portal, BIS knowledge portal) for standards updates and notifies RITES administrators. Approved updates are ingested



through a controlled change management process, no standard update modifies the active validation ruleset without RITES approval preserving consistency for in-flight DPR appraisals.

4.9 Benchmarking and Reference Data Repository

The Benchmarking Repository stores sector-wise, geography-wise, and time-period-wise statistical benchmarks derived from historical DPR data: cost-per-kilometre ranges for road types by terrain, bridge construction cost per span, per-km electrification costs, and comparable metrics across all sectors. Benchmarks are expressed as statistical distributions (P10/P50/P90) rather than point values, supporting probabilistic comparison. All benchmarks are versioned, traceable to source DPRs, and require RITES approval before being activated in the validation engine.

4.10 Data Access Layer and API Exposure

The Data Access Layer exposes curated datasets through a REST API (for transactional queries), a GraphQL endpoint (for flexible analytical queries from dashboards), and distributed data processing framework-based batch export APIs (for bulk data access by authorized systems). All data access is authenticated, authorized per RBAC roles, and logged. The developer sandbox provides access to synthetic/masked datasets with the same schema as production, so third-party integrators can build and test without accessing real project data.

5. Functional Module Design and Implementation Approach

This section covers the design and implementation approach for all functional modules each corresponding to a distinct domain capability. All modules are implemented as microservices, deployed independently, and integrated through the API Gateway and event bus. Module configurations, templates, validation rules, workflows, and scoring criteria are configuration-driven and adjustable by RITES administrators without code changes.

5.1 Governance, Users and Roles (RBAC)

5.1.1 Role Definition and Access Control Framework

A hierarchical RBAC framework for defined roles:

- System Administrator
- Project Administrator
- DPR Author / Consultant
- Department Reviewer
- Technical Expert
- Financial Reviewer
- Appraisal Committee Member

- Approver
- Viewer (read-only access)

Permissions are granular: read, create, edit, validate, approve, and export are distinct permissions assignable per resource type. Role assignments are managed by RITES-designated administrators through a self-service administration console.

5.1.2 Audit Logging and Traceability

Every user action login, data access, field edit, document upload, validation run, workflow transition, score change generates an immutable audit event stored in search and analytics engine. Audit logs capture user identity, session ID, timestamp, action type, resource identifier, and before/after values for data changes. Audit reports are available to RITES administrators through the Monitoring Dashboard.

5.2 Master Data Management

5.2.1 Organizational and Functional Masters

Organizational masters capture RITES' internal structure (divisions, verticals, regional offices) and external stakeholder organizations (project proponents, consultants, contractors). Functional masters define approval hierarchies, delegation matrices, and communication preferences. All master data is managed through a governed MDM console accessible to authorized RITES administrators.

5.2.2 Sector and Project Type Masters

Sector masters define the infrastructure sectors supported by PARAKH, with associated sub-sector classifications, applicable standards, and required SME roles. Project type masters define project categories within each sector (e.g., New Railway Line, Gauge Conversion, Doubling within the Railways sector), each with its own template configuration and validation ruleset.

5.2.3 Site and Context Masters (Terrain, Climate, Soil, Hydrology)

Site context masters encode geographic and environmental context attributes - terrain classification (plain, rolling, hilly, steep), agro-climatic zones, seismic zones (BIS IS 1893), flood-prone area designations, and soil classification zones that drive context-aware validation logic. These are derived from authoritative national datasets and updated through the Continuous Knowledge Update Framework.

5.2.4 Standards and Reference Libraries

Standards libraries provide structured parameter definitions (name, unit, valid range, standard reference, sector applicability) that are consumed by the validation engine. Each entry carries a version, effective date, and RITES approval timestamp. Changes are change-controlled and audit-logged.



5.3 DPR Template Creation Engine

5.3.1 Configurable Template Framework (Chapters, Sections, Tables)

The Template Engine provides RITES administrators with a visual template builder to define DPR structures as a hierarchy of Chapters, Sections, Sub-sections, and Input Fields. Each field is typed (text, number, date, geo-coordinate, file attachment, table, or dropdown from a master) and carries metadata: label, help text, unit, validation reference, and mandatory/optional/conditional classification.

5.3.2 Sector-Specific Template Configuration

Templates are sector-specific, reflecting the distinct content requirements of, for example, a Railway New Line DPR versus an Urban Transport DPR. A template library ships with pre-configured templates for all sectors, developed in collaboration with RITES SMEs during Phase 1. Templates are versioned; active DPRs reference the template version at submission, ensuring consistency across the appraisal lifecycle.

5.3.3 Mandatory, Optional and Conditional Inputs

Fields are classified as Mandatory (always required), Optional (enhances appraisal quality but not blocking), or Conditional (required only when a triggering condition is met, e.g., 'Seismic Design Provisions are mandatory when project site is in Zone III or IV'). Conditional logic is defined through the Ontology and Dependency Framework and evaluated in real time during DPR authoring.

5.4 Role-to-DPR Mapping and Stakeholder Coordination

Each DPR is assigned a Stakeholder Map at initiation defining which roles are responsible for each chapter or section (e.g., Structural Engineer for Bridge Design, Financial Analyst for Cost Estimates). The system tracks submission status per stakeholder, issues automated reminders at configurable intervals, and flags overdue submissions to the project coordinator. A consolidated submission dashboard provides RITES reviewers with real-time visibility of DPR completeness by contributor.

5.5 Dictionary and Standards Repository

The Dictionary provides a controlled vocabulary for all technical terms used in DPR authoring defining terms, units, abbreviations, and context-specific meanings. The Standards Repository provides structured access to engineering codes and manuals. Both are searchable through the UI and accessible to the validation engine as lookup references. The Dictionary and Standards Repository are maintained as versioned datasets approved by RITES, with the Continuous Knowledge Update Framework monitoring for new or revised standards.

5.6 Ontology and Dependency Framework

5.6.1 Entity-Relationship Modelling for DPR Parameters

Ontology defines entities and their relationships within the DPR domain: a Project has Alignments; an Alignment traverses GeologicalZones; a Bridge requires FoundationDesign which depends on

SoilBearingCapacity. These relationships are modeled in OWL (Web Ontology Language) and stored in a graph database component, supporting reasoning over complex multi-parameter dependencies.

5.6.2 Context-Aware Dependency Mapping

Dependencies are context-sensitive: the same parameter may have different required inputs depending on site conditions, project type, or design choice. The Dependency Engine evaluates the context at runtime reading site master data, project type, and already-captured parameters to determine which additional parameters are activated, which validation rules apply, and which sections become mandatory.

5.6.3 Site-Condition-Driven Validation Logic

Site conditions (terrain, seismic zone, flood frequency, soil type) are used to activate sector-specific validation rules.

The Benchmarking Dashboard enables cross-project and cross-sector comparative analysis: cost intensity by sector and geography over time; quality score distributions; appraisal cycle time trends; and anomaly frequency by parameter type. These analytics inform RITES' policy decisions on appraisal methodology updates, benchmark threshold revisions, and sector-specific guideline improvements.

5.7 Historical DPR Repository and Benchmarking

The Historical DPR Repository is PARAKH's reference archive: a structured, access-controlled knowledge base that converts every completed appraisal into reusable intelligence for future reviews. The module serves three distinct functions: it provides the archival infrastructure for searching and retrieving historical DPRs; it maintains a statistical benchmarking dataset derived from those DPRs, expressed as sector-wise, geography-wise, and time-period-wise distributions; and it surfaces the most comparable approved projects when an appraiser is reviewing a new DPR. Every DPR that reaches the Approved state is automatically ingested into the Repository through an event-triggered pipeline, with no manual curation step required. The Repository grows continuously, and the accuracy of AI advisories, benchmark comparisons, and anomaly detection all improve as the corpus expands.

5.7.1 Institutional Knowledge Base and DPR Quality Tagging

Archive Design and Search

The Historical DPR Repository stores all completed DPRs, including their structured parameter data, appraisal records, observations, scoring results, and attached documents, in a multi-layer storage architecture. Structured data (parameters, costs, scores, observations) is held in a relational database with geospatial extensions for spatial indexing. Full-text content (parsed chapter text, appraiser comments, advisory narratives) is indexed in search and analytics engines, supporting keyword and phrase search across the entire corpus. Semantic search is provided via vector similarity search extension using embedding representations of DPR chapters and parameter descriptions, supporting retrieval by conceptual similarity rather than keyword match: for example, retrieving DPRs describing "soft alluvial soil conditions near river crossings" without requiring an exact phrase match.

Search is faceted: appraisers can filter by sector, project type, state/district, terrain class, cost range, appraisal outcome (Approved / Returned / Rejected), appraisal year, and quality score band. All search



and retrieval operations are governed by RBAC users access only DPRs within their sector and organisation scope; cross-sector access for benchmarking and analytics is granted to designated RITES SME roles.

DPR Quality Tagging

Every DPR that completes appraisal is tagged with a structured quality profile along the following dimensions, derived from the appraisal record:

1. **Completeness Score:** percentage of mandatory and conditional fields populated, at submission
2. **Technical Quality Score:** weighted aggregate of rule pass rate, standard deviation of engineering parameters from sector norms, and number of critical technical observations raised
3. **Financial Quality Score:** cost reasonableness score (percentage of line items within benchmark P10–P90 range), contingency adequacy, and schedule-cost consistency
4. **Appraisal Outcome:** Approved, Conditionally Approved, Returned for Revision, Rejected, with the revision round at which approval was granted
5. **DPR Quality Band:** a composite classification (Excellent / Good / Acceptable / Poor) computed from the above scores using a sector-specific weighted formula maintained in MDM

These quality tags are stored alongside the DPR in the Repository and serve as labels for ML model training, benchmark stratification, and near-peer retrieval filtering. RITES administrators can adjust the quality band formula (weights and thresholds) through the MDM interface; the Repository recomputes tags for all affected historical DPRs on the next scheduled batch run.

Access Control and Versioning

Each DPR version is stored immutably. The Repository holds the complete version history of every DPR including draft, returned, and revised versions so that appraisers can inspect how a DPR evolved across revision cycles. Access to returned and rejected DPRs is restricted to RITES appraiser roles and above; project proponents can only access their own project's history.

5.7.2 Cross-Project and Cross-Sector Benchmarking

Benchmark Dataset Construction

The Benchmarking Repository is derived from the approved DPR corpus through a scheduled Spark batch job that runs nightly. For each sector, project type, terrain class, and geography tier (state / agro-climatic zone), the job extracts key cost and engineering parameters from all approved DPRs in the Good or Excellent quality band and computes:

- **Statistical distributions (P10 / P50 / P90) for cost line items:** cost per km, cost per structure, cost per unit area, earthwork rate, bridge cost per linear metre span, and equivalent sector-specific metrics
- **Statistical distributions for key engineering parameters:** carriageway width, gradient, vertical clearance, design speed, axle load, material specifications, seismic design coefficient, as applicable per sector
- **Peer group size:** the number of DPRs contributing to each distribution cell, surfaced in the UI so appraisers know whether a benchmark is statistically sound or based on too few samples



Benchmarks are effective-dated: each distribution carries the cohort period (e.g., FY 2022–2024 approved DPRs), the CPWD cost index applicable to that period, and a cost-inflation adjustment factor so that historical benchmarks from older cohorts are normalized to the current CPWD/MoRTH schedule before comparison. When a DPR is submitted, the Benchmarking Service fetches the benchmark distribution for its sector, terrain class, project-size band, and current effective period and computes the percentage deviation of each cost line item from the P50 benchmark, flagging items beyond the configured threshold as anomalies.

Benchmarking Dashboard

The Benchmarking Dashboard provides RITES management and SMEs with a self-service analytics interface (open-source BI platform) over the benchmark dataset:

- **Cost Intensity Trends:** unit cost trajectories by sector and geography over time, useful for identifying cost escalation patterns and setting revised thresholds
- **Quality Score Distributions:** histogram of appraisal quality scores by sector and year, showing whether overall DPR quality is improving across the project proponent ecosystem
- **Appraisal Cycle Time Analytics:** average and percentile cycle times (submission-to-approval) by sector and revision round, surfacing which sectors or project types consistently require multiple revision cycles
- **Anomaly Frequency Analysis:** frequency and severity distribution of validation findings by parameter type, identifying systematic gaps in project proponent submissions (e.g., consistently missing geotechnical investigation data for bridge DPRs in hilly terrain)
- **Comparative Sector View:** side-by-side comparison of key metrics across sectors, supporting RITES management in prioritising capacity-building interventions

All dashboard views are exportable to PDF and Excel. RITES administrators can configure automated monthly benchmark refresh reports delivered to designated distribution lists.

5.7.3 Near-Peer Retrieval and Pattern Identification

Near-Peer Retrieval

When an appraiser opens a DPR for review, the repository automatically surfaces the five most similar approved DPRs the "near-peer set" based on a composite similarity score computed across:

- **Structured similarity:** sector match (exact), terrain class match, project-size band match (total project cost within $\pm 50\%$), district/state adjacency (geographic proximity scored via geospatial database extension)
- **Semantic similarity:** cosine similarity between the vector similarity search extension embedding of the new DPR's project description and those of approved DPRs in the corpus, capturing conceptual similarity in project scope and site conditions

The near-peer set is displayed in the Appraisal Dashboard alongside the DPR under review. For each peer project, the appraiser can see: project name, sector, approval year, key cost metrics, appraisal quality band, and a link to the full peer DPR (subject to access control). This peer context accelerates the appraiser's calibration allowing them to immediately see whether the new DPR's cost structure, technical parameters, and design choices are consistent with comparable approved projects.

Near-peer retrieval is also used by the Anomaly Detection Engine and the AI Advisory Service. Anomaly scores are computed relative to the near-peer distribution, not just the sector-wide distribution, giving a more precise comparison for projects with distinctive site conditions. AI advisories that reference benchmark deviations cite the near-peer set size and its parameter range, giving the appraiser full provenance for every comparative claim.

Pattern Identification

The Repository feeds a scheduled pattern analysis job that identifies recurring quality failure patterns across the corpus:

1. **Systematic omissions:** parameter groups that are frequently missing or incomplete for a given sector-terrain combination (e.g., hydrological analysis consistently absent in irrigation DPRs from specific states)
2. **Correlated anomalies:** cost line items that tend to be anomalous together (e.g., earthwork and slope protection costs co-inflated in hilly highway DPRs), indicating potential coordination between line items during DPR preparation
3. **Revision triggers:** observations that most frequently cause DPRs to be returned for revision, informing RITES' DPR preparation guidelines and training programmes for project proponents

Pattern insights are surfaced in the Benchmarking Dashboard and exported quarterly as a Pattern Intelligence Report for RITES quality improvement initiatives. The pattern data also feeds the Completeness Validation Engine's dependency rules if a systematic omission pattern is identified, RITES administrators can promote the missing parameter to a Conditional Mandatory field for the affected sector-terrain combination, hardening the validation rules based on empirical evidence from the corpus.

5.8 DPR Authoring and Submission Interface

5.8.1 Structured Data Entry and Document Upload

The Authoring Interface renders the DPR template as a guided, section-by-section form. Numeric fields display unit labels and valid range hints. Lookup fields integrate with master data and standards dictionaries. GIS fields open an embedded map viewer (open-source mapping library) for alignment drawing or coordinate capture. Document attachment fields accept PDF, DWG, IFC, and image formats with size limits and type validation. Real-time field-level validation provides instant feedback as the user enters data, reducing back-and-forth during formal validation.

5.8.2 Version Control and Submission Workflows

Every save of a DPR creates an immutable version snapshot. Authors can review version history, compare versions, and revert to earlier states. The submission workflow enforces a multi-stage gate:

Draft → Submitted → Under Review → Observations Raised → Revised → Approved/Rejected.

Workflow transitions require authenticated user actions and generate events consumed by the notification and audit systems. Bulk submission of multiple DPR sections by different stakeholders is coordinated through the Stakeholder Coordination module.

5.9 Completeness Validation Engine

5.9.1 Section-Level and Input-Level Completeness Checks

The Completeness Validation Engine evaluates every submitted DPR against its template definition, checking that all Mandatory fields are populated, all Conditional fields triggered by applicable conditions are populated, and all required document attachments are present. Completeness is reported as a percentage score per section and overall, with a drill-down report identifying specific gaps. The engine can be run on demand by authors during preparation and is automatically triggered upon formal submission.

5.9.2 Stakeholder Submission Tracking

For multi-contributor DPRs, completeness tracking is disaggregated by contributor role and section ownership. A Completeness Dashboard shows RITES reviewers the submission status of each contributor's assigned sections in real time, so bottlenecks are visible before the formal review cycle begins.

5.10 Technical Validation Engine

5.10.1 Rule-Based and Standards-Driven Validation

RITES shall provide domain inputs, applicable sector engineering standards, benchmark thresholds, and SME validation support as the authoritative source for all technical validation rules in the platform. GlobalLogic's Technical Validation Engine translates these RITES-approved inputs into configurable rule logic, parameter definitions, threshold ranges, and scoring rubrics. All validation rules, formulas, scoring parameters, and threshold values shall be reviewed by RITES SMEs in the UAT/staging environment and formally approved by RITES before production deployment. Final approval authority on all technical validation rules and codal reference mappings rests with RITES.

The Technical Validation Engine executes a library of validation rules derived from engineering standards, RITES' appraisal methodology, and sector-specific technical norms against DPR parameter values. Rules are expressed in a structured rule language (rule engine or a custom DSL compiled to executable Java bytecode), allowing RITES domain experts to author and modify rules through a governed rule management console without requiring software development. Each rule carries a reference to the standard clause it implements, supporting full traceability of validation outcomes.

5.10.2 Contextual and Dependency-Driven Checks

Technical validation is context-aware: rules are activated based on project type, sector, site conditions, and design choices, as determined by the Ontology and Dependency Framework. This prevents inapplicable rules from generating false positives and ensures that high-risk scenarios trigger the specific checks warranted by that context.

5.10.3 Engineering Parameter Validation Across Sectors

Parameter validation covers the key engineering dimensions of all sectors: geometric design parameters (alignment, gradient, curvature) for transport; structural parameters (loads, spans, material strengths) for bridges and buildings; hydraulic parameters (discharge, waterway, freeboard) for water



and flood management; electrical parameters (load demand, power factor, cable sizing) for power; and demand parameters (traffic, ridership, freight) for transport sectors. A sector-specific validation suite is configured in collaboration with RITES SMEs during Phase 3 and is subject to RITES approval before activation.

5.11 Consistency Engine

5.11.1 Cross-Section Consistency Verification

The Consistency Engine checks that values entered in different sections of a DPR are mutually consistent. For example: the project length in the Executive Summary must match the alignment length in the Engineering Design chapter; the earthwork quantities in the Cost Estimate must be consistent with the alignment profile; the number of major bridges declared must match the number of bridge design chapters submitted. Cross-section consistency rules are defined by RITES SMEs and maintained in the rule library.

5.11.2 Financial-Technical-Schedule Cross-Validation

A dedicated cross-validation suite reconciles the financial, technical, and schedule dimensions of a DPR: cost-per-unit-of-construction-type must be within the benchmarked range for the sector, terrain, and material; construction schedule durations must be feasible relative to crew productivity and site access assumptions; contingency provisions must be within policy-defined ranges. Deviations from benchmarks are flagged as observations with severity levels (Minor, Major, Critical) for RITES reviewers' consideration.

The Consistency Engine additionally performs narrative-to-structured data cross-validation by consuming outputs from the NLP Document Parsing pipeline. Free-text sections, Executive Summary, Project Justification, and site description chapters are parsed by the NLP service to extract numerical and categorical claims (alignment length, project cost, number of structures, employment generation, traffic volume). These extracted values are compared against the corresponding structured input fields in the DPR data model. Discrepancies beyond a configurable tolerance (e.g., executive summary states a 42 km alignment but the Engineering Design chapter records 39 km, a 7.1% deviation) are flagged as Narrative Inconsistency observations, with the conflicting paragraph reference and the structured field path identified for the RITES reviewer.

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5.12 Anomaly Detection Engine

5.12.1 Statistical and ML-Based Outlier Detection

The Anomaly Detection Engine uses a combination of statistical methods (z-score, IQR-based outlier detection) and machine learning models (unsupervised anomaly detection model, Autoencoder neural networks) trained on historical DPR data to identify parameters that are statistically anomalous relative to comparable projects. Models are trained on the Gold-layer benchmark datasets and are sector-, project-type-, and terrain-specific. All models are trained on-premises using open-source frameworks (open-source ML and deep learning frameworks) and served through the model serving layer.

5.12.2 Pattern Recognition and Benchmark Comparisons

Beyond point-value outlier detection, the Anomaly Engine identifies pattern-level anomalies: cost distribution across work categories that deviates significantly from historical patterns for similar projects; material quantity ratios inconsistent with engineering norms; or schedule activity sequences that violate standard construction logic. Pattern anomalies are presented as observation flags with supporting benchmark comparisons, so RITES reviewers can make informed judgements rather than accepting AI flags uncritically.

5.13 Module 15: Review Workflow and Observations Management

5.13.1 Configurable Appraisal Workflow and Scoring Frameworks

The Review Workflow engine implements RITES' appraisal process as a configurable multi-stage workflow:

Initial Screening → Technical Review → Financial Review → Environmental and Social Review → Consolidated Appraisal → Final Approval.

Each stage is assigned to specific RITES roles. PARAKH supports multiple configurable scoring frameworks that can be defined, versioned, and activated independently by RITES administrators without code changes. The example frameworks include:

- Technical Completeness Framework weights completeness coverage and standards compliance
- Financial Viability Framework weights cost reasonableness, benchmark deviation, and contingency adequacy
- Sector-Specific Appraisal Framework applies sector SME-defined rubrics for Railway, Highway, Port, Airport, and Urban Transport DPRs etc.

Scoring frameworks defining evaluation criteria, weights, and scoring rubrics for each DPR dimension are configurable by RITES administrators and are subject to RITES' approval authority. Multiple frameworks can be run concurrently on the same DPR to produce composite or comparative appraisal views.

5.13.2 Issue Tracking, Iterative Revisions and Closure

Observations raised by RITES reviewers are tracked as structured issue records: observation text, severity, section reference, standard citation, assigned respondent, due date, and status (Open,



Responded, Accepted, Closed). The DPR proponent responds to each observation with a revision or a justified clarification. RITES reviewers can accept, partially accept, or re-open responses. Observation history forms part of the permanent DPR appraisal record.

5.14 Reporting and Appraisal Dashboard

5.14.1 Decision-Support Dashboards and KPI Views

The Analytics Dashboard provides RITES management and reviewers with real-time visibility across the DPR pipeline: DPRs by sector and appraisal stage; average appraisal cycle time by sector; validation pass rates; observation resolution rates; and benchmark deviation trends. Dashboards are built using an open-source BI platform (open source), with drill-down capability from KPI aggregates to individual DPR records. The final metrics to be showcased as part of the dashboard to be discussed RITES SMEs in the initial phase.

5.14.2 Consolidated Appraisal Summaries and Risk Reports

The Consolidated Appraisal Report is a system-generated document that aggregates all validation outcomes, observation records, scoring results, benchmark comparisons, and machine-generated findings into a structured appraisal summary. The report is generated in PDF and structured data formats, signed digitally by the RITES Approver, and stored as a permanent record in the DPR repository. A Risk Summary identifies key risk factors found during appraisal.

5.15 Simulation and Sensitivity Analysis

5.15.1 What-If Analysis and Scenario Modelling

The Simulation Module allows RITES analysts to define scenario parameters varying cost escalation rates, traffic growth assumptions, construction duration, or financing conditions and evaluate their impact on project viability indicators: NPV, IRR, benefit-cost ratio, and schedule completion date. Scenarios are run in an isolated compute environment and do not modify the live DPR record. Scenario results are stored for comparison and presented through interactive visualization.

5.15.2 Route Alignment and Cost Optimization Analysis

For transport infrastructure DPRs, the platform provides a GIS-integrated alignment comparison tool that evaluates alternative route alignments against cost, land acquisition impact, environmental sensitivity, and traffic effectiveness criteria. Cost optimization suggestions are generated by the AI analytics engine by comparing the proposed alignment's cost profile against optimized alignments for similar terrain and project type from the historical repository.

6. AI and Advanced Analytics Strategy

This section describes the AI approach for PARAKH, how machine-generated recommendations are produced, constrained, explained, and governed. The idea is to provide the different AI layers and approach towards data governance, explainable AI, guardrails, evaluation and fine tuning techniques etc. covered in below sub-sections.

6.1 AI as Assistive and Advisory Layer - Design Philosophy

The core design principle of PARAKH's AI layer is that it is a structured analytical tool, not an autonomous decision-maker. Every output from the AI layer, whether a completeness flag, a cost-variance alert, or a consistency observation, is presented to the appraiser as an advisory, clearly labelled with a confidence tier (High, Medium, or Low). The appraiser is required to review each advisory and may override any finding; the system enforces no restriction on overrides but records every decision immutably.

The AI advisory pipeline operates across three complementary tiers. The first tier comprises deterministic, rule-based checks: mandatory field presence, cross-field logical rules, and sector-specific engineering standards drawn from RDSO, IRC, BIS, MoRTH, and NBC specifications. These checks are 100% traceable; every flag includes the exact rule identifier, the input value, and the standard reference. The second tier applies statistical and machine learning models, unsupervised anomaly detection models for cost anomalies, regression models for cross-parameter consistency which surface soft-signal issues that rule engines cannot capture. The third tier uses NLP and semantic analysis to interpret unstructured text in uploaded documents, extracting entities and detecting contradictions between document-level claims and structured form data. The proportion of deterministic versus ML-based checks across these tiers is not prescribed by RITES; the final distribution will be optimized during the Milestone 1 System Study in consultation with RITES SMEs to maximize accuracy, explainability, traceability, and auditability and is subject to RITES' formal approval in the SRS.

Every AI output carries a confidence score computed from the model's internal probability estimates. Low-confidence advisories are visually differentiated in the appraisal interface and trigger a mandatory acknowledgement step before the appraiser can proceed. All AI decisions including which model version generated the output, the input features used, and the confidence score are stored in an immutable advisory log alongside the DPR version they relate to. When an appraiser overrides an advisory, both the AI finding and the human judgment are persisted, creating a rich feedback dataset for model retraining.

GlobalLogic's AI approach for PARAKH starts from the RFP's explicit constraint that the platform must not replace expert judgment. Machine-generated outputs surface relevant information, flag potential issues, and produce analytical insights; they do not make final appraisal decisions. Every AI output is labelled as machine-generated, carries a confidence indicator, provides a reasoning trace, and requires an explicit reviewer acknowledgement before influencing the appraisal record. This design satisfies the functional requirements and the OWASP AI security standards. All AI models deployed within PARAKH are open-source or custom-trained models running on infrastructure under RITES' control. No third-party managed inference APIs are used to process RITES project data, which preserves data sovereignty, meets the confidentiality obligations and gives RITES full auditability over model behaviour.

6.2 Hallucination Guard: Deterministic Codal Reference Architecture

For all engineering code, standard, and codal reference outputs, PARAKH employs a strict deterministic lookup architecture. The system maintains a Validated Reference Library a closed, versioned corpus of engineering standards ingested from authoritative sources (RDSO manuals, IRC codes, BIS standards, NBC 2016, MoRTH specifications, AAI circulars, ICAO Annexes, MoPSW guidelines, NDMA/IS 1893 seismic parameters) and stored as structured, queryable records in the

Standards Repository. When the platform cites a standard clause, it exclusively queries this deterministic lookup engine using exact-match or structured-index lookups. The LLM is categorically prohibited from generating, inventing, or inferring a standard reference number, clause number, or parameter value through probabilistic next-token prediction.

The hallucination guard operates as a two-stage pipeline.

- In Stage 1, the LLM performs synthesis, summarisation, and reasoning tasks describing anomalies, generating draft observations, interpreting project context producing outputs that are inherently advisory and carry explicit confidence scores.
- In Stage 2, any standard or codal reference in an LLM output is intercepted by the Reference Validator component, which performs an exact lookup against the Validated Reference Library. If the reference exists and the cited clause or value matches the stored authoritative record, the citation is passed through unchanged with a “Validated Reference” tag rendered in the UI. If the reference does not exist in the validated library, or if the cited value conflicts with the stored authoritative value, the citation is blocked, replaced with a “Reference not validated manual verification required” flag, and the LLM output is presented to the appraiser with an explicit warning.

This architecture ensures that deterministic rule-based engines (not LLMs) handle all pass/fail engineering and codal compliance judgements, while LLMs operate strictly as synthesis and summarisation assistants.

6.3 AI Processing Pipeline: Seven-Layer Architecture

Uploaded DPR documents PDFs, DOCX files, and scanned attachments enter a preprocessing pipeline before any NLP analysis is applied. The document parsing library extracts raw text and metadata from all supported formats, including OCR for scanned PDFs via OCR engine (with an infrastructure-domain trained model for improved accuracy on engineering drawings and tables). The extracted text is segmented by section using a combination of heuristic rules (headings, page layout signals) and a fine-tuned classification model trained on RITES DPR templates..

Named entity recognition (NER) models extract structured information from free-text sections: project names, geo-coordinates, cost figures, material specifications, standard references (e.g., IRC:6-2017, IS:1893), and stakeholder designations. Extracted entities are cross-validated against structured form fields discrepancies between a cost figure mentioned in the narrative and the value entered in the cost schedule are surfaced as consistency observations. Domain terminology is normalised using the PARAKH ontology, which maps synonyms and abbreviations across RDSO, IRC, BIS, NBC, and AAJ vocabularies to canonical concept identifiers.

The NLP pipeline processes historical and newly submitted DPR documents to extract structured data from unstructured text. The pipeline is built on NLP processing library (for tokenization, NER, and dependency parsing) and transformer model library (for document classification, semantic similarity, and table extraction). Domain-specific models are fine-tuned on infrastructure engineering vocabulary using the historical DPR corpus. OCR processing uses OCR engine with layout analysis for scanned DPRs. The extraction pipeline achieves structured output for key sections (project summary, cost estimate, engineering parameters, survey data) with quality scores that flag low-confidence extractions for human review.



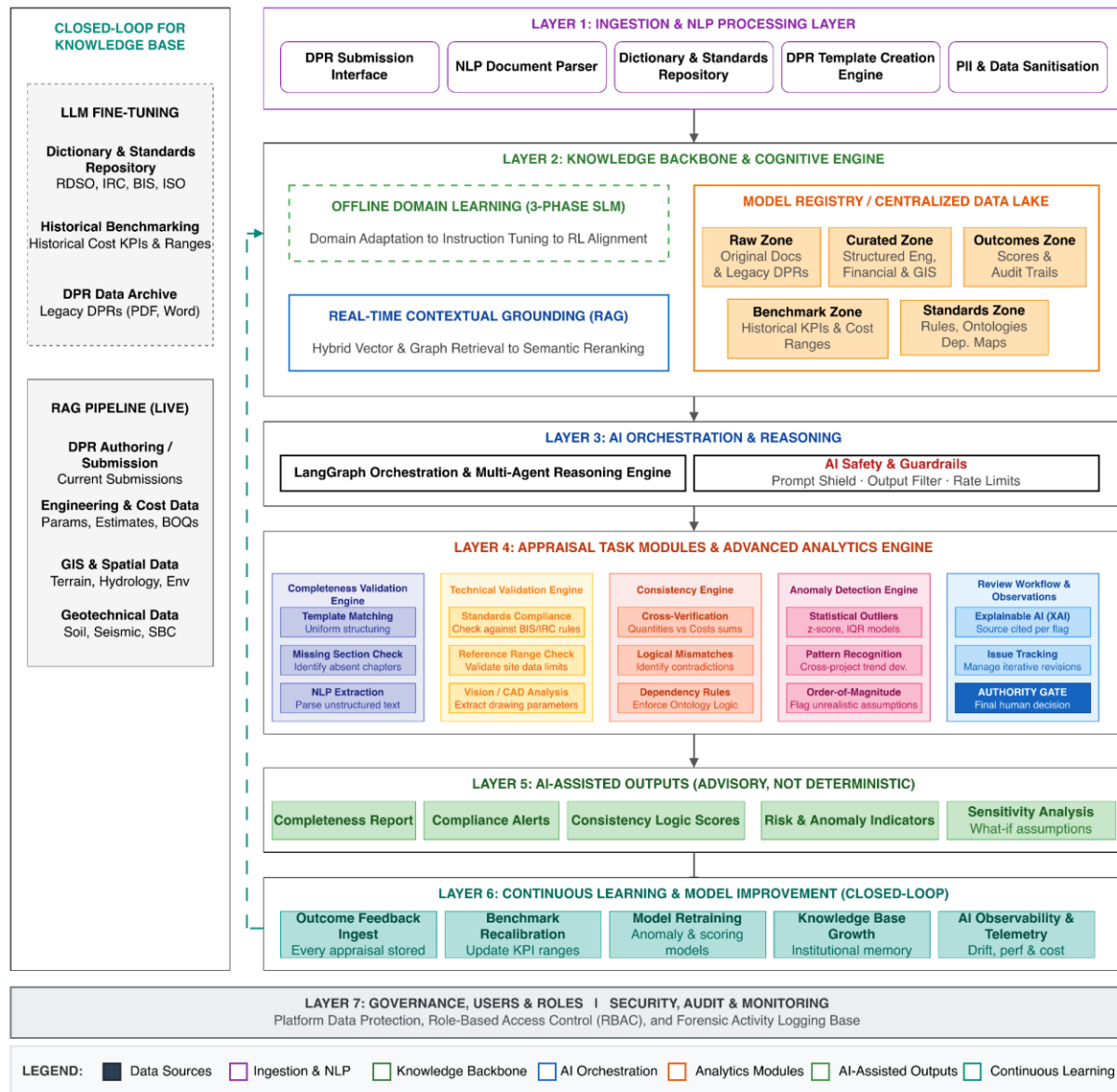


Figure 6.3 - AI Architecture and NLP Document Ingestion Pipeline

The PARAKH framework utilizes a bifurcated cognitive approach: LLM fine-tuning develops the system's specialized engineering reasoning and technical vocabulary, while a real-time RAG pipeline provides factual grounding from current regulatory codes and site-specific evidence. Together, they eliminate AI hallucinations, ensuring every high-stakes infrastructure appraisal is logically sound, objectively verified, and backed by an immutable audit trace for human oversight.

Step 1: DPR Preparation & Ingestion Layer (Layer 1)

The data lifecycle initiates at the universal gateway for all project inputs. Whether handling historical archives or active submissions, data is standardized before processing:

- **DPR Authoring / Submission Interface:** Acts as the primary multimodal pipeline, capturing CAD drawings, BIM models, and unstructured DPRs via API or batch uploads.

- **NLP Document Parser:** Extracts and structures granular text, tables, and financial parameters (BOQs).
- **Dictionary & Standards Repository:** Enforces terminology control and standards harmonization, prioritizing Indian regulatory precedence over global standards.
- **DPR Template Creation Engine:** Maps extracted data against mandatory schemas, performing initial data quality validation and canonical formatting.

Step 2: Knowledge Backbone & Centralized Data Lake (Layer 2)

Normalized data descends into the cognitive core, structured around a centralized Data Lake and Model Registry. Data is categorized into distinct zones to feed the dual AI engines:

- **Centralized Data Lake:** Data is routed into strict functional buckets: Raw Historical Repositories and Benchmarking KPIs, Curated Master Data Management, and Standards Zone for Ontology & Dependencies.
- **Offline Domain Learning and Hybrid Cognitive Architecture:** Historical and standards data are processed through a 3-phase learning lifecycle: Domain-Adaptive Pre-training (DAPT), Instruction Fine-Tuning, and Reinforcement Learning Alignment (DPO, GRPO, PPO) to build the model's foundational engineering knowledge. For ongoing model refinement in later phases, the framework uses Parameter-Efficient Fine-Tuning (PEFT) techniques such as Low-Rank Adaptation (LoRA), which updates domain knowledge with a fraction of the compute required for full retraining. The result is a Hybrid Architecture (RAG plus Fine-Tuned LLM), where the fine-tuned model provides the engineering reasoning layer and the RAG pipeline provides real-time, evidence-based factual grounding.
- **Real-Time Contextual Grounding (RAG):** Active project data bypasses the SLM and flows into the RAG pipeline, utilizing Hybrid Vector & Graph Retrieval to act as the live, factual evidence base for the current appraisal.

Step 3: AI Orchestration & Reasoning (Layer 3)

With data structurally and cognitively mapped, the Multi-Agent Reasoning Engine takes command. Acting as the central intelligence controller, it synthesizes learned engineering logic with project-specific facts. Governed by **M4 (Role-to-DPR Mapping)**, it dynamically routes data payloads and assigns multi-agent workflows to the appropriate, specialized appraisal tasks based on organizational permissions.

Step 4: Appraisal Task Modules & Evaluation Layer (Layer 4)

The orchestrated workflow executes a suite of high-stakes, specialized audits:

1. **Completeness Validation Engine:** Validates submissions against M3 templates to identify missing chapters or absent parameters.
2. **Technical Validation Engine:** Cross-checks inputs against reference ranges, M5 standards (BIS/IRC), and performs deep CAD/vision analysis.
3. **Consistency Engine:** Verifies cross-chapter engineering logic, ensuring mathematical alignment (e.g., quantities vs. cost summaries) using M6 dependency rules.
4. **Anomaly Detection Engine:** Deploys deterministic statistics (z-score, IQR) and pattern recognition to flag order-of-magnitude errors and cost risks.



5. **Review Workflow & Observations:** Culminates the layer by generating Explainable AI (XAI) source citations, maintaining the issue tracker, and acting as the Authority Gate for final human decision-making.

Step 5: Decision Support & AI-Assisted Outputs (Layer 5)

The framework translates the deep technical audits into advisory, non-deterministic outputs for senior leadership:

- **Reporting and Appraisal Dashboard:** Generates consolidated risk indicators, anomaly alerts, and completeness flags in a single executive view.
- **Simulation & Sensitivity Analysis:** Empowers appraisers to run "what-if" models against the data, testing project viability against shifting assumptions (e.g., changing material costs).

Step 6: Continuous Learning & Model Improvement (Layer 6)

To ensure the framework evolves, an automated Closed-Loop Feedback system captures the outcomes of every human-verified appraisal. This data is fed back into the M8 Benchmark Zone to continually recalibrate KPI thresholds, retrain anomaly detection models, and update the institution's knowledge base.

Step 7: Foundational Security & Governance (Layer 7)

The entire vertical data pipeline is anchored by a secure infrastructure base. Governed by M1 (Governance, Users & Roles) and M7 (Security, Audit & Monitoring), this layer enforces rigorous Role-Based Access Control (RBAC), protects absolute data sovereignty, and maintains forensic activity logging for every action taken within the platform.

6.3 Machine Learning Models for Anomaly and Outlier Detection

Anomaly detection in PARAKH operates on below two levels:

- Parameter-level outlier detection
- Cross-parameter consistency modelling

For parameter-level detection, the platform trains an **unsupervised anomaly detection model** per sector-and-project-type combination using approved historical DPRs from the RITES archive as the baseline distribution. Unsupervised anomaly detection model is selected for its ability to handle mixed-type feature vectors and its insensitivity to the scale of individual features both important properties given the heterogeneous engineering parameters across 13 infrastructure sectors.

Statistical methods complement the ML approach for sectors or parameter categories where the historical corpus is smaller. Z-score and interquartile range (IQR) methods are applied per sector benchmark distribution, with thresholds configurable by the PARAKH administrator. Items flagged by statistical methods are labelled distinctly from ML-flagged items so the appraiser understands the basis of each observation. For newly introduced sectors or project types with no historical precedent, the system defaults to rule-based checks only and explicitly indicates that ML benchmarking is not yet available for that category.

Anomaly detection is implemented as an ensemble of models: an unsupervised anomaly detection model (fast, effective for high-dimensional parameter spaces), Autoencoder neural networks (deep learning framework, for detecting distributional anomalies in cost and quantity profiles), and gradient-



boosted classifiers trained on labelled anomaly examples from historical DPR reviews. Each model is sector and project-type-specific, trained on the Gold-layer benchmark dataset. Model performance is evaluated through held-out test sets and compared against RITES SME-validated ground truth. Models are registered in model registry and experiment tracking platform with full versioning, training metadata, and performance metrics.

6.4 Benchmarking and Predictive Analytics

Cost benchmarking is delivered by the Cost Benchmarking Service, which operates against a versioned rate schedule. The rate schedule holds sector unit rates for materials, labour, plant and equipment sourced from CPWD Schedule of Rates, MoRTH specifications, state PWD rates, and Gol project benchmarks each effectively-dated and tagged with a geographic region and terrain class. When a DPR is submitted for appraisal, every cost line item is matched against the applicable rate schedule entry; the percentage deviation from the benchmark is computed and any item exceeding the configured threshold is flagged in the advisory report.

Peer-project comparison is enabled by the historical DPR corpus stored in the Data Lake. The benchmarking service queries the corpus for approved DPRs in the same sector, terrain class, and project-size band, and presents statistical summaries (median, P75, P95) for key cost heads and engineering parameters. This contextual comparison helps appraisers distinguish between a genuinely anomalous cost and one that merely reflects a legitimate regional or site-specific variation. The comparison results are included in the advisory report with provenance references to the comparable projects.

Predictive analytics extend benchmarking to forward-looking scenarios. The Simulation and Sensitivity Analysis module (Module 14) allows appraisers to run what-if scenarios varying input assumptions such as material cost indices, traffic growth rates, or inflation factors and observe the projected impact on project viability metrics. These scenarios draw on econometric trend models trained on historical Gol construction cost indices and sector-specific price series, and they are always presented as advisory tools to support human judgment rather than as definitive forecasts.

The predictive analytics module uses gradient-boosted regression models (ensemble and gradient-boosted regression models) trained on historical DPR cost and schedule data to generate predicted ranges for key project parameters, total project cost, construction duration, land acquisition area, peak manpower requirement as reference anchors during DPR review. Predictions are presented as probability ranges, not point estimates, clearly communicating model uncertainty. For demand forecasting (traffic, ridership, freight), established econometric models (AADT growth regression, four-step transport models) are implemented alongside ML-based alternatives, with RITES selecting the preferred methodology per project type.

6.5 Explainable AI (XAI) and Audit Trails for AI Outputs

Every ML-generated flag in PARAKH is accompanied by a SHAP (SHapley Additive exPlanations) explanation. Feature attribution values decompose the model's output into per-feature contributions, showing exactly which input parameters drove the anomaly score for a given DPR. The Explainability Service serialises SHAP (SHapley Additive exPlanations) values into both a feature importance chart (rendered in the appraisal UI) and a human-readable natural language summary that the appraiser reads without needing to interpret charts.

Rule-based checks emit a full rule trace alongside every finding: the rule identifier, the governing standard, the actual input value, the permissible range or required value, and the pass/fail result. This trace is presented as a collapsible detail panel in the advisory report, allowing the appraiser to verify the basis of the finding directly against the referenced standard without leaving the application.

All advisories, explanations, and rule traces are stored immutably in the Advisory Log, keyed to the specific DPR version and model version that generated them. When an appraiser overrides an advisory either accepting a flagged item with justification or dismissing a finding as inapplicable - the override is recorded alongside the original AI output. The complete provenance chain (AI output → appraiser decision → justification text → timestamp → actor) is visible to committee members during review and is included in the final appraisal report. This creates an end-to-end audit trail that satisfies both internal governance requirements and any future regulatory scrutiny of AI-assisted decisions.

All ML model outputs are accompanied by SHAP feature attribution analysis, identifying the top contributing features to each prediction or anomaly flag. For rule-based validation outputs, the specific rule, standard clause, and parameter values that triggered the outcome are recorded. SHAP (via the SHAP Python library) is used as the primary explainability framework, including for text classification outputs from NLP models where SHAP's KernelExplainer provides model-agnostic feature attributions. This satisfies section 6.8.5's requirement for traceable, explainable outputs and the OWASP AI risk control requiring human-interpretable AI reasoning.

6.6 AI Model Training, Retraining and Governance

The continuous learning loop converts every appraiser decision into a training signal - with no annotation team required.

Learning threshold milestones:

- 1,000 PDFs ingested
- RAG advisory system live (Day 1 of operations); 100 reviewed DPRs
- First held-out evaluation set assembled; 300 contrastive pairs (overrides)
- First DPO fine-tuning run feasible; 500 pairs per sector
- Sector-specific fine-tuning viable; 2,000 reviewed DPRs near-peer retrieval reaches high effectiveness.

These thresholds represent GlobalLogic's technical estimates based on industry benchmarks for LLM fine-tuning at comparable scale; they will be validated and confirmed during the SRS workshop with RITES at Milestone 1.

RAG quality improves continuously with every approved DPR auto-ingested, regardless of whether a model training run has occurred. The initial training corpus for PARAKH's ML models is assembled from three sources: approved historical DPRs from the RITES archive (the primary source of ground-truth labels), synthetic sector-specific datasets generated by domain experts to cover edge cases and underrepresented project types, and publicly available GoI project databases for supplementary benchmarking data. All datasets are versioned and stored in the Data Lake's silver layer with full lineage metadata, ensuring that any future audit can trace a model's training data to its source.

Model lifecycle is managed through ML experiment tracking system: experiment tracking, model versioning, artifact storage, and promotion workflow (Staging → Production). All model training uses

data from the approved Gold-layer benchmark dataset. Retraining is triggered by scheduled pipelines (quarterly for anomaly models, annually for prediction models) and by data drift detection alerts from the model monitoring system. No model update is deployed to production without documented performance evaluation and explicit RITES approval, as required by section 6.8.3. Model training and inference run on RITES-controlled infrastructure; no training data or model artefacts leave the RITES environment.

6.7 Compliance: AI Recommendations vs. Expert Authority

PARAKH's governance framework establishes a clear and unambiguous boundary between AI advisory capability and human expert authority. The platform's AI layer is positioned as a computational aid that increases appraisal consistency and reduces the burden of manual cross-checking, not as a system that determines appraisal outcomes. This positioning is reflected in the user interface, the appraisal report template, and the platform's data model, which separates AI-generated observations from appraiser findings as distinct entities. Every AI-generated observation in the appraisal report is prefixed with an 'AI Advisory' label and a confidence tier (High / Medium / Low). The final appraisal recommendation Approve, Return for Revision, or Reject is authored exclusively by the human appraiser based on their professional judgment. The system enforces no restriction on the appraiser's ability to differ from AI advisories; the only enforcement is that every override must be accompanied by a recorded justification (free text, minimum 20 characters), which becomes part of the permanent DPR record.

The committee review stage provides an additional layer of human oversight. Committee members can see both the AI advisories and the appraiser's responses to them, creating a transparent record of how AI inputs were weighed in the appraisal process. This architecture aligns with the principles of responsible AI deployment in public infrastructure decision-making: AI amplifies the appraiser's analytical capability while the accountability for the appraisal decision remains unambiguously with the qualified human expert.

The platform enforces a clear separation between AI-generated outputs and human appraisal decisions. AI recommendations are displayed in a distinct 'AI Insights' panel, clearly labelled as machine-generated. Workflow transitions from validation stages require explicit human action - a reviewer clicking 'Acknowledge and Proceed' or 'Override with Justification'. Override actions require a mandatory justification text, are audit-logged, and are visible to RITES supervisors. This architecture ensures that appraisal outcomes are always attributable to a human decision-maker, maintaining RITES' institutional accountability for all appraisal outcomes.

6.8 AI Model Origin, Hosting, and Data Sovereignty Declaration

GlobalLogic explicitly declares the following regarding AI/ML model origin, hosting, and data flow:

1. **Production Inference:** All LLM inference against RITES DPR data is performed on open-source models (open-source language model family) deployed within RITES-controlled or RITES-designated hosting infrastructure. The inference engine is an open-source LLM serving framework (open-source, Apache 2.0). No token of RITES data is sent to any third-party managed API in the production environment.
2. **Development and QA Environments:** GlobalLogic's managed Dev and QA environments use managed AI inference service as the LLM backend. These environments process



exclusively synthetic DPR data and mock project records. No real RITES DPR, project, or organisational data is ingested into a managed AI inference service environment at any time.

3. **Model Fine-tuning:** Domain adaptation (QLoRA) is performed using open-source tooling (Unsloth/PEFT, PyTorch). Fine-tuned model weights are stored in RITES' name and transferred as project deliverables.
4. **Model Licences:** open-source language model (open-source licence (commercial use permitted)); open-source language model (Apache 2.0); open-source language model (Apache 2.0). All licences permit deployment without royalty.
5. **Security Controls:** Model serving endpoints are not exposed to the internet; accessible only over RITES internal network or VPN during development. open-source LLM serving framework API endpoints are behind the API Gateway with JWT authentication and RBAC enforcement. All inference calls are logged with model ID, version, token count, and latency in the immutable audit trail.

Model Portability, Exit and Migration Plan: The selected open-source LLMs are served via open-source LLM serving framework, which is model-agnostic. No vendor is in a position to impose subscription dependence, library licensing constraints, or forced upgrades. Specific commitments:

- **Model weights are self-hosted on RITES-controlled infrastructure;** RITES owns all fine-tuned weights and LoRA adapters as part of the all-IP-vests-with-RITES commitment.
- **Model deprecation resilience:** if any model is deprecated by its originating research group, the open-source LLM serving framework inference layer supports drop-in replacement with any compatible open-source model requiring only weight-swap and regression testing, with no architectural change, no API contract change, and no licence renegotiation.
- **Exit/migration from the entire AI stack requires zero proprietary unlocking:** all data is in open formats (relational database, Iceberg, object storage layer), all APIs follow OpenAPI 3.0, and all model artefacts are transferable to any successor agency.
- **No component in the AI stack creates library-ransom risk:** the platform continues to function at full capability if any single open-source model is swapped, updated, or replaced, as the Hallucination Guard and RAG pipeline are model-agnostic by design.

6.9 Enterprise Cost Tracking, Audit, and Governance

To address enterprise management aspects beyond core functionality, the platform incorporates specific features for operational oversight:

1. **AI Cost Tracking and Optimization:** The observability framework monitors infrastructure resource consumption (GPU hours, CPU usage, and object storage costs). Critically, it tracks LLM inference cost per transaction by counting token usage for all AI advisory service calls. This fine-grained data is aggregated to provide cost-per-DPR-appraisal metrics, allowing RITES to monitor and optimize operational expenditure, especially for high-cost AI components.
2. **Enterprise Audit Trails:** The existing audit logging is extended to create a comprehensive, forensic record necessary for external regulatory and financial audits. This includes immutable logging of all financial-technical consistency checks, access to historical DPR data, all changes

to master data, and the full provenance chain for every AI-assisted decision. The audit log is accessible via an API and dashboard for compliance analysts.

3. **Governance Framework Integration:** Key governance processes are enforced programmatically: mandatory four-eyes principle (two human approvals) for all changes to validation rules and standards libraries and version control for all sector-specific templates and scoring frameworks, ensuring that changes to the core appraisal methodology are traceable, controlled, and approved by RITES leadership.

7. Integration and Interoperability Architecture

This section covers integration with external platforms: PM GatiShakti, GIS systems, BIM tools, and national enterprise systems.

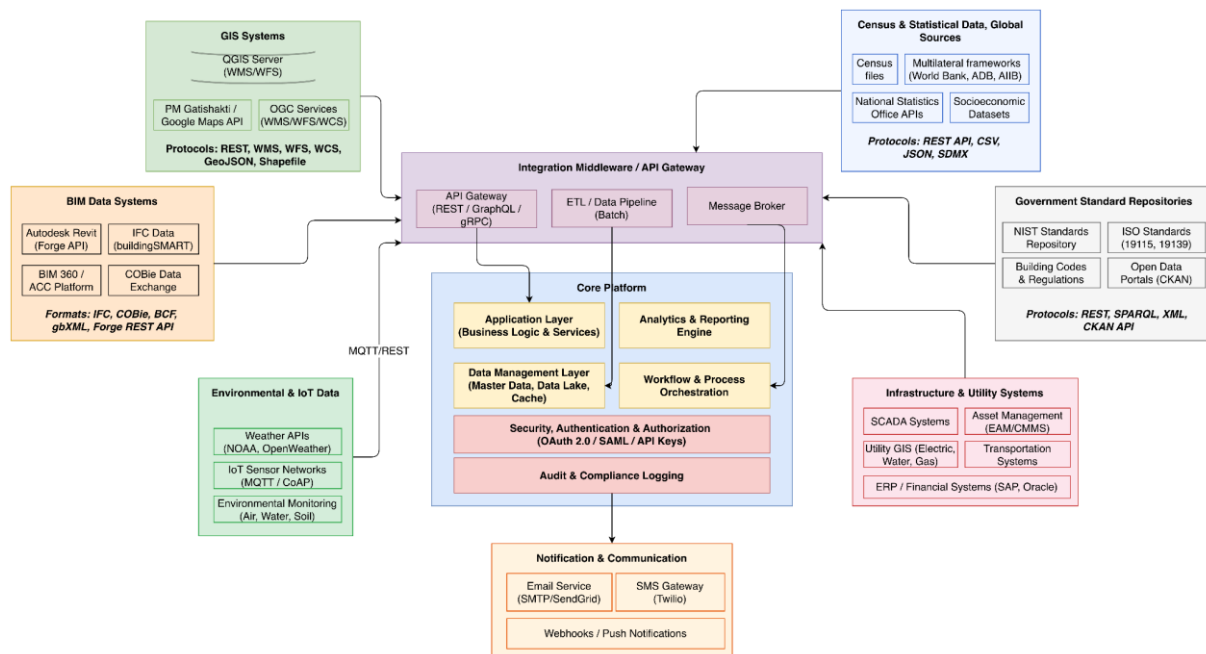


Figure 7: Integration Architecture

7.1 Integration with GIS Platforms and Spatial Systems

The GIS Service integrates with PM GatiShakti's NIC-hosted API using the OGC-standard WMS and WFS protocols. For each project boundary submitted by an appraiser, the service fetches up to eight constraint layers: forest cover, flood zone, seismic zone, Coastal Regulation Zone (CRZ), national highway alignment, railway corridor, power transmission lines, and protected wildlife areas etc. The response is a structured GeoJSON payload that the platform intersects with the project boundary polygon using server-side spatial analysis (geospatial database extension). Conflicts are classified as HARD block (requires statutory clearance before DPR approval), advisory (appraiser must acknowledge), or informational (no action required).

The GIS Service implements graceful degradation for PM GatiShakti unavailability. When the external API times out or returns an error, the service falls back to a RITES-hosted geospatial tile and feature server instance that holds a periodic snapshot of the nationally significant constraint layers. The UI



surfaces a 'Degraded Mode' indicator showing the snapshot date, and the appraiser is offered a manual GIS input option to override or supplement the automated check. This design means GIS constraint analysis does not block the appraisal workflow when external services are unavailable. Note: RITES will need to obtain non-production PM GatiShakti API credentials from NIC before the Phase 2 milestone to enable full integration testing.

7.2 Integration with PM GatiShakti National Platform

PARAKH implements full integration with the PM GatiShakti National Master Plan portal for spatial data enrichment of DPRs across all 13 infrastructure sectors, consuming WMS/WFS geospatial layers for infrastructure constraint analysis and project spatial context.

Integration with PM GatiShakti is implemented through its published REST APIs for project registration, geospatial layer access (WMS/WFS), and project status update exchange. PARAKH consumes GatiShakti's infrastructure layer data (existing projects, planned corridors, utility lines) to enrich DPR geospatial context and to check for alignment conflicts with existing or planned national infrastructure. The integration is designed with a graceful degradation pattern: if GatiShakti services are unavailable, PARAKH caches the last-successfully-fetched layers (with staleness indicators visible to users) and continues operating in read-only spatial mode. Credentials and non-production API endpoints will be obtained from RITES as per the integration schedule.

7.3 Integration with BIM and Engineering Design Systems

PARAKH accepts BIM models in IFC 2x3 and IFC 4 format (the open, non-proprietary standard supported by all major BIM authoring tools). Files up to 2 GB are supported via chunked multipart upload through the Document Service, which validates the MIME type and runs an antivirus engine virus scan before writing to the object storage layer. The upload is fully async: the author receives an immediate acknowledgement with a processing-queued indicator, and an in-app notification (via SSE) is pushed when parsing completes. This eliminates the error-prone manual re-entry of structural quantities into the cost schedule - a step that currently introduces rounding discrepancies and version drift between the engineering drawings and the financial projections in a DPR.

Parsing is performed by the IFC Parser service using the open-source IFC processing library. The parser extracts the full structural tree walls, slabs, columns, beams, MEP components, openings along with quantities (volumes, areas, lengths per element type), materials, and property sets. The parsed element catalog is published as an ifc.parsed event, consumed by the Quantity Takeoff Engine. Where an IFC model is absent - a common scenario for early-stage DPRs - the author completes structural parameters manually; BIM integration is an accelerator, not a hard dependency.

The Quantity Takeoff Engine maps extracted BIM quantities against the MDM sector unit rate schedule to compute cost estimates per line item. Computed values are pre-populated into the corresponding DPR cost schedule fields, clearly marked as 'Auto-filled from BIM | Review required' in the UI. Authors can accept, override, or delete any BIM-derived value; every interaction is logged with full provenance (field path, BIM source object key, parser version, author action, timestamp). Appraisers see this provenance chain during review, so they can trace any cost figure back to its BIM origin.



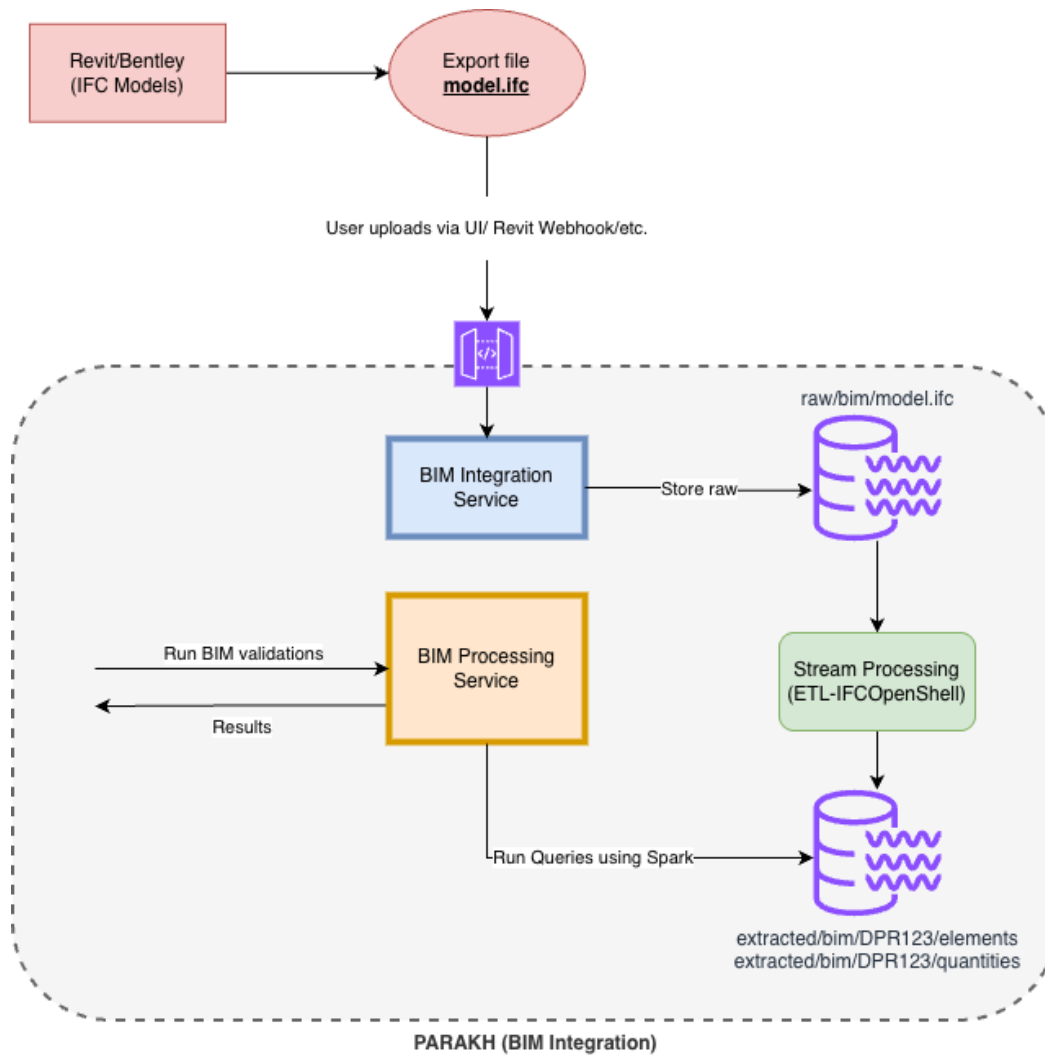


Figure 7.3: Integration with BIM

7.4 Integration with Other Enterprise and National Systems

PARAKH's integration catalogue is extensible through the API Gateway. Planned integrations include: RITES' internal ERP/project management system (for project metadata synchronization), email and notification systems (SMTP/push notifications for workflow alerts), and national data repositories (NRSC for satellite imagery, IMD for meteorological data, GSI for geological survey data). All integrations use REST APIs with OAuth2.0 or API key authentication, with integration specifications documented as OpenAPI schemas and maintained in the Integration Catalogue deliverable.

7.5 API Gateway: Logging, Throttling, and Access Control

The API Gateway enforces API governance: every API call is authenticated (JWT/API key), authorized against RBAC policies, rate-limited per consumer tier, and logged to search and analytics engine for audit and analytics. API versioning (v1, v2) is managed at the gateway, supporting backward-compatible evolution of the API surface. Deprecated API versions remain operational for a minimum

of six months after the deprecation notice is published; notices are surfaced via a Sunset response header and announced in the Developer Portal changelog. Migration guides are provided before any version retirement, and consumers on deprecated versions receive advisory alerts in the portal. Developer portal capabilities (developer portal) provide API documentation, testing consoles, and credential management for authorized third-party integrators.

7.6 Sandbox and Developer Environment for Third-Party Integration

A dedicated Developer Sandbox environment provides authorized third-party systems and agencies with a full-featured PARAKH integration environment populated with synthetic, schema-identical data. The sandbox is accessible over RITES-defined network controls (VPN or allow-listed IPs) and provides all production API endpoints with independent rate limits and credential sets. Mock services simulate national platform integrations (GatiShakti, NRSC) for integration testing without dependency on production national systems. Sandbox availability targets 99% during business hours.

7.7 Interoperability Testing and Compliance Validation

Interoperability compliance is validated through a structured test suite: OpenAPI conformance tests (Dredd/Schemathesis) verify that all API responses match published schemas; OGC compliance tests validate geospatial tile and feature server service outputs against OGC Compliance & Interoperability Testing Initiative (CITE) frameworks; IFC validation uses buildingSMART's IFC validation service. Integration acceptance criteria specifying named systems, test data, and expected outcomes are defined in the Integration Test Plan and agreed with RITES.

8. Deployment, Infrastructure and Hosting Strategy

This section describes how PARAKH is deployed: the containerization strategy, environment structure (Dev, UAT, Production), infrastructure sizing, and disaster recovery. The platform runs unchanged on cloud, NIC MeghRaj, or on-premises; the production infrastructure is RITES-designated.

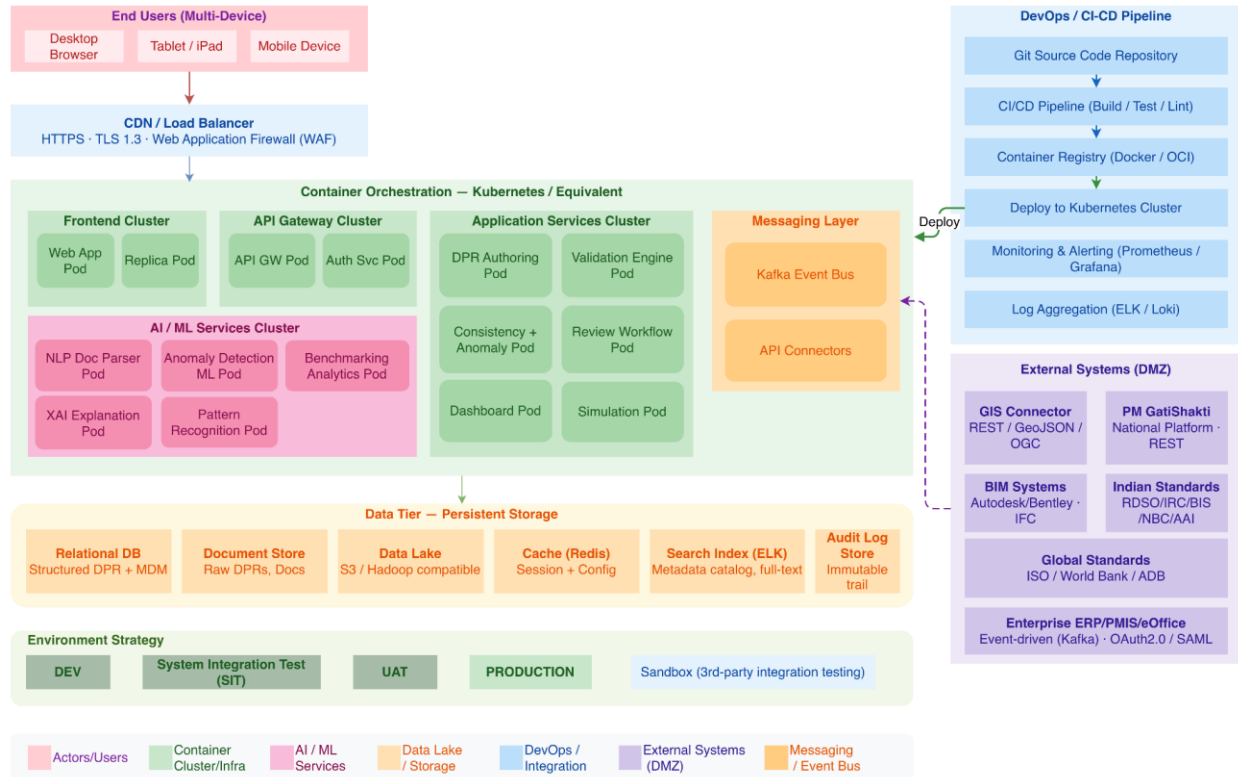


Figure 8 - Deployment Architecture

8.1 Deployment Model: Cloud, On-Premise and Hybrid Options

PARAKH is designed as a deployment-agnostic platform. The containerized, container orchestration platform-based architecture runs identically on a public cloud (any major cloud provider), a government-empanelled cloud (NIC MeghRaj, NICSI, or empanelled CSPs under MeitY), a private cloud, or bare-metal on-premise infrastructure. GlobalLogic conducts all development and testing phases on provisioned cloud infrastructure. Post-UAT, the platform is deployed to RITES-designated infrastructure. Infrastructure sizing recommendations, Helm charts, Terraform IaC scripts, and deployment runbooks are provided to RITES as deliverables.

8.2 Containerization and Virtualization Approach

All application components are containerized using container runtime with multi-stage builds to minimize image size and attack surface. Container images are scanned for vulnerabilities using Trivy in the CI pipeline before promotion. container orchestration platform manages container orchestration: pod scheduling, health checks (liveness and readiness probes), resource quotas, horizontal pod autoscaling, and rolling deployments with zero-downtime capability. Helm charts package all container orchestration platform manifests for reproducible deployments across environments. Namespace-level isolation separates development, UAT, and production workloads on shared clusters.

8.3 Infrastructure Sizing and Optimization Principles

Infrastructure sizing is optimized per the RFP's explicit requirement to avoid unnecessary overhead. The microservices architecture allows each component to be sized independently: the NLP document processing service (GPU-optional) is sized for the expected DPR ingestion throughput; the API serving layer is sized for concurrent user load; the Data Lake compute (Spark) is sized for batch processing windows. AI/ML inference components are designed to run on CPU by default, with GPU acceleration only for the OCR and NLP training pipelines, which are scheduled batch jobs rather than real-time services. Detailed sizing recommendations are provided as part of the Phase 1 Architecture Document deliverable.

8.4 Environment Strategy: Development, UAT, Production

Four environments are maintained throughout the project: Development (Globallogic infrastructure, for active development and unit testing), Integration/Test (GlobalLogic infrastructure, for integration and system testing), UAT (production-like environment with anonymized/synthetic data, for RITES acceptance testing), and Production (RITES-designated infrastructure, post-UAT). Feature flags control which features are active in each environment. Environment parity is enforced through identical Helm chart configurations; differences are limited to resource sizing and external endpoint credentials. Test data in UAT is synthetically generated to match production data volumes and distributions without containing real project data.

8.5 Disaster Recovery and Business Continuity Planning

Disaster Recovery targets (RTO and RPO) will be defined and agreed with RITES during project initiation; the architecture below supports aggressive recovery objectives. database streaming replication provides real-time data durability; object storage erasure coding provides object storage fault tolerance. Automated database backups (daily full, hourly incremental) are stored in a geographically separated backup location. container orchestration platform cluster recovery procedures and application deployment (from Helm charts in source control) are documented and tested during the Phase 6 commissioning sprint. The DR plan is delivered as part of the Operations Runbook and tested annually during the AMC period.

8.6 Post-UAT Handover to RITES-Designated Infrastructure

Upon UAT sign-off, GlobalLogic executes a structured infrastructure handover: provisioning the production environment on RITES-designated infrastructure using IaC scripts; migrating UAT-validated data to production; configuring monitoring, alerting, and backup services; performing Go-Live smoke tests; and providing a Go-Live Readiness Certificate. RITES operations team receives hands-on training during the handover sprint.

9. Testing and Quality Assurance Strategy

Testing and quality assurance across unit, integration, performance, security, and UAT levels are described here. VAPT is conducted at pre-UAT and pre-Go-Live stages.

9.1 Testing Framework and Levels (Unit, Integration, System)

GlobalLogic adopts a test pyramid approach: a broad base of automated unit tests (Java/Python testing framework, >80% code coverage target), a middle layer of automated API integration tests (REST Assured, Pytest-httpx), and a focused set of system tests (browser testing framework for frontend, Postman/Newman for API flows). All tests are version-controlled alongside application code and executed automatically in the CI/CD pipeline on every commit. Test results, coverage reports, and quality gate statuses are visible to the RITES project team through the project management dashboard.

9.2 Performance and Load Testing

Performance testing uses k6 (open-source, scriptable load testing) to simulate concurrent user loads across key platform workflows: DPR submission, validation engine execution, dashboard rendering, and geospatial map tile loading. Performance baselines are established during milestone 4 and validated against agreed NFRs before UAT. Load tests simulate at least 3× the expected peak concurrent user load to validate headroom. Results are reported as p50/p95/p99 response times, error rates, and resource utilization profiles.

9.3 Security Testing and Vulnerability Assessment

Security testing is integrated throughout the SDLC: SAST (SonarQube, Semgrep) on every commit; dependency scanning (OWASP Dependency-Check, Trivy) in CI; DAST (OWASP ZAP) on every deployment to the Integration environment; container image scanning (Trivy) before image promotion. Formal VAPT is conducted at pre-UAT stage and pre-Go-Live stage. The VAPT scope covers application, API, data, infrastructure, AI/ML components, and all external integrations. All Critical and High vulnerabilities are remediated before UAT commencement. VAPT reports and remediation certificates are submitted to RITES.

9.4 User Acceptance Testing (UAT) Planning and Execution

UAT is conducted in a production-like environment with RITES-nominated users and SMEs across all 13 sectors and key functional roles. A UAT Test Plan is prepared and agreed with RITES, defining test scenarios, test data, acceptance criteria, and the UAT review panel. GlobalLogic provides a dedicated UAT support team for the UAT period. UAT sign-off requires RITES' formal acceptance against agreed criteria, documented in the UAT Acceptance Report.

9.5 API Conformance and Interoperability Testing

API conformance is validated using Schemathesis (contract-based, property testing) against the published OpenAPI 3.0 specifications. OGC service compliance is validated using the OGC CITE test suite for WMS 1.3, WFS 2.0, and WMTS. IFC file processing is validated against buildingSMART IFC validation schemas. Integration test results for all named external systems are included in the System Integration Test Report submitted to RITES before UAT.

9.6 Defect Management and Resolution Process

Defects are tracked in a shared defect management system (CI/CD platform Issues or Jira, per RITES' preference) with severity classification (Critical/Major/Minor/Irritant) aligned to the SLA categories. Resolution timelines follow the SLA matrix. A weekly defect review meeting with the RITES project team tracks open defects, resolution progress, and re-test status. Defect root cause analysis reports are prepared for Critical defects and shared with RITES for system improvement insights.

10. Implementation Methodology and Project Delivery Plan

The delivery methodology and schedule are set out here. Nine months of build, test and stabilization stages run across five milestones (M1 at D+1, M2 at D+3, M3 at D+4, M4/Go-live at D+7 and M5 at D+9), followed by 15 months of Operations and Maintenance.

10.1 Delivery Approach: Agile Methodology

GlobalLogic employs an Agile delivery model calibrated for government enterprise projects of this nature. The macro-level delivery follows the milestone structure defined in the RFP, each milestone has defined deliverables and payment triggers that require formal RITES approval. Within each milestone, development work is organized in Agile sprints with defined sprint goals, backlog prioritization, daily standups, sprint reviews (open to RITES project team participation), and retrospectives. This provides RITES with predictable milestone-level governance and delivery visibility while maintaining the iterative agility needed to incorporate RITES SME feedback on validation rules, templates, and workflows without costly change cycles.

Detailed project timeline with milestones and deliverables

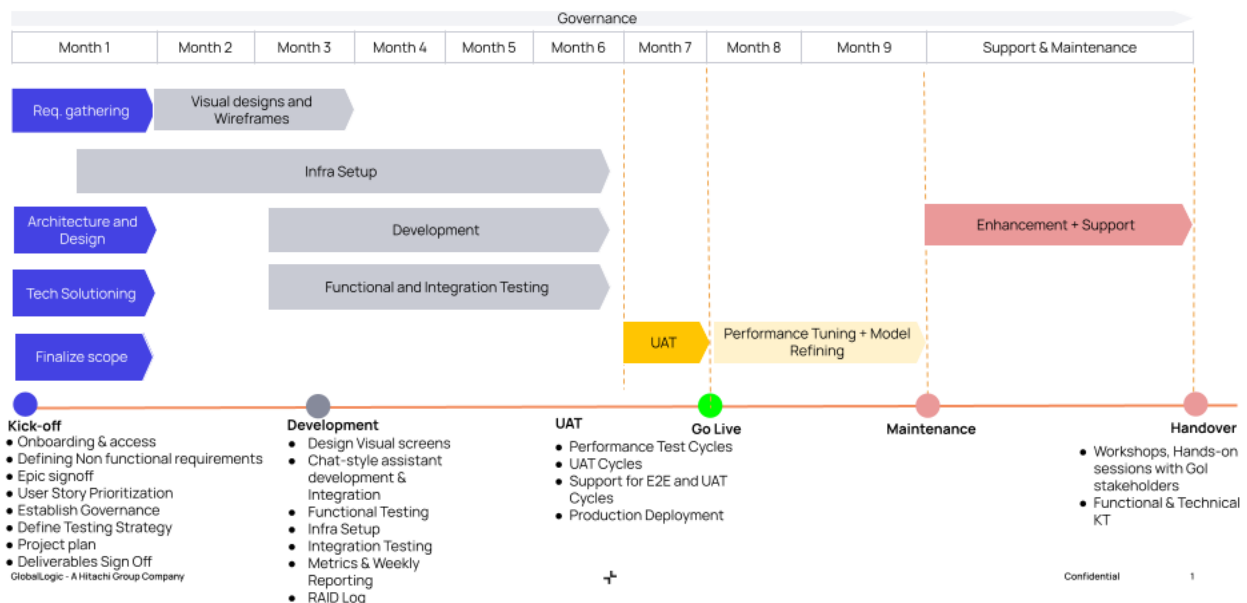


Figure 10.1: Project Timeline

10.2 Milestone Chart



Reference: RITES/ITES/APR/2026 · D0 = Contract Award Date · Total Contract Value = 100% · M6 maintenance runs D+9 → D+24 (15 months)

The following work schedule sets out the project's operational cadence on a week-by-week basis. It specifies the engineering, development, and analytical activities planned for each period, along with the deliverables, validation reports, and milestones that RITES can expect for review and sign-off at the end of each week.

Note: This schedule is created based on the information provided in the RFP and is subject to change once the project reaches the discovery / requirement gathering stage

WORK SCHEDULE

AI-Based DPR Appraisal Platform · RITES Ltd. · Tender No.: RITES/ITES/APR/2026

Contract Duration: 24 Months (9 Months Development + 15 Months Maintenance) · Timeline Reference: From Date of Letter of Award (D)

Month	Week	Task Category	Detailed Task Description	RFP Reference	Key Deliverables
PHASE 1: DEVELOPMENT & STABILIZATION <i>Months 1 – 9 · Milestones M1 through M5</i>					
Month 1	Week 1	Project Kick-off & Mobilisation	› Conduct project kick-off meeting with RITES stakeholders › Establish project governance structure › Finalise project charter and communication plan setup PMO	Sec 6.6 – Roles & Responsibilities; Sec 3.1 – Platform Design	› Signed Project Charter › PMO setup confirmation › Stakeholder RACI matrix



Month 1	Week 2	Requirements Gathering	› Conduct requirements workshops with RITES SMEs across domains (civil, transport, finance, environment) › Document AS-IS appraisal process › Capture functional & non-functional requirements	Sec 3 – Scope of Work; Sec 6.6 – RITES Role	› Requirements workshop proceedings › Draft Functional Requirements document
Month 1	Week 3	System Architecture Design	› Design modular API-driven platform architecture › Define Data Lake architecture › Design microservices framework › Prepare system context & component architecture › Define API standards (REST/GraphQL/gRPC)	Sec 3.1 – Platform Design & Architecture; Annexure 12	› Draft System Architecture Document › Data Lake design blueprint › API Framework specification
Month 1	Week 4	SRS Finalisation & AI/ML Strategy	› Finalise Software Requirements Specification (SRS) › Define AI/ML strategy including NLP/RAG model selection › Prepare implementation roadmap with milestone schedule › Present to RITES for approval	Sec 3 – Scope; Sec 3.7 – AI & Analytics; Milestone 1 (D+1M)	› Approved SRS › AI/ML Strategy document › Implementation Roadmap › DPR Framework for all sectors
Month 2	Week 1	Core Platform Development	› Set up development environment › Implement base platform infrastructure using containerisation › Develop user authentication and RBAC framework › Implement core database schema	Sec 3.1 – Platform Design; Sec 3.8 – Deployment & Testing	› Development environment setup › Base platform skeleton › Authentication module (OAuth2.0/SAML)
Month 2	Week 2	DPR Preparation Module – Ph.1	› Develop DPR authoring module with configurable templates › Implement structured data capture › Develop document management and version control system	Sec 3.2 – DPR Preparation & Authoring; Annexure 12	› DPR Template Engine (v1) › Structured data capture forms › Document management module (v1)
Month 2	Week 3	Validation Engine – Ph.1	› Develop rule-based validation engine for DPR inputs › Implement automated completeness checks › Build basic consistency validation › Implement initial workflow engine	Sec 3.3 – Validation & Appraisal Engine; Annexure 12	› Rule-based validation engine (v1) › Completeness check module › Basic workflow engine



Month 2	Week 4	Data Lake & AI Components – Ph.1	› Set up Data Lake infrastructure › Implement initial data ingestion pipelines › Develop AI-enabled document parsing and NLP-based summarisation › Build RAG-based validation support › Develop initial dashboards	Sec 3.4 – Data Management; Sec 3.7 – AI & Analytics	› Data Lake infrastructure (initial) › Document parsing AI module › NLP summarisation module › Initial dashboards
Month 3	Week 1	Core Platform Integration & Testing	› Integrate DPR authoring, validation engine and Data Lake components › Develop API framework › Conduct module-level unit testing › Fix defects › Prepare for Milestone 2 delivery	Sec 3.6 – Integration; Sec 3.8 – Deployment & Testing	› Integrated core platform › API Framework (integration-ready) › Unit test reports
Month 3	Week 2	Dashboard & Reporting Module	› Develop decision-support dashboards › Implement DPR processing status views › Build basic reporting capabilities › Integrate SLA monitoring dashboard foundation	Sec 3.5 – AI Analytics; Sec 6.3.7 – SLA Reporting	› Decision-support dashboards (v1) › Reporting module (v1)
Month 3	Week 3	QA Preparation	› Conduct internal QA and integration testing of Draft Tool Version 1 › Conduct performance baseline testing › Document known issues and resolution plan	Sec 3.8 – Deployment, Testing & Commissioning	› QA test reports › Performance baseline report
Month 3	Week 4	M2 Submission Prep	› Demonstrate core platform with DPR preparation module, document management, basic workflows, Data Lake, rule-based validation, AI capabilities and API framework	Milestone 2 (D+3M); Sec 6.10; Sec 6.11 – Payment (25%)	› Draft Tool Version 1 (accepted) › Demonstration report › Defect log and resolution plan
Month 4	Week 1	Sector-Specific Module Development	› Develop sector-specific DPR modules (transport, civil, structural, environment, finance) › Configure sector templates, validation rules and scoring frameworks › Implement sector-specific ontology	Sec 3.2 – DPR Preparation; Sec 3.3 – Validation; Annexure 12	› Sector-specific DPR modules (transport, civil, environment, finance) › Sector validation rule sets



Month 4	Week 2	Advanced Validation & Appraisal Engine	› Implement cross-validation across DPR sections › develop configurable scoring and appraisal frameworks › Build dependency-driven validation › Implement gap identification and compliance checks › Integrate benchmarking engine	Sec 3.3 – Validation & Appraisal; Annexure 12	› Advanced validation engine (v2) › Benchmarking engine › Configurable scoring framework
Month 4	Week 3	Advanced AI Capabilities	› Develop anomaly detection models › Implement AI-assisted appraisal engine › Build pattern recognition for DPR parameters › Develop explainable AI output module › Ensure AI accuracy benchmarks per RFP	Sec 3.7 – AI & Advanced Analytics; Sec 6.4C – AI Accuracy	› Anomaly detection module › Assisted appraisal AI engine › Explainable AI outputs › AI accuracy baseline report
Month 4	Week 4	Data Lake Expansion & GIS Integration	› Expand Data Lake with historical DPR ingestion › integrate external datasets (GIS, geotechnical, environmental, socio-economic) › Implement metadata cataloguing and data governance › Initial GIS platform integration	Sec 3.4 – Data Management; Sec 3.6 – Integration	› Expanded Data Lake › Historical DPR dataset ingested › GIS integration (Phase 1) › Metadata catalogue
Month 5	Week 1	RBAC, Audit Logs & Security Hardening	› Implement full Role-Based Access Control (RBAC) › Develop comprehensive audit trail and logging › Implement API gateway with throttling › Conduct security review against OWASP › Implement OAuth2.0/SAML	Sec 3.6 – Integration; Sec 6.3 – SLA; Sec 6.7 – IPR	› RBAC module › Audit log system › API gateway setup › Security assessment report
Month 5	Week 2	Advanced Analytics & Scenario Evaluation	› Develop scenario analysis and alternative configuration evaluation › Build advanced analytics dashboards › Implement historical benchmarking visualisations › Develop AI confidence monitoring module	Sec 3.5 – AI Analytics; Sec 6.3.7 – SLA Reporting	› Scenario analysis module › Advanced analytics dashboards › AI Confidence monitoring module
Month 5	Week 3	Integration Testing & Performance Validation	› Conduct end-to-end integration testing of Tool Version 2 › Test GIS and external system integrations › Conduct performance and load testing › Validate AI accuracy against benchmarks › Fix critical defects	Sec 3.8 – Deployment & Commissioning; Sec 6.4C – AI Accuracy	› Integration test report › Performance test report › AI accuracy validation report › Defect resolution log



Month 5	Week 4	M3 Submission Prep	› Demonstrate enhanced multi-sector platform with advanced validation, appraisal engine, expanded Data Lake, advanced AI capabilities, GIS integration and RBAC	Milestone 3 (D+5M); Sec 6.10; Sec 6.11 – Payment (25%)	› Tool Version 2 (accepted) › All sector modules deployed › Advanced AI features live › RBAC and audit logs active
Month 6	Week 1	Full Platform Integration & Production Readiness	› Complete integration of all platform modules › Finalise all API integrations (GIS, PM GatiShakti, BIM where applicable) › Implement sandbox/developer environment › Conduct comprehensive system integration testing	Sec 3.6 – Integration; Sec 3.8 – Deployment & Testing	› Fully integrated platform › Sandbox environment › Integration test completion certificate
Month 6	Week 2	Analytics, Decision Support & Reporting Finalisation	› Finalise decision-support dashboards and scenario analysis tools › Implement full analytics suite › Complete AI Hallucination and Confidence Audit Report mechanism › Finalise SLA monitoring dashboard	Sec 3.5 – AI Analytics; Sec 6.3.7 – SLA Reporting	› Final analytics dashboards › Operational SLA dashboard › AI Audit Report template
Month 6	Week 3	Security Audit & Compliance Validation	› Conduct comprehensive security audit › Validate OWASP compliance › Test against AI/agentic security risks › Conduct penetration testing › Review data rights and confidentiality controls › Validate DPI alignment	Sec 6.2 – Team; Sec 6.7.9 – Data Security; Annexure 12	› Security audit report › OWASP compliance certificate › Penetration test report › DPI compliance checklist
Month 6	Week 4	UAT Preparation & Environment Setup	› Prepare production/UAT environment › Develop UAT plan and test cases covering all modules › Brief RITES SMEs and nominated users on UAT process › Set up performance monitoring and incident management tools	Sec 3.8 – Deployment & Commissioning; Sec 6.3 – SLA	› UAT environment configured › UAT plan and test cases › Performance monitoring setup
Month 7	Week 1	UAT – Phase 1	› Conduct UAT for DPR Preparation, Authoring and Validation modules › RITES SMEs validate templates, ontology, sector rules and scoring › Document defects and change requests › Implement P1/P2 defect fixes	Sec 3.2 – DPR Preparation; Sec 3.3 – Validation; Sec 6.6 – RITES Testing	› UAT Phase 1 report › Defect log › Change request register



Month 7	Week 2	UAT – Phase 2	› Conduct UAT for AI capabilities, analytics dashboards, GIS integrations and external system connections › Validate AI accuracy, anomaly detection and benchmarking outputs › Test scenario analysis module	Sec 3.5 – AI Analytics; Sec 3.6 – Integration; Sec 3.7 – AI	› UAT Phase 2 report › AI accuracy validation by RITES SMEs › Integration acceptance sign-off
Month 7	Week 3	UAT Defect Resolution & Final Fixes	› Resolve all UAT-identified defects › Implement approved change requests › Re-test resolved issues › Finalise API documentation › Prepare OpenAPI-compliant integration specifications › Conduct final performance validation	Sec 3.1 – Platform Architecture; Sec 3.8 – Deployment	› Resolved defect log › API documentation › OpenAPI specifications › Final performance validation report
Month 7	Week 4	M4 Submission Prep	› Submit Final Version of Tool for RITES acceptance › Demonstrate fully integrated production-ready platform with all modules, GIS/API integrations, analytics, scenario analysis, security validation and UAT completion certificate	Milestone 4 (D+7M); Sec 6.10; Sec 6.11 – Payment (20%)	› Production-ready platform (accepted) › UAT Completion Certificate › Interoperability compliance artifacts › All integrations live
Month 8	Week 1	Production Deployment & Go-Live	› Deploy fully accepted platform to production › Conduct go-live readiness checks › Configure monitoring and alerting › Establish incident management process › Notify RITES of go-live	Sec 3.8 – Deployment; Sec 6.3 – SLA; Sec 6.10	› Production deployment completion report › Go-live sign-off › Incident management process document › Monitoring activated
Month 8	Week 2	Performance Tuning & Issue Resolution	› Monitor system performance post go-live › Fine-tune AI models based on live data › Optimise query performance › Address data migration issues	Sec 6.3 – SLA; Sec 6.4 – Liquidated Damages; Sec 6.4C – AI Accuracy	› Post go-live performance report › AI model tuning log
Month 8	Week 3	Training – Administrators & Power Users	› Conduct structured training for RITES administrators and power users › Train on platform configuration, RBAC management, template administration, data ingestion and system monitoring	Sec 3.9 – Training, Documentation & Change Management	› Admin training completion certificates › Training material (admin guide) › Recorded training sessions



Month 8	Week 4	Training – End Users & Stakeholders	› Deliver end-user training to RITES domain staff across sectors › Conduct hands-on DPR preparation, validation and appraisal workflow training › Distribute user manuals › Establish helpdesk support process	Sec 3.9 – Training, Documentation & Change Management	› End-user training completion records › User manual › Helpdesk setup document
Month 9	Week 1	System Stabilisation & AI Model Refinement	› Monitor system under full operational load › Resolve outstanding issues › Retrain and fine-tune AI models using accumulated live data › Improve anomaly detection accuracy › Address AI hallucination incidents	Sec 3.7 – AI & Analytics; Sec 6.4C – AI Accuracy; Sec 6.3.7	› AI model refinement report › AI Confidence & Hallucination Audit Report (Month 1) › System stability report
Month 9	Week 2	Technical & User Documentation	› Prepare comprehensive technical documentation including system architecture, API documentation, database design, deployment guide and security guide › Complete user documentation including user manuals and FAQs	Sec 3.9 – Training, Documentation & Change Management	› Technical documentation package › User manuals (all modules) › API documentation › Deployment and security guide
Month 9	Week 3	Knowledge Transfer & Change Management	› Conduct knowledge transfer sessions with RITES IT team › Hand over source code at defined milestone › Prepare operational runbooks › Support change management activities for wider organisational adoption	Sec 3.9 – Change Management; Sec 6.7.1 – IPR & Source Code Handover	› Knowledge transfer completion certificate › Source code handover › Operational runbooks › Change management report
Month 9	Week 4	M5 Submission Prep	› Submit Stabilized System for RITES acceptance › Demonstrate system with resolved issues, tuned performance, refined AI models, completed training, full documentation and operational support readiness › Obtain formal sign-off	Milestone 5 (D+9M); Sec 6.10; Sec 6.11 – Payment (10%)	› Stabilized system acceptance certificate › All documentation delivered › Training completion records › Operational readiness certificate

PHASE 2: OPERATIONS & MAINTENANCE *Months 10 – 24 · Milestone M6 (2% per quarter × 5 quarters)*



Months 10–12	Quarterly	Maintenance & Monitoring – Q1	› Monitor deployed platform for performance, availability and SLA compliance › Resolve incidents within defined SLA timelines › submit monthly SLA reports and quarterly AI audit reports › Implement approved enhancements › Conduct quarterly review with RITES	Milestone 6 – O&M; Sec 6.3 – SLA; Sec 6.4 – Penalties	› Monthly SLA reports (Months 10–12) › Q1 AI Confidence & Hallucination Audit Report › Incident resolution logs › Quarterly performance review report
Months 13–15	Quarterly	Maintenance & Monitoring – Q2	› Continue platform monitoring and maintenance › fine-tune AI models based on evolving RITES usage patterns › Implement bug fixes and patch releases › Support any data expansion requirements › Submit Q2 SLA and audit reports	Milestone 6 – O&M; Sec 6.3 – SLA; Sec 3.10 – O&M Support	› Monthly SLA reports (Months 13–15) › Q2 AI Audit Report › Patch release notes › Data expansion report
Months 16–18	Quarterly	Maintenance & Monitoring – Q3	› Ongoing platform operations › Conduct periodic system health checks › Address any accumulated technical debt › Evaluate platform scalability for increased DPR volumes › Submit Q3 operational reports	Milestone 6 – O&M; Sec 3.10 – O&M Support	› Monthly SLA reports (Months 16–18) › Q3 AI Audit Report › System health assessment › Scalability evaluation report
Months 19–21	Quarterly	Maintenance & Monitoring – Q4	› Continue operations and maintenance › Support RITES in onboarding additional user departments › Implement any regulatory/compliance updates › Conduct mid-maintenance period review with RITES	Milestone 6 – O&M; Sec 3.10 – O&M Support	› Monthly SLA reports (Months 19–21) › Q4 AI Audit Report › Onboarding support records › Mid-period review meeting minutes
Months 22–24	Quarterly	Maintenance & Monitoring – Q5 / Contract Closure	› Final quarter of maintenance › Prepare end-of-contract handover documentation › Conduct final knowledge transfer and system audit › ensure all licenses and subscriptions are in order › Submit final quarterly report › Prepare for contract closure or renewal	Milestone 6 – O&M; Sec 6.11 – Payment (2% per quarter); Sec 6.17 – Exit	› Monthly SLA reports (Months 22–24) › Q5 AI Audit Report › End-of-contract handover document › Final system audit report › Contract closure certificate

Note: D = Date of Letter of Award (LOA). Payment milestones are subject to formal acceptance by RITES. Penalties apply for delays attributable to the Service Provider.

10.4 Sprint Planning and Release Strategy

Sprints are maintained throughout the project. Sprint planning sessions (attended by the GlobalLogic delivery lead and the RITES project manager) define sprint goals and backlog items from the prioritized

product backlog. Sprint reviews include a working demo of completed features, with RITES team members invited to provide direct feedback. Release candidates are deployed to the Integration environment at the end of each sprint; release to UAT occurs at Milestone 4 upon system integration test completion.

10.5 Milestone Schedule and Deliverables Mapping

Each milestone requires formal RITES acceptance before the work for the next milestone commences and before the corresponding payment is due. Phase milestone acceptance criteria are defined in the SRS and agreed with RITES in Milestone 1. GlobalLogic will submit milestone completion reports with evidence artefacts (deployed systems, test reports, delivered documents) for RITES review within the agreed acceptance period.

10.6 Risk Management Framework

Key project risks identified and mitigated:

- **SME availability for knowledge transfer:** Mitigation: Early engagement with RITES SMEs in Phase 1 for parallel knowledge capture; structured template and rule approval process with defined turnaround SLAs.
- **National platform API availability:** Mitigation: GatiShakti integration designed with degraded-mode fallback; integration test plan includes simulator mode for unavailable national services.
- **Scope evolution during development:** Mitigation: Agile sprints enable incremental scope refinement; formal Change Request process with impact assessment governs scope changes post-SRS sign-off.
- **RITES infrastructure readiness for Post-UAT deployment:** Mitigation: RITES infrastructure provisioned and tested for compatibility two months before planned Go-Live.
- **Historical DPR data quality for ML training:** Mitigation: Historical DPR corpus quality assessed in Phase 1; IDP pipeline quality thresholds defined; human-review queue for low-confidence extractions.

11. Project Governance and Management Structure

The project governance structure is described here: team organisation, the RITES-GlobalLogic collaboration framework, and the RASCI matrix. The Project Manager is permanently onsite at RITES Corporate Office, Gurugram, from Day 1.

11.1 Proposed Project Organization and Team Structure

GlobalLogic proposes a co-located and distributed delivery model. The Project Delivery Unit is led by a Project Manager. The delivery team comprises: Architecture Team (Solution Architect, AI Architect); Platform Engineering Team (backend engineers - Java enterprise Java framework and Python REST

API framework); AI/ML Team (ML Engineers, NLP specialists, MLOps engineer); Frontend Team (frontend framework developers, UX designer); GIS Team (geospatial tile and feature server, geospatial database extension, open-source mapping library); QA and DevSecOps Team; and Integration Team. Sector SMEs from GlobalLogic's domain practice (infrastructure, transport, environment) engage RITES' SMEs during knowledge transfer phases.

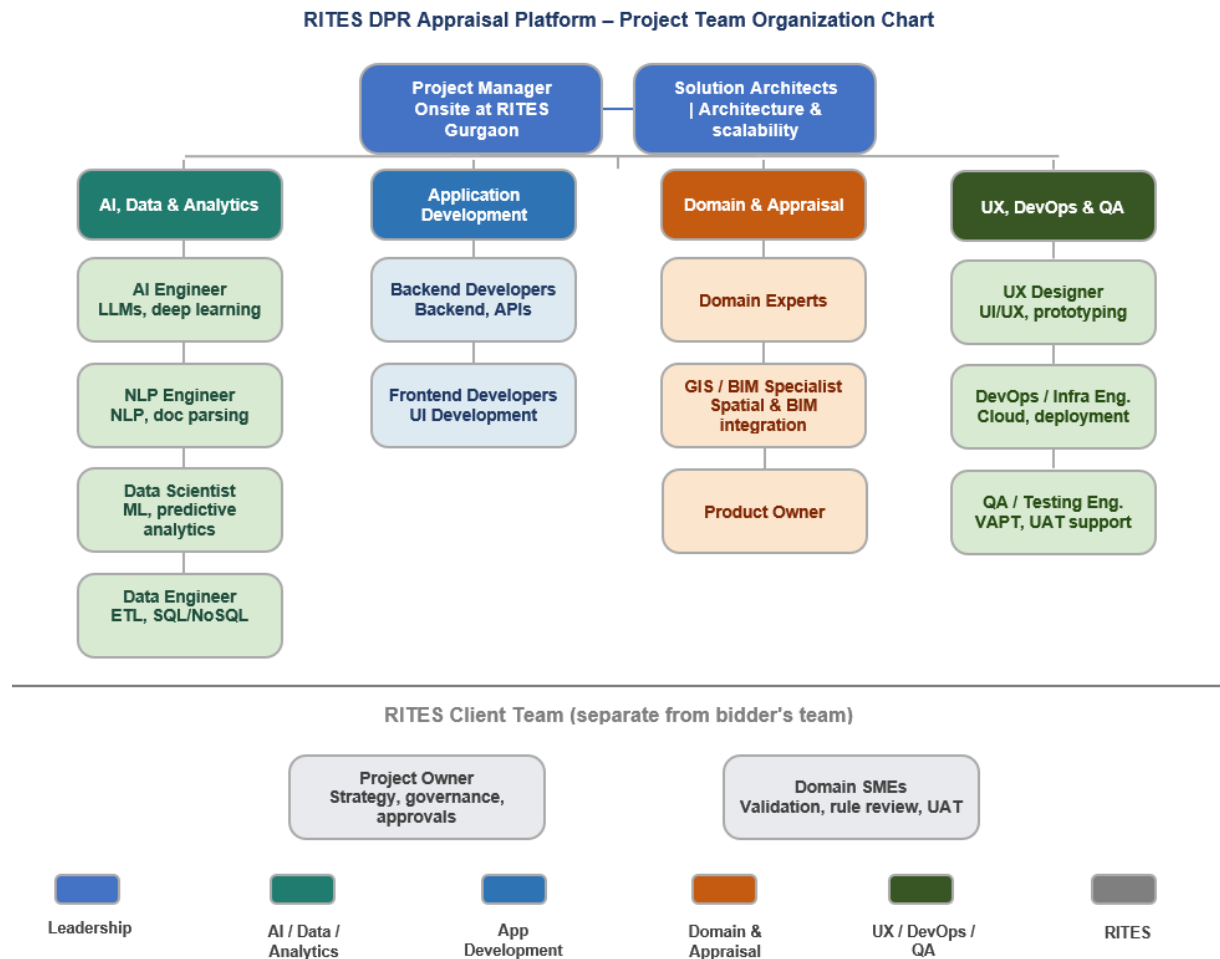


Figure 11.1: Project Team Structure

11.2 RITES & GlobalLogic Collaboration and Communication Framework

The collaboration model includes: weekly project status meetings (Project Manager level, virtual); sprint reviews (open to RITES team, on-site or virtual); monthly steering committee meetings (senior leadership, on-site at RITES Gurugram); a shared project management platform (JIRA/CI/CD platform, with RITES read access); a shared document repository (for deliverable drafts, review, and approval); and a dedicated integration Slack/Teams channel for day-to-day queries. All formal communications (change requests, milestone submissions, acceptance requests) are conducted in writing.

Program Governance

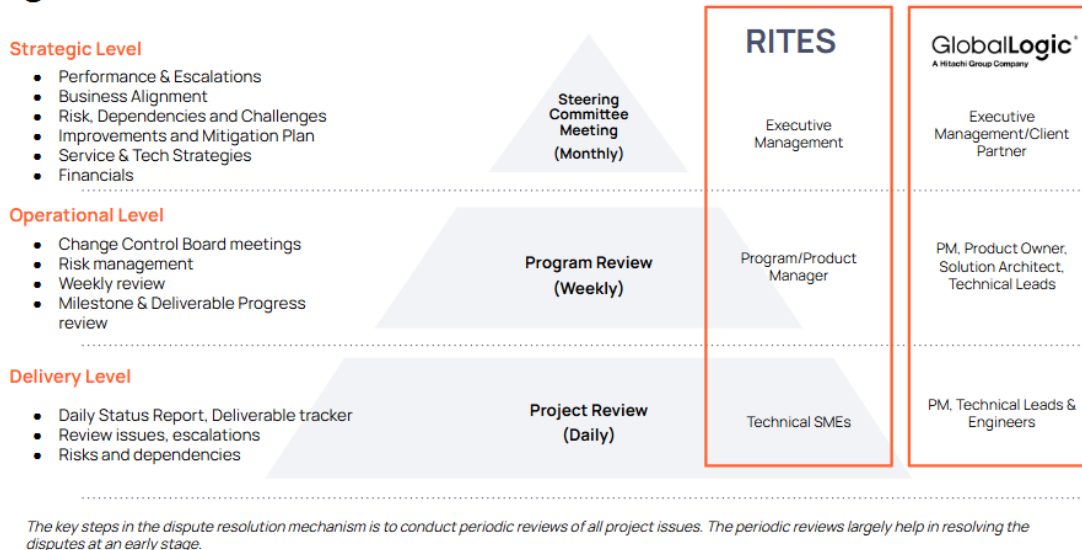


Figure 11.2: Program Governance

11.3 Change Request and Scope Management Process

Scope changes post-SRS sign-off are managed through a formal Change Request (CR) process: CR raised by either party in writing; GlobalLogic provides an impact assessment (scope, timeline, cost) within 5 business days; RITES reviews and approves/rejects; approved CRs are incorporated into the project plan with amended milestone dates if applicable. Refinements within the agreed scope (e.g., validation rule adjustments, UI improvements) are managed as backlog items without a CR, provided they do not extend the project timeline or increase the contract value.

11.4 RASCI Matrix

GlobalLogic proposes the following RASCI matrix to clarify accountability across project roles, reduce ambiguity at role boundaries, and maintain clear communication lines throughout delivery.

LEGEND:

R = Responsible – The party that executes/performs the task

A = Accountable – The party ultimately answerable for the task's outcome (approver/owner)

S = Supported – The party that provides support/resources to the responsible party

C = Consulted – The party whose input/expertise is sought (two-way communication)

I = Informed – The party kept updated on progress/decisions (one-way communication)

Task	RITES	GlobalLogic
Overall program management of the solution & risk management	S,C,I	R,A
Elaborate Business Requirements & sign off on product features	R, A	S,C, I
Solution Architecture, Technology Decisions, and Product Backlog Management and Prioritization	S, C, I	R, A
Create, document, and groom Epics & User Story Backlog	S, C, I	R, A
Development of the Solution as defined in this statement of work (SOW)	S, C, I	R, A
Reviewing and approving acceptance criteria	R, A	S, C, I
Conduct demos of User Stories	S, C, I	R, A
Delivery of software at the end of each sprint	S, C, I	R, A
Ensure external Interfaces are compatible as per requirements and solution architecture	R,A	S,C,I
Resolve external dependencies	R,A	S,C,I
Integration and E2E testing	S, C, I	R, A
User Acceptance Test	R,A	S,C,I
Performance testing	S, C, I	R,A
Infrastructure Devops procurement post UAT	R,A	S, C, I



12. Training, Documentation and Change Management

User training, technical documentation, and change management are covered here. All documentation is a milestone acceptance condition. RITES will receive system manuals, API documentation, architecture decision records, and a training programme covering all user personas defined in Section 1.4.

12.1 User Training Plan (System Users, Administrators, Reviewers)

A structured Training Plan is developed in Phase 1 and executed across three cohorts:

1. **End Users (DPR authors, project coordinators)** - hands-on training in DPR authoring, submission, and collaboration workflows
2. **RITES Reviewers and SMEs** - training in the appraisal workflow, observations management, scoring, and AI insights interpretation
3. **Platform Administrators** - training in template configuration, master data management, RBAC administration, monitoring, and routine operations. Training is delivered through a combination of instructor-led workshops (on-site at RITES Gurugram), self-paced e-learning modules (embedded in the platform), and reference video recordings. A 'Train the Trainer' component enables RITES to extend training to new users independently

12.2 Technical Documentation (Architecture, APIs, Data Models)

Technical documentation deliverables include: System Architecture Document (component diagram, deployment diagram, data flow diagrams); API Documentation (OpenAPI 3.0 specs for all exposed APIs, with examples); Database Schema Documentation (ERD, table descriptions, indexing strategy); Data Lake Design Document (layer definitions, pipeline descriptions, data lineage maps); AI/ML Model Documentation (training data, feature definitions, performance metrics, feature attribution analysis); VAPT Reports and Remediation Certificates; and SBOM (dependency inventory, license register, vulnerability register). All documentation is version-controlled in source control alongside code.

12.3 User Documentation and Manuals

User-facing documentation includes: User Manuals per role (author, reviewer, administrator); quick reference guides for common workflows; and in-app contextual help (tooltips, guided tours, FAQ panels). All user documentation is provided in English. Documentation is maintained as living documents updated with each platform release during the AMC period.

12.4 Change Management and Adoption Strategy

Successful platform adoption requires managing the organizational transition from manual to platform-supported appraisal. GlobalLogic proposes a Change Management approach aligned with established frameworks: stakeholder impact assessment in Phase 1; communications plan (regular platform updates, success stories, executive briefings); a Super User network within RITES to champion adoption and provide peer support; phased rollout (starting with a pilot sector and expanding progressively); and a feedback mechanism for users to report usability issues and improvement



requests directly to the product team. Adoption metrics (active user count, DPRs submitted through the platform, validation pass rates) are tracked and reported monthly during the AMC period.

13. Operations and Maintenance Support Plan

This section covers the 15-month Operations and Maintenance period following the 2 month of stabilization stage. SLA commitments on uptime, latency, and incident response take effect post stabilization phase. The support structure and escalation tiers are defined in Section 13.1.

13.1 Post-Deployment Support Structure and SLAs

GlobalLogic provides Operations and Maintenance (O&M) support for 15 months post stabilization phase as specified in the RFP. The support structure comprises a dedicated L1 helpdesk (ticket logging, initial diagnosis, user query resolution), L2 application support (functional and application-layer issue resolution), and L3 platform engineering (infrastructure, database, security, AI/ML). L1 Oncall helpdesk will support 24*7. L2 and L3 support will be available during business hours (9 am to 5 pm) Monday to Friday excluding public holidays.

Key Considerations: During O&M period, the enhancement capacity of the support team will be limited to **76 person days of efforts per month**. Any enhancements beyond this capacity will be considered in the subsequent months or taken up as part of change requests.

SLAs strictly follow the RFP: Critical 15-minute response, 4-hour resolution; Major 30-minute response, 8-hour resolution; Medium 2-hour response, 2-working day resolution; Low 6-hour response, 5-working day resolution.

13.2 Incident Management and Escalation Process

1. Purpose

The purpose of this Incident Management and Escalation Process is to provide a structured and consistent approach for identifying, managing, communicating, resolving, and preventing incidents that may impact the operation of an AI-based platform. The process is designed to minimize business disruption, protect customer experience, maintain system integrity, and ensure continuous service improvement.

Unlike traditional software platforms, AI-based systems introduce additional risks related to model behavior, data quality, inference performance, hallucinations, bias, drift, security vulnerabilities, and third-party AI service dependencies. Therefore, incident management must address both conventional technology failures and AI-specific operational challenges.

This process applies to all components of the AI platform, including: User-facing applications and interfaces, AI and machine learning models, Large Language Models (LLMs), Prompt orchestration services, APIs and integration layers, Data ingestion and processing pipelines, Vector databases and

knowledge repositories, Cloud infrastructure and hosting environments, Security and identity management systems, Monitoring and observability platforms, Third-party services and external dependencies

2. Incident Definition

An incident is any unplanned interruption, degradation, security concern, performance issue, or abnormal behavior that affects the platform's ability to deliver expected services.

Examples of incidents include:

Platform and Infrastructure Incidents: Service outage, Application crash, Database failure, Network connectivity issues, Cloud infrastructure disruption, Deployment failures

AI-Specific Incidents: Significant reduction in model accuracy, Hallucinated or incorrect responses, Prompt injection attacks, Model drift, Retrieval-Augmented Generation (RAG) failures, Corrupted embeddings or vector stores, Excessive inference latency, Unexpected model behavior

Security Incidents: Unauthorized access attempts, Credential compromise, Data leakage, Malware detection, API abuse, Denial-of-service attacks

Data Incidents: Missing or corrupted datasets, Data pipeline failures, Incomplete synchronization, Incorrect training data ingestion, Knowledge base inconsistencies

3. Incident Detection and Reporting

Incidents may be detected through multiple channels, including:

- Automated monitoring systems
- Infrastructure alerts
- Security monitoring tools
- AI model observability platforms
- Customer support tickets
- Internal testing teams
- Business users
- Third-party vendor notifications

Any employee, customer, partner, or automated system identifying an incident must report it through the designated incident management platform.

Each incident record should contain:

- Unique Incident ID
- Date and time detected
- Reporter details
- Affected service or component
- Incident description



- Business impact assessment
- Severity classification
- Current status
- Assigned owner
- Escalation history
- Resolution details

All incidents must be logged and tracked through closure.

4. Incident Classification

Incidents shall be classified according to their impact on business operations, customers, security, compliance, and platform functionality.

Severity 1 – Critical

Incidents resulting in complete service disruption or significant business impact.

Examples: Full platform outage, Complete AI service unavailability, Major security breach, Exposure of sensitive information, Failure of critical business processes

Severity 2 – High

Incidents causing major degradation but allowing partial operation.

Examples: Significant increase in response latency, Failure of key AI workflows, Major model accuracy degradation, Service interruption affecting multiple customers

Severity 3 – Medium

Incidents with moderate impact and available workarounds.

Examples: Failure of non-critical features, Limited customer impact, Intermittent service degradation, Localized integration failures

Severity 4 – Low

Minor issues with limited operational impact.

Examples: Cosmetic defects, Reporting inaccuracies, Documentation issues, Enhancement requests

Severity may be revised as additional information becomes available.

5. Incident Response Process

Step 1: Incident Identification and Validation

Upon receiving an alert or report, the Operations Team validates whether the issue constitutes a legitimate incident.

Initial activities include: Confirming symptoms, Verifying monitoring alerts, Assessing affected systems, Determining customer impact, Establishing initial severity

If confirmed, an Incident Manager is assigned.

Step 2: Incident Assessment

The Incident Manager coordinates a rapid assessment to understand: Scope of impact, Business consequences, Number of affected users, Potential security implications, Dependency failures, AI model involvement

The assessment determines the appropriate technical teams and stakeholders required for resolution.

Step 3: Containment

The objective of containment is to prevent further impact while investigation continues.

Potential containment activities include: Isolating affected services, Disabling problematic model versions, Rolling back deployments, Switching to backup systems, Redirecting traffic, Restricting external access, Suspending affected workflows

For AI-specific incidents, containment may involve: Disabling unsafe prompts, Reverting to a previous model version, Temporarily disabling RAG functionality, Restricting specific use cases, Implementing response filters

Step 4: Investigation and Diagnosis

Technical teams perform detailed analysis to identify the root cause.

Investigation activities may include: Log analysis, Infrastructure monitoring review, Database inspection, Security audit review, Model behavior analysis, Prompt execution tracing, Vector database validation, API transaction analysis

For AI-related incidents, investigation should assess: Model drift indicators, Data quality issues, Prompt changes, Knowledge base updates, Embedding integrity, Third-party model provider status, Hallucination patterns, Retrieval effectiveness

Findings should be continuously documented.

Step 5: Resolution and Recovery

Once the root cause is identified, corrective actions are implemented.

Following implementation, validation testing is performed to confirm:

- Service functionality
- Performance stability
- Security integrity
- Data accuracy
- AI response quality

Services are restored only after successful validation.

Step 6: Monitoring and Stabilization

After recovery, systems remain under enhanced observation to ensure stability.

Monitoring activities include: Performance tracking, Error monitoring, Security surveillance, User experience validation, AI output verification

Additional safeguards may be implemented to prevent recurrence during the stabilization period.

Step 7: Incident Closure

An incident may be closed when:

- Service is restored
- Impact has been eliminated
- Root cause is documented
- Stakeholders are informed
- Required follow-up actions are identified

The Incident Manager ensures all documentation is complete before closure.

6. Escalation Process

Escalation ensures that incidents receive appropriate attention and expertise as their complexity or business impact increases.

Level 1 – Operations and Support Team

Responsibilities include: Incident intake, Initial triage, Monitoring review, Basic troubleshooting, Incident logging

Level 1 personnel attempt immediate remediation where possible.

Level 2 – Engineering Teams

Includes: Platform Engineering, DevOps Engineering, AI/ML Engineering, Data Engineering, Application Development Teams

Responsibilities include:

- Advanced troubleshooting
- Root cause investigation
- Technical remediation
- Recovery execution

Issues beyond operational capabilities are escalated to this level.

Level 3 – Technical Leadership

Includes: Engineering Managers, AI Practice Leads, Security Leads, Architecture Teams, Product Leadership

Responsibilities include:

- Strategic decision-making
- Risk assessment
- Cross-functional coordination
- Prioritization of recovery efforts
- Customer impact management

This level becomes involved when incidents present significant business or technical risk.

Level 4 – Executive Leadership

Includes: Chief Technology Officer, Chief Information Officer, Executive Sponsors, Business Leadership

Responsibilities include:

- Executive oversight
- Regulatory decision-making
- External communications approval
- Business continuity decisions
- Customer and partner engagement

Executive involvement is required for incidents with significant business, legal, financial, or reputational implications.

7. Communication Management

Effective communication is critical throughout the incident lifecycle.

Communication objectives include:

- Maintaining stakeholder awareness
- Providing accurate status updates
- Managing customer expectations
- Coordinating technical teams



- Supporting executive decision-making

Stakeholders may include:

- Internal support teams
- Engineering teams
- Product owners
- Business leaders
- Customers
- Vendors and partners
- Regulatory bodies where applicable

Communications should focus on:

- Incident summary
- Business impact
- Current status
- Actions underway
- Known risks
- Expected next steps

Only authorized personnel should communicate externally regarding active incidents.

8. Post-Incident Review and Continuous Improvement

All major incidents should undergo a formal Post-Incident Review (PIR) or Root Cause Analysis (RCA).

The review should identify:

- What happened
- Why it happened
- Contributing factors
- Detection effectiveness
- Response effectiveness
- Escalation effectiveness
- Communication effectiveness
- Preventive improvements

Special attention should be given to AI-related causes such as:

- Model drift
- Prompt design weaknesses
- Knowledge base quality issues
- Data governance gaps
- AI safety control failures

The review will result in documented corrective and preventive actions that are tracked through completion.



9. Governance and Compliance

Incident records, investigation findings, communications, and corrective actions shall be maintained in accordance with organizational governance, audit, security, and compliance requirements.

Periodic reviews of incident trends should be conducted to identify recurring issues, operational risks, and opportunities for platform improvement.

Through disciplined incident management, structured escalation, and continuous learning, the AI platform can maintain operational resilience, ensure responsible AI usage, improve service reliability, and deliver a consistent experience to customers and stakeholders.

Incidents are logged through a shared helpdesk system (RITES-accessible), classified by severity, assigned to the appropriate support tier, and tracked to resolution with SLA compliance monitoring. Automated alerting (metrics collection system AlertManager → PagerDuty/email) notifies the L2/L3 team of Critical incidents within 5 minutes of detection, before a ticket may even be raised. A daily incident summary is shared with the RITES project manager; a monthly SLA compliance report is submitted to RITES management.

13.3 System Monitoring, Performance Reporting and Updates

The AI Platform shall be continuously monitored using a comprehensive observability framework designed to provide real-time visibility into platform health, performance, reliability, security, and operational efficiency. The monitoring framework combines metrics collection, centralized logging, distributed tracing, alerting, and reporting capabilities to ensure proactive identification and resolution of issues before they impact business operations.

Continuous Monitoring and Observability

The platform operations team shall maintain 24x7 monitoring of all production environments through an integrated observability stack. Monitoring shall cover infrastructure components, application services, APIs, AI services, databases, data pipelines, integration layers, and supporting cloud resources.

Key operational metrics shall be collected and analyzed, including but not limited to:

- API response times and latency trends
- Transaction success and failure rates
- Application and service availability
- Error rates and exception frequencies
- CPU, memory, storage, and network utilization
- Database performance indicators
- AI model response times and throughput
- Data pipeline processing status
- Security and authentication events
- Capacity utilization and resource consumption
- Service availability and operational performance indicators



Metrics shall be aggregated through the platform's metrics collection system and visualized through centralized observability dashboards. These dashboards shall provide both operational and executive-level views to support technical troubleshooting, service management, and business oversight.

Log Management and Analysis

All platform components shall generate structured operational, application, audit, and security logs that are centrally collected and stored within the log aggregation platform. Logs shall be retained in accordance with agreed retention policies and made available for troubleshooting, forensic investigations, compliance reviews, and performance analysis.

The operations team shall continuously monitor logs for:

- System failures and exceptions
- Application errors
- Security incidents
- Integration failures
- Performance degradation indicators
- Infrastructure anomalies
- Unauthorized access attempts

Automated alerting mechanisms shall notify support teams when predefined thresholds or abnormal patterns are detected.

Distributed Tracing and Root Cause Analysis

The platform shall utilize distributed tracing capabilities to track requests and transactions across multiple services and components. This enables end-to-end visibility into user requests, API calls, backend services, AI workflows, and external integrations.

Distributed tracing data shall support:

- Performance optimization
- Bottleneck identification
- Dependency analysis
- Incident investigation
- Root cause determination
- Service reliability improvements

The operations team shall leverage tracing information during incident management activities to accelerate diagnosis and resolution.

Dashboard Access and Visibility

RITES administrators shall be provided read-only access to designated operational dashboards. These dashboards shall provide visibility into key platform health indicators, service performance metrics, infrastructure utilization, active alerts, and historical trends.

Access controls shall ensure that dashboard visibility is restricted to authorized personnel while maintaining appropriate security and governance requirements.

Performance Reporting

A comprehensive performance report shall be prepared and shared with RITES on a monthly basis. The report shall provide an overview of platform operations and service performance during the reporting period.

The report shall include, at a minimum:

- Platform availability and uptime statistics
- Summary of incidents and operational events
- Incident trends and root cause categories
- Service performance analysis
- Resource utilization trends
- Capacity planning observations
- Security event summaries
- Monitoring and alerting effectiveness
- Open operational risks and mitigation actions
- Planned maintenance activities
- Platform enhancement and optimization initiatives

Where applicable, the report shall also include recommendations for performance tuning, infrastructure optimization, capacity expansion, and operational improvements.

Platform Updates and Maintenance

The platform shall be regularly maintained to ensure continued security, reliability, and performance. Maintenance activities may include operating system patches, infrastructure updates, security fixes, dependency upgrades, bug fixes, monitoring enhancements, and minor application releases.

All planned maintenance activities shall be executed during agreed maintenance windows to minimize disruption to business operations. Advance notification shall be provided to RITES prior to the commencement of any scheduled maintenance activity. Notifications shall include the scope of work, affected services, anticipated impact, maintenance schedule, and rollback procedures where applicable.

Following completion of maintenance activities, validation checks shall be performed to confirm service availability, system stability, and successful implementation of updates. Any issues



identified during post-maintenance verification shall be addressed through established incident management procedures.

This monitoring, reporting, and maintenance process shall support the ongoing operational stability, transparency, and continuous improvement of the AI Platform throughout the contract period.

13.4 License and Subscription Management

The License and Subscription Management process establishes the governance, control, monitoring, and reporting mechanisms required to manage all software licenses, subscriptions, and associated entitlements utilized within the AI Platform. The objective of this process is to ensure uninterrupted platform operations, maintain compliance with software licensing obligations, provide transparency into software usage, and minimize operational and financial risks associated with expired or unmanaged licenses.

License Governance and Ownership

GlobalLogic shall be responsible for the procurement, administration, tracking, and management of all licenses and subscriptions required for the development, testing, integration, and deployment of the AI Platform during the period specified in the RFP through the completion of the User Acceptance Testing (UAT) phase.

The scope of license management shall include all proprietary software components, cloud-based subscriptions, development tools, monitoring and observability platforms, security tools, middleware components, database technologies, AI and machine learning services, and any third-party software required for successful delivery of the solution.

A designated owner shall maintain oversight of all software assets and ensure that licensing obligations are fulfilled throughout the lifecycle of the project.

License Inventory Management

A centralized license inventory shall be maintained throughout the project lifecycle. The inventory shall serve as the authoritative repository for all software assets and subscription-based services used within the solution.

The inventory shall capture the following information for each licensed component:

- Software or service name
- Vendor and publisher information
- License type and edition
- Subscription plan details
- License quantity and allocation
- Purchase and activation dates
- Expiration and renewal dates
- Usage restrictions and entitlements



- Associated system or platform component
- License owner and administrator
- Support and maintenance coverage details

To ensure full transparency and traceability, all licensed software components shall be documented within the Software Bill of Materials (SBOM). The SBOM shall be maintained as a living document and updated whenever new software components are introduced, existing components are upgraded, or licensed products are removed from the platform.

License Compliance and Auditing

Periodic reviews shall be conducted to verify that all software usage remains compliant with applicable licensing terms and conditions. The review process shall ensure that deployed software does not exceed licensed entitlements and that all software components are being utilized in accordance with vendor agreements.

Compliance reviews shall include:

- Verification of active licenses
- Validation of license allocations
- Identification of unused or underutilized licenses
- Assessment of license utilization trends
- Review of software version compliance
- Verification of subscription status

Any identified compliance risks or licensing gaps shall be documented and addressed through corrective action plans.

Subscription Monitoring and Renewal Management

GlobalLogic shall continuously monitor all active subscriptions and software licenses to ensure uninterrupted availability of platform services. Automated tracking mechanisms shall be utilized wherever possible to monitor expiration dates, support coverage periods, and subscription validity.

To facilitate continuity planning and timely decision-making, formal renewal notifications shall be provided to RITES at least ninety (90) days prior to the expiration of any software license or subscription requiring customer action, transfer, extension, or procurement.

Renewal notifications shall include:

- License or subscription details
- Expiration date
- Business impact of non-renewal
- Renewal options and recommendations
- Estimated renewal costs, where applicable
- Dependencies on affected platform components

This advance notification process enables adequate time for budget approvals, procurement activities, and operational planning.

Open-Source Software Management

The platform architecture shall prioritize the use of open-source technologies wherever practical and appropriate. The objective is to reduce long-term licensing costs, avoid vendor lock-in, increase architectural flexibility, and support sustainable platform operations.

Open-source components shall undergo security, compliance, and technical evaluations prior to adoption. All approved open-source software shall be recorded within the SBOM together with version information, source repositories, applicable licenses, and known dependencies.

The majority of platform components shall utilize open-source technologies, thereby eliminating recurring licensing costs associated with those components while maintaining enterprise-grade functionality and scalability.

Reporting and Change Management

License inventory status shall be reviewed as part of major release governance activities. Whenever a major release introduces new software components, upgrades existing technologies, or modifies platform dependencies, the license inventory and SBOM shall be updated accordingly.

Periodic reporting shall provide visibility into:

- Active licenses and subscriptions
- Upcoming renewals
- Compliance status
- Newly introduced software components
- Open-source dependency usage
- Licensing risks and mitigation actions

Through disciplined license governance, proactive renewal management, comprehensive inventory tracking, and strategic use of open-source technologies, the platform shall maintain licensing compliance, operational continuity, and cost efficiency throughout the solution lifecycle

.13.5 Platform Evolution and Continuous Improvement

During the AMC period, GlobalLogic implements approved enhancements from the Enhancement backlog (subject to RITES' change approval process), expands knowledge library coverage as new DPRs are appraised, retrains AI/ML models on growing historical data, and incorporates updates to engineering standards. A quarterly platform review meeting with RITES management assesses system performance, identifies improvement opportunities, and prioritizes the enhancement backlog for the next quarter.

AMC Scope and Change Request Boundary: The O&M/AMC scope includes: routine bug fixes, performance optimisations, security patches, minor configuration changes, incremental validation rule

updates reflecting RITES-approved standard revisions, user interface refinements, and minor feature enhancements within the approved SRS scope that do not materially extend the platform's functional boundary. The following are expressly outside AMC scope and shall be treated as formal Change Requests, requiring RITES' written approval and separate agreed pricing:

- New functional modules not specified in the original RFP or approved SRS
- Substantial new sector-specific validation template sets beyond the initial 13 sectors defined in RFP
- AI model retraining triggered by major policy overhauls, new codal revisions, or benchmark changes that materially alter the training corpus scope (incremental fine-tuning on new RITES-approved DPR batches remains within AMC scope)
- Integration with new external government systems or APIs not specified in the RFP (e.g., new ministry data sources, additional e-governance platforms)
- Any scope addition that materially extends the platform's functional boundary beyond the approved SRS. Change Requests are governed by the formal change management process, including impact assessment, written approval by RITES, and agreed milestones and pricing adjustments.

Continuous Learning Governance - All continuous improvement updates including model retraining, validation rule changes, benchmark threshold revisions, and configuration changes to AI scoring parameters follow a mandatory five-step governed pipeline before production deployment:

1. **Proposal:** Change is proposed via the Enhancement Backlog, triaged by GlobalLogic's Platform Lead, and prioritized by RITES
2. **Development:** Change is implemented and version-controlled in Git with full diff traceability and linked to the originating backlog item
3. **Testing:** Updated models and rules are tested against a sector-specific regression test suite (minimum 200 labelled DPR samples per affected sector) and validated against existing SLA accuracy benchmarks
4. **RITES Validation:** RITES SMEs validate the updated capability in the UAT/staging environment; RITES provides written sign-off (email or platform workflow)
5. **Production Promotion:** Only upon written RITES approval is the change promoted to production via the ArgoCD GitOps pipeline with automated rollback capability.

An immutable audit trail maintained in the Audit and Compliance module (Section 5.13) records every model version, change rationale, test result summary, RITES approver identity, and production promotion timestamp. No model retraining output or rule change may be deployed to production without completing steps 3–4 in full.

14. Innovations, Value-Adds and Flexibility Provisions

Capabilities proposed beyond the base RFP scope, at no additional cost, are listed here. These cover DPI integration readiness and sector extensibility beyond the initial infrastructure sectors.

14.1 Proposed Enhancements Beyond Base RFP Scope

GlobalLogic proposes the following value add to RITES current feature sets:

AI Assistant to generate content for DPR Creation: GlobalLogic proposes an AI authoring assistant embedded within the DPR Builder that helps project proponents draft section content, reducing blank-page effort and improving first-submission quality. The assistant generates contextualised draft text by retrieving content from semantically similar approved DPRs in the historical corpus, keeping all suggestions grounded in real precedent rather than general AI synthesis. Guardrails ensure the feature cannot introduce inaccurate codal references or fabricated benchmarks: any engineering standard or threshold cited in a suggestion is validated deterministically against the authoritative standards library before it reaches the author, and AI-generated text is visually distinguished from author-written content at all times. The author retains full editorial control, the assistant accelerates drafting, accountability remains with the proponent.

14.2 Future-Proofing: Emerging National Platforms and DPI Readiness

PARAKH is designed for DPI-era integration readiness. The API-centric design, OpenAPI-compliant interfaces, OGC-compliant GIS services, and canonical data models position the platform to connect to future national digital infrastructure initiatives, including the emerging Digital Infrastructure for Infrastructure Governance (DIIG) ecosystem, without architectural rework. The event streaming backbone can be extended to publish DPR appraisal events to national data exchange platforms as they mature. The Data Lake architecture supports future federation with national data lakes through standardized data sharing protocols.

14.3 Sector Extensibility: Scaling Beyond 13 Sectors

PARAKH's configuration-driven architecture, where templates, validation rules, parameter libraries, and benchmarks are configurable definitions rather than hard-coded logic, allows RITES to extend the platform to additional infrastructure sectors beyond the initial 13 without platform code changes. Adding a new sector requires configuration of templates and parameter libraries (RITES SME-driven), authoring of validation rules (using the rule management console), and assembly of benchmark data, all activities within RITES' administrative capabilities following handover. This extensibility protects RITES' investment and ensures the platform remains relevant as RITES' consultancy scope evolves.

Annexure A Proposed Technology Stack and Tools

All technology product names, versions, and library choices referenced in this proposal are consolidated in the table below. Prose sections throughout this proposal use capability-focused descriptions to remain vendor-neutral; this table is the authoritative technology stack definition.

Layer / Domain	Technology / Tool	Rationale
Container Orchestration	Kubernetes (K8s) + Helm	Cloud-agnostic, open-source, industry-standard; enables HPA and rolling deployments.
CI/CD & GitOps	GitLab CI + ArgoCD + SonarQube + Trivy	End-to-end DevSecOps pipeline with SAST, DAST, and container scanning.
API Gateway	Kong (open source)	OAuth2/OIDC, rate limiting, logging, developer portal; no vendor lock-in.
Event Streaming	Apache Kafka 3.x	High-throughput, fault-tolerant event bus; enables async processing and audit streaming.
Schema Registry	Apicurio Registry (Red Hat / CNCF-aligned)	Open-source Avro schema registry for Kafka topics; enforces BACKWARD_TRANSITIVE schema compatibility; prevents malformed events propagating downstream. Self-hosted OSS available at zero cost; Red Hat subscription optional.
Frontend	React.js 18 + TypeScript + MapLibre GL + PWA	SPA, responsive, offline-capable; MapLibre for OGC-compliant map rendering.
Backend - Enterprise	Java 25 + Spring Boot 4.0.5	Transactional integrity, mature ecosystem, high-throughput API serving.
Backend - AI/ML	Python 3.11 (or latest) + FastAPI	Native ML library ecosystem; async, performant API serving for inference.
Database - Transactional	PostgreSQL 15 (or latest) + PostGIS + Patroni	Open-source, ACID-compliant, geospatial-capable; Patroni for HA failover.
Database - Document	PostgreSQL JSONB	Flexible schema for heterogeneous DPR section data.
Database – Graph	Neo4j Community Edition 5.x	Open-source graph database (Apache 2.0 compatible for CE); models ontology, parameter dependencies, and codal reference relationships. Community Edition is free with no per-node licence cost. Enterprise clustering optional for production.



Database – Vector	Qdrant 1.x	Open-source vector database (Rust, Apache 2.0); high-performance ANN search with payload filtering for semantic DPR retrieval and RAG. On-premises deployable. Used for dedicated vector workloads requiring higher throughput than pgvector.
Cache	Redis 7	Session management, API caching, leaderboard/queue data structures.
Data Lake - Storage	MinIO (S3-compatible)	Open-source, on-prem deployable; erasure-coded for fault tolerance.
Data Lake - Table Format	Apache Iceberg	Open table format; schema evolution, time-travel, partition pruning.
Data Lake - Processing	Apache Spark 3.5	Batch and micro-batch transformation; integrates with Iceberg and Airflow.
Data Lake - Orchestration	Apache Airflow 2.x	DAG-based pipeline orchestration; versioned, observable.
Data Catalog	OpenMetadata	Open-source; asset discovery, lineage, quality dashboards.
Search	OpenSearch 2.x + pgvector	Full-text and faceted search for DPR content, metadata, and audit events. pgvector retained for lightweight embedding queries within PostgreSQL where dedicated Qdrant capacity is not required. OpenSearch also handles search analytics and anomaly detection on audit log streams.
GIS Server	GeoServer 2.x / ArcGIS / ESRI + PostGIS	OGC-compliant (WMS, WFS, WMTS, WCS); open-source, widely adopted.
BIM Processing	IfcOpenShell	Open-source IFC 4.x processor; vendor-neutral.
Identity & Auth	Keycloak 24	OAuth2.0, OIDC, SAML 2.0; LDAP/AD federation; open-source.
NLP & Document AI	spaCy 3.x + HuggingFace Transformers + Tesseract 5	Open-source NLP stack; domain fine-tuning on infra vocabulary.



Document Processing	Apache Tika 2.x	Open-source multi-format document extraction engine (Apache 2.0); extracts raw text and metadata from 1,500+ MIME types including PDF, DOCX, XLSX, ODP, and scanned image PDFs. First stage of the DPR ingestion pipeline before Tesseract OCR and spaCy NLP processing.
Cross-Browser Testing	BrowserStack Automate + App Automate	Commercial cloud-based cross-browser and cross-device test automation. Used for automated regression testing of the PARAKH web UI across Chrome/Firefox/Safari/Edge and mobile apps (Android/iOS) across a real-device matrix. Subscription-based; included in project cost.
AI Models	Self-hosted open-source LLMs: Mistral 7B/Mixtral 8x7B (Apache 2.0), Meta Llama 3.1 8B/70B (Llama Community Licence), Google Gemma 2B/7B (Apache 2.0), served via vLLM inference engine on RITES-controlled GPU infrastructure (production). AWS Bedrock (Llama 3.1 70B/8B) used solely in GlobalLogic Dev/QA environments with synthetic/mock data, no RITES DPR data processed on Bedrock.	Self-hosted deployment ensures complete RITES data sovereignty, no DPR data leaves RITES-controlled boundaries. Open-source licences (Apache 2.0/Llama Community) permit commercial deployment without royalty. vLLM provides high-throughput inference via PagedAttention. Bedrock accelerates GlobalLogic Dev/QA without incurring RITES data-sovereignty risk. Model selection: Llama 3.1 70B for complex appraisal reasoning; 8B for fast NLP classification and entity extraction tasks.
ML Frameworks	scikit-learn + PyTorch + XGBoost + LightGBM	Open-source; CPU-primary inference; GPU optional for training.
ML Inference Server	vLLM for LLM Inference and for Classical ML - NVIDIA Triton Inference Server	Model-agnostic serving; supports ONNX, PyTorch, scikit-learn models.
Explainability	SHAP + LIME	Open-source XAI; feature attribution and local explanations.
MLOps	MLflow, LangSmith	Experiment tracking, model registry, artifact management; open-source.



Observability	Prometheus + Grafana + Loki + Tempo + Opik + Evidently AI	Full-stack metrics, logs, and traces; open-source; no vendor lock-in.
Secrets Management	HashiCorp Vault	Dynamic secrets, encryption-as-a-service; open-source community edition.
Security Testing	OWASP ZAP + Trivy + OWASP Dependency-Check	DAST, container, and dependency scanning integrated in CI.
Infrastructure as Code	Terraform + Ansible + Helm	Multi-cloud IaC; configuration management; K8s package management.
MDM Platform	Custom	Custom-built MDM module in phase 1. Later, to be evaluated for any COTS product.
Dashboards / Analytics UI	Apache Superset	Open-source BI; interactive dashboards; SQL-based exploration.

Annexure B Infrastructure Bill of Materials

This annexure presents the technical Infrastructure Bill of Materials (BOM) for the PARAKH platform, which mandates an unpriced BoM in the Technical Bid. This annexure contains zero pricing information; all costs are declared exclusively in Volume III (Financial Proposal / Annexure 07). Items are grouped by category: Compute, Database, Storage, Networking, AI/ML, Document Processing, Notifications/Integration, Testing, Security, and Licences. Infrastructure sizing is based on GlobalLogic's technical estimates for the Dev/QA/UAT phases; production sizing will be finalised with RITES during the Milestone 1 SRS workshop.

#	Component / Service	Category	Specification / Configuration	Phase	Purpose in PARAKH
1	App / Orchestration Servers (managed container orchestration service)	Compute	ARM-based compute instance: m7g.xlarge (x2 Dev, x3 QA), m7g.2xlarge (x4 UAT); managed container orchestration platform; Horizontal Pod Autoscaler (HPA) enabled	Dev / QA / UAT	container orchestration platform control plane and application microservice pods

2	Vector Database	Database	vector database 1.x on r7g.large Graviton3: x1 Dev, x1 QA, x2 UAT (replication); Apache 2.0 open-source	Dev / QA / UAT	Semantic similarity search; RAG retrieval; near-peer DPR comparison
3	Graph Database	Database	graph database 5.x: t4g.micro Dev, r7g.large QA, m7g.large UAT; CE is free; Enterprise optional for production clustering	Dev / QA / UAT	Ontology and dependency graph; codal reference linkage
4	Relational Database	Database	relational database RDS + geospatial database extension: db.t4g.medium Dev, db.t4g.large QA, db.m7g.large Multi-AZ UAT; Patroni for HA failover	Dev / QA / UAT	Transactional DPR data; structured appraisal records; geospatial data
5	Distributed Cache	Cache	in-memory cache ElastiCache: cache.t4g.micro Dev, cache.t4g.small QA, cache.t4g.small x3 cluster UAT	Dev / QA / UAT	Session management; API response caching; queue data structures
6	Object Storage	Storage	object storage layer (S3-compatible, open-source) + EBS GP3: 300 GB Dev, 600 GB QA, 1 TB UAT; S3 Intelligent-Tiering for active objects; S3 Glacier for audit log archival (7-year retention)	Dev / QA / UAT	DPR PDFs, IFC models, GeoJSON, scanned images; audit archive
7	Message Queue / Event Bus	Messaging	event streaming platform (managed or self-hosted on managed container orchestration service); schema registry for Avro schema governance; exactly-once semantics via transactional event delivery	Dev / QA / UAT	DPR lifecycle events; async validation pipeline; audit event streaming
8	Load Balancer + WAF	Networking	application load balancer + cloud provider WAF with OWASP managed ruleset; SSL termination; L7 routing	Dev / QA / UAT	HTTPS ingress; web application firewall; DDoS protection



9	Monitoring and Observability Stack	Observability	observability dashboard + metrics collection system + Loki + Tempo: t4g.medium x1 Dev, x2 QA, x3 UAT; Opik + Evidently AI for AI/ML pipeline monitoring	Dev / QA / UAT	Metrics, logs, distributed traces; SLA dashboards; alerting
10	LLM Inference	AI / ML	Production: self-hosted open-source LLM serving framework (open-source, Apache 2.0) serving open-source language model family on RITES-designated GPU nodes. Dev/QA (GlobalLogic infra): open-source language models on-demand	Dev (managed AI service) / Production (self-hosted)	16-module DPR appraisal inference; NLP entity extraction; summarisation
11	Embedding Models	AI / ML	nomic-embed-text-v1.5 + nomic-embed-vision-v1.5 (Nomic AI, US, MIT) CPU-primary inference; GPU-optional for batch corpus ingestion or for vision embedding model	Dev / QA / UAT	Vector embeddings for vector database; semantic search; RAG retrieval
12	LLM Fine-tuning (Dev phase only)	AI / ML	QLoRA / GaLore via Unsloth + PEFT on spot GPU instances (g5.2xlarge or equivalent GPU instance); Dev phase only; approx. 200 GPU-hours	Dev only	Domain adaptation of foundation model on infrastructure engineering corpus
13	Historical DPR Ingestion Pipeline	Document Processing	Document parsing library 2.x (multi-format extraction) + Tesseract 5 OCR (scanned PDF fallback); ~1,000 DPRs x ~400 pages average; one-time ingestion at project start	One-time (Dev)	Digitise and structure RITES legacy DPR corpus into the Data Lake
14	CDN (Content Delivery Network)	Networking	CDN service; static web and mobile assets; GIS map tile caching; geo-distributed edge nodes	Dev / QA / UAT	Mobile application performance; responsive GIS map tile delivery



15	SMS / OTP Gateway	Notifications	MSG91 or SMS notification service; approx. 5,000 SMS per month (UAT/Production scale)	Dev / QA / UAT	OTP-based authentication; workflow status alerts
16	Email Notification Service	Notifications	email relay service; approx. 50,000 emails per month (UAT/Production scale)	Dev / QA / UAT	Appraisal workflow notifications; milestone alerts; report distribution
17	Push Notification Service	Notifications	Firestore Cloud Messaging (FCM) for Android; Apple Push Notification Service (APNs) for iOS; zero marginal cost at planned volumes	Dev / QA / UAT	Mobile companion app workflow push notifications
18	eSign / Digital Signature API	Integration	NIC eSign (Aadhaar-based digital signature); REST API integration; approx. 200 DPR signings per month; annual NIC access subscription	UAT / Production	DPR digital signature; formal appraisal decision approval
19	Cross-Browser and Cross-Device Test Platform	Testing	cross-browser test automation platform (web UI automation across Chrome/Firefox/Safari/Edge) + App Automate (Android/iOS device matrix); subscription	Dev / QA	Automated regression testing across browser and device matrix
20	Vulnerability Assessment and Penetration Testing (VAPT)	Security	CERT-In empanelled agency; scope: web application, API layer, infrastructure, AI/ML components, container images; mandatory at pre-UAT stage and pre-Go-Live; cost included in GlobalLogic bid price	Pre-UAT; Pre-Go-Live	Government security compliance per MeitY/NIC/STQC requirements
21	DPDP Act 2023 Compliance Audit	Security / Compliance	Independent audit agency; data mapping, consent framework design, Data Protection Officer (DPO) advisory review	Pre-Go-Live	Digital Personal Data Protection Act 2023 compliance



22	SAST / SCA Scanning Tools	Security	Semgrep OSS (SAST) + OWASP Dependency-Check (SCA) + Trivy (container); integrated in CI/CD pipeline pipeline; zero licence cost (open-source)	Continuous (all phases)	Static analysis and supply-chain vulnerability scanning in CI/CD
23	Google Play Developer Account	Licence	One-time developer account registration; account maintained in RITES' name for perpetual app store access	Pre-UAT	Mandatory for Android companion app distribution on Google Play Store
24	Apple Developer Program	Licence	Annual developer programme membership; account ownership transferred to RITES at contract completion	Pre-UAT	Mandatory for iOS companion app distribution on Apple App Store
25	SSL / TLS Wildcard Certificate	Licence	*.rites-parakh.gov.in (or equivalent GOI-designated domain); annual renewal; cloud-managed TLS certificate available at zero cost if RITES hosts on cloud provider	Pre-Go-Live	HTTPS for all platform endpoints and mobile apps
26	Mobile Code-Signing Certificate	Licence	Android APK signing certificate + iOS distribution certificate; annual renewal; maintained for full contract duration	Pre-UAT	App store compliance and device trust for companion mobile apps

- End of Technical Proposal -

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